Self-similar blowup in wave models from industry

and its stability via the method of Donninger et al.

Research notes (AI-assisted draft, multi-agent)

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Abstract

We survey wave-type models from engineering in which finite-time singularities form with self-similar structure. The models are electrostatic MEMS with inertia, thermal explosion under Maxwell–Cattaneo heat conduction, strings and membranes on softening foundations (φ^4/Klein–Gordon), nonlinear acoustics, director fields (wave maps) and relativistic membranes. For each we analyse the stability of the self-similar blowup with the method of Donninger and coauthors (similarity coordinates, semigroup theory, spectral analysis, modulation). For every ODE-type blowup with scaling exponent $s > -1$, the linearisation turns out to have a universal spectrum $\{1 - \ell - 2n\} \cup \{-2s - 2 - \ell - 2n\}$, whose only non-negative points come from time translation and Lorentz boosts. For the hyperbolic MEMS equation $w_{tt} + \gamma w_t - \Delta w = -\lambda w^{-q}$, $q > 1$, in dimensions $d \leq 3$, this yields (with full spectral analysis and a proof sketch of the nonlinear step) that touchdown $w \simeq c(T-t)^{2/(q+1)}$ with a flat profile in the light cone is stable modulo symmetries, also in the presence of damping, with rate $(T-t)^{\omega}$, $\omega < \min\{1, 2(q-1)/(q+1)\}$. Non-trivial focused touchdown profiles are found numerically in dimensions 2 and 3; numerically they carry additional unstable modes. We also identify where the method fails or the model changes the singularity type: the full Cattaneo model has a parabolic blowup rate, Arrhenius kinetics prevents blowup, and quasilinear acoustic and membrane models remain open. The spectra at the ODE profiles are cross-checked by high-precision collocation and, in selected cases, Evans-function tests. Direct simulations of the clamped MEMS device confirm the touchdown exponent, the constant c and the convergence rate 2/3. They also show that exponents fitted at moderate gaps are biased low.

Contents

1 Introduction **3**

2 Industrial wave models with self-similar blowup: a survey **4**
2.1 Electrostatic MEMS with inertia: touchdown as quenching 5
2.2 Hyperbolic heat conduction and thermal explosion 5
2.3 Softening foundations and focusing Klein–Gordon equations 6
2.4 Nonlinear acoustics: derivative-quadratic wave equations 6
2.5 Nematic liquid crystals, σ-models and wave maps 7
2.6 Relativistic strings, membranes and Born–Infeld 7
2.7 Further models 7
2.8 The Donninger et al. method: literature 7
2.9 Novelty assessment 9

3 The Donninger et al. method for ODE-type blowup **10**
3.1 Setting and self-similar variables 10
3.2 ODE blowup and a universal spectrum 11
3.3 The five steps 12

3.4 Lower-order terms . . . 13
3.5 What the method gives, and what it does not . . . 14

4 Hyperbolic MEMS: stability of self-similar touchdown 14
4.1 The model . . . 14
4.2 ODE touchdown and its symmetry orbit . . . 15
4.3 Self-similar profiles . . . 16
4.4 Similarity variables and the free operator . . . 18
4.5 Spectral analysis . . . 19
4.6 Nonlinear stability of ODE touchdown . . . 20
4.7 Damping and other integrable perturbations . . . 21
4.8 Physical interpretation . . . 22
4.9 Numerical checks . . . 23
4.10 Relation to Gimperlein–He–Lacey . . . 23
4.11 Caveats and open points . . . 24

5 Hyperbolic thermal explosion: exponential nonlinearities 24
5.1 Modelling: Cattaneo flux and Frank-Kamenetskii source . . . 25
5.2 Reduced model: ODE blowup, symmetries, similarity variables . . . 26
5.3 Linearised operator and its spectrum . . . 27
5.4 Nonlinear stability of the ODE blowup (reduced model) . . . 29
5.5 An exact check in one dimension: Liouville's equation . . . 30
5.6 The full Cattaneo model: a different blowup . . . 31
5.7 Arrhenius kinetics: no blowup . . . 33
5.8 Status, literature and caveats . . . 33

6 The power family, Klein–Gordon and lower-order perturbations 34
6.1 Industrial models and the unified family . . . 34
6.2 ODE blowup and the excluded range $-1 \le p \le 1$ . . . 35
6.3 The linearised operator and the free flow . . . 35
6.4 The spectral theorem in all angular sectors . . . 36
6.5 What is proved in the literature . . . 37
6.6 Lower-order perturbations: damping and restoring forces . . . 38
6.7 Non-ODE self-similar profiles . . . 40
6.8 Numerical confirmation . . . 40
6.9 Summary and caveats . . . 41

7 Geometric models: wave maps and relativistic membranes 42
7.1 Wave maps: origin, scaling, and what is industrially relevant . . . 42
7.2 Co-rotational wave maps and similarity variables . . . 42
7.3 The spectral problem and how mode stability is proved . . . 43
7.4 Numerically computed spectrum . . . 45
7.5 Known stability theorems . . . 46
7.6 Relativistic membranes and the scalar Born–Infeld equation . . . 46
7.7 Assessment and open problems . . . 48

8 Nonlinear acoustics and derivative nonlinearities 48
8.1 Models . . . 49
8.2 Shocks versus ODE-type blowup . . . 49
8.3 The Riccati wave equation: symmetries, tilted blowup, similarity variables . . . 50
8.4 Free flow, spectrum, and the neutral modes . . . 51
8.5 Nonlinear stability of the Riccati blowup . . . 52

8.6 The null-form model: blowup is touchdown of a free wave . . . 53
8.7 Westervelt and Kuznetsov: degenerate self-similar blowup . . . 54
8.8 The viscous term is not a lower-order perturbation . . . 55
8.9 Numerical confirmation . . . 55
8.10 Summary and caveats . . . 56

9 Numerical verification **57**
9.1 Spectra of the linearised operators . . . 57
9.2 Touchdown dynamics of the hyperbolic MEMS equation . . . 60
9.3 Asymmetric data: flat or boosted profile? . . . 63
9.4 Exponential nonlinearity in one dimension . . . 63
9.5 Discrepancies, caveats and reproducibility . . . 64

10 Summary, engineering consequences and open problems **65**
10.1 What the analysis says about the devices . . . 65
10.2 Open problems suggested by these notes . . . 67

1 Introduction

Many engineering devices are governed, at least in an inertial regime, by semilinear or quasilinear wave equations whose solutions can lose regularity in finite time. Examples are the *touchdown* (pull-in) of an electrostatically actuated MEMS membrane, *thermal runaway* under hyperbolic (Maxwell–Cattaneo) heat conduction, the collapse of strings and membranes on softening foundations, the degeneration of high-intensity acoustic waves, and singularities of director fields or relativistic membranes. In each case the most robust singularity is *self-similar*: near the singular point (T, x_0) the solution has the form $(T-t)^{-s}\,\Psi\big(\frac{x-x_0}{T-t}\big)$ (or a logarithmic analogue). For the engineer the relevant questions are: what is the singularity rate and the local profile, and are they *robust* under perturbations of the data and of lower-order model terms (damping, restoring forces, inhomogeneous material parameters)?

Over the last fifteen years Donninger and coauthors have developed a systematic way to answer the robustness question rigorously [26, 38, 39, 40, 41, 20, 107, 61, 29]. One passes to similarity coordinates in the backward light cone, which turns the self-similar solution into a stationary state of an autonomous evolution equation. One then shows that the linearised operator generates a semigroup on a high-order Sobolev space, with spectrum consisting of finitely many unstable eigenvalues that come from symmetries, plus a stable part with a gap. Finally one closes a nonlinear argument by modulating the symmetry parameters. These notes apply this programme to a list of wave models from engineering and applied physics.

Results in brief.

1. *A universal spectrum for ODE-type blowup* (Theorem 3.2). If $u_T(t) = \kappa(T-t)^{-s}$ is a spatially homogeneous blowup solution with scaling exponent $s > -1$, then the potential in the linearisation is forced to be $V_0 = (s+1)(s+2)$. In the light cone the spectrum is $\{1-\ell-2n\} \cup \{-2s-2-\ell-2n\}$, $\ell, n \in \mathbb{N}_0$, with the time translation ($\lambda = 1$) and Lorentz boosts ($\lambda = 0$, $\ell = 1$) as the only non-negative modes, and spectral gap $\min\{1, 2s+2\}$. This covers focusing powers ($s = \frac{2}{p-1}$), the MEMS force w^{-q} ($s = -\frac{2}{q+1}$), exponential sources ($s = 0$), and, as the borderline $s = -1$, the derivative-quadratic acoustic toy model (Section 8).

2. *Hyperbolic MEMS* (Section 4). Touchdown of $w_{tt} + \gamma w_t - \Delta w = -\lambda w^{-q}$, $q > 1$, is asymptotically self-similar with rate $w \simeq c(T-t)^{2/(q+1)}$ and a flat profile in the light cone. For $d \le 3$ it is nonlinearly stable (proof sketch, Theorem 4.10) modulo time translation and

Lorentz boosts, with convergence rate $(T-t)^\omega$, $\omega < \min\{1, \frac{2(q-1)}{q+1}\}$ ($= \frac{2}{3}$ for the Coulomb force). Damping does not change this. In $d = 1$ the ODE profile and its boosts are the only self-similar profiles. In $d = 2, 3$ we find numerically a sequence (conjecturally infinite) of focused profiles; the first two have extra unstable directions. Once the proof sketches are completed (see the caveats in Section 4.11), these results put the formal and numerical picture of [57, 73] on a rigorous footing.

3. *Thermal explosion* (Section 5), *softening foundations and Klein–Gordon* (Section 6), *wave maps and membranes* (Section 7), *nonlinear acoustics* (Section 8). See Table 12 for the status of each model.

4. *Independent numerics* (Section 9). The spectra at the ODE profiles are confirmed by collocation in high precision. Direct simulations of the MEMS equation with clamped boundary conditions reproduce the touchdown rate, the constant c and the convergence to the flat profile at the predicted rate $(T-t)^{2/3}$. For asymmetric data they show a flat profile at the first touchdown point and boosted profiles with $a = T'(x_0)$ elsewhere on the blowup curve.

How to read these notes. Section 2 surveys the models and the literature, including a novelty check. Section 3 isolates the common part of the method. The model sections can then be read independently. Each one labels its statements as known, proved or sketched here, or conjectural/numerical.

Provenance and status. These notes were drafted by a team of AI agents: one for the literature, one for each model class and one independent numerical verifier. A coordinating agent cross-checked their results and assembled the text. The key formulas were checked symbolically or numerically by scripts in `code/`. The nonlinear stability arguments are *proof sketches* in the style of the cited works, not complete proofs. Several references could only be verified from abstracts (network access to arXiv was blocked); such places are marked `% TODO verify` in the sources. The text should be read as a detailed research plan and a set of verified computations, not as a finished paper.

2 Industrial wave models with self-similar blowup: a survey

All models collected below are, after nondimensionalisation, of the form

$$\varepsilon^2 \partial_t^2 u + \gamma\, \partial_t u - \Delta u + m^2 u = F(u) \quad \text{or} \quad F(u, \partial u), \tag{2.1}$$

possibly with a fourth-order or quasilinear principal part. Near a singularity at (T, x_0) the time scale $T - t$ is short compared with the relaxation time ε^2/γ, so after rescaling t the principal part is the wave operator and $\gamma\partial_t u$, $m^2 u$ are lower order (in the similarity variables of Section 3 they carry the factors $e^{-\tau}$ and $e^{-2\tau}$). The prototype singularity is the spatially homogeneous ("ODE") solution u_T of $u'' = F(u)$, which is self-similar: $u_T = \kappa(T-t)^{-s}$ or $u_T = -2\log(T-t) + \text{const}$. We say *blowup* if $|u| \to \infty$ and *quenching* (in MEMS: *touchdown*) if u stays bounded while $F(u)$ and $\partial_t u$ become unbounded; in similarity variables the two cases differ only in the sign of the scaling exponent s ($s < 0$ for quenching). For each model class we record the equation and its industrial origin, known blowup/quenching results, self-similar solutions and stability results; a summary is given in Table 1.

2.1 Electrostatic MEMS with inertia: touchdown as quenching

In the small-aspect-ratio limit, the deflection $u \in (-1, 0]$ of an electrostatically actuated elastic membrane suspended above a rigid ground plate (gap $1 + u$) satisfies

$$\varepsilon^2 u_{tt} + u_t - \Delta u = -\frac{\lambda f(x)}{(1+u)^2} \quad \text{in } \Omega, \qquad u = 0 \text{ on } \partial\Omega, \tag{2.2}$$

where ε^2 is the ratio of inertial to damping forces, $\lambda \propto V^2$ measures the applied voltage and f the dielectric profile; see the monographs [108, 49, 83], the survey [91], and [51, 50] for the hyperbolic model (2.2). Variants include a nonlocal version, in which λ is replaced by $\lambda/(1 + \alpha \int_\Omega (1+u)^{-1}\, dx)^2$ (device in series with a capacitor) [81, 83], a fringing-field correction [102], and plate models with $\Delta^2 u$ [90, 91]. At nanometre gaps, van der Waals and Casimir forces (scaling like w^{-3} and w^{-4} in the gap w) compete with the electrostatic force w^{-2} and can cause collapse even without applied voltage [110, 3]. With $w := 1 + u$ and in the inertia-dominated regime one is led to the model

$$w_{tt} - \Delta w = -\lambda\, w^{-q}, \qquad q > 1 \quad (q = 2 \text{ electrostatic},\ q = 3 \text{ van der Waals},\ q = 4 \text{ Casimir}). \tag{2.3}$$

ODE quenching. Equation (2.3) has the self-similar solution

$$w_T(t) = \kappa_q\, (T - t)^\sigma, \qquad \sigma = \tfrac{2}{q+1}, \qquad \kappa_q^{q+1} = \frac{\lambda(q+1)^2}{2(q-1)} \qquad (q = 2:\ \sigma = \tfrac{2}{3},\ \kappa_2^3 = \tfrac{9\lambda}{2}), \tag{2.4}$$

for which $w_T \to 0$ and $\partial_t w_T \to -\infty$ as $t \to T$. In the notation of Section 3, $s = -\sigma$ and $V_0 = F'(w_T)(T-t)^2 = 2q(q-1)/(q+1)^2$; the roots of $a^2 + a - V_0 = 0$ are $a_1 = 1 - \sigma$ and $a_2 = -(2 - \sigma)$, so the candidate eigenvalues are $1 - \ell - 2n$ and $-2(q-1)/(q+1) - \ell - 2n$. The only non-negative eigenvalues are 1 (time translation) and 0 with $\ell = 1$ (Lorentz boosts), exactly as for focusing power nonlinearities. (This is the formal count; it is verified in the model sections.)

Known results. Local well-posedness, global existence for small λ and finite-time touchdown for large λ are known for (2.2) and its variants [15, 51, 66, 81, 50, 82, 102]; [50] shows numerically that the dynamic pull-in threshold lies below the static one and depends on ε, and [82] studies the quenching behaviour of $u_{tt} = u_{xx} + \lambda(1-u)^{-2}$ in one dimension. For (2.3) with radial data, Gimperlein, He and Lacey [57] (see also [73]) obtain *formally* a sequence of self-similar quenching solutions; formal asymptotics and numerics indicate that radial solutions increasing in $|x|$ quench at the origin like (2.4), that the trivial profile is stable up to the time-translation mode, and that the non-trivial profiles are unstable. For the parabolic MEMS problem, quenching profiles are constructed rigorously in self-similar variables, e.g. [92]. We found *no* rigorous stability result for the hyperbolic touchdown (2.4) (see Section 2.9).

2.2 Hyperbolic heat conduction and thermal explosion

The Maxwell–Cattaneo–Vernotte law $\tau \boldsymbol{q}_t + \boldsymbol{q} = -k\nabla\theta$ [13, 117] (reviews: [76, 114]) replaces Fourier's law when the time scale is comparable with the thermal relaxation time τ (pulsed laser heating, fast reactions). Combined with the energy balance $\rho c\, \theta_t + \nabla \cdot \boldsymbol{q} = Q(\theta)$ with an Arrhenius or Frank-Kamenetskii source $Q(\theta) = \delta e^\theta$ (thermal explosion, combustion [4]), elimination of $\boldsymbol{q}$ gives

$$\tau \rho c\, \theta_{tt} + \rho c\, \theta_t - k\Delta\theta = Q(\theta) + \tau Q'(\theta)\, \theta_t. \tag{2.5}$$

The "hyperbolic Gelfand problem" usually studied is the semilinear model

$$u_{tt} + \gamma u_t - \Delta u = \delta e^u, \qquad u_T(t) = \log \tfrac{2}{\delta} - 2\log(T - t), \tag{2.6}$$

i.e. (2.5) without the last term. A short computation shows that this omission matters: for the full system the flux of a spatially homogeneous state is divergence free, so homogeneous solutions obey the first-order ODE $\rho c\,\theta_t = Q(\theta)$ and blow up like $\theta \approx -\log(T-t)$ (parabolic Frank-Kamenetskii type), and near such a state $\tau Q'(\theta)\theta_t$ is of the same order as $\tau\rho c\,\theta_{tt}$. Thus the wave-type ODE blowup in (2.6) is a feature of the semilinear model (and of Liouville-type equations), not of the Cattaneo system itself; we did not find the singularity formation of (2.5) treated in the literature.

For (2.6): in 1+1 dimensions $u_{tt}-u_{xx}=e^u$ is the classical Liouville equation, whose general solution is explicit. Kichenassamy [84] constructs, by Fuchsian reduction and a Nash–Moser argument and in any dimension, solutions blowing up on a prescribed spacelike hypersurface with logarithmic behaviour, and according to the abstract treats Cauchy data close to constants (we have only seen the abstract). In one dimension, Godin [63] shows for mixed problems that solutions are either global or blow up on a spacelike curve, and Azaiez–Masmoudi–Zaag [1] determine the blowup rate at non-characteristic points for low-regularity data.

2.3 Softening foundations and focusing Klein–Gordon equations

A string or beam on a softening (Duffing-type) elastic foundation, after scaling, gives

$$u_{tt} - u_{xx} + u = |u|^{p-1}u \quad (p=3) \qquad \text{or} \qquad u_{tt} + u_{xxxx} + u = u^3, \tag{2.7}$$

the focusing Klein–Gordon (ϕ^4) equation in the string case. Since the cubic law is a truncated Taylor expansion, blowup signals a loss of stability (snap-through, collapse) rather than physically infinite displacement. Fourth-order suspension-bridge models show finite-time blowup with wide oscillations, which is not self-similar [54, 53]. For (2.7) the ODE profile is $\kappa(T-t)^{-s}$, $s=2/(p-1)$, $\kappa^{p-1}=s(s+1)$ ($\sqrt{2}/(T-t)$ for $p=3$), and the mass term is an $e^{-2\tau}$ perturbation. Blowup rates for focusing NLKG are due to Killip, Stovall and Visan [85] and Hamza–Zaag [71] (one dimension or radial). Donninger and Wallauch [43] show for a class of strongly perturbed supercritical equations in $d=3$ that the ODE profile of the unperturbed equation still describes stable blowup. The damped analogue $u_{tt}+\gamma u_t-\Delta u=|u|^{p-1}u$ is covered by Lyapunov-functional methods in similarity variables, which give the blowup rate for lower-order perturbations [69, 70]; in one dimension, solutions converging exponentially to the ODE profile are constructed in [68].

2.4 Nonlinear acoustics: derivative-quadratic wave equations

High-intensity focused ultrasound (HIFU) and lithotripsy are modelled by the Westervelt and Kuznetsov equations [78]

$$p_{tt} - c^2\Delta p - b\Delta p_t = k\,(p^2)_{tt}, \qquad \psi_{tt} - c^2\Delta\psi - b\Delta\psi_t = \partial_t\big(|\nabla\psi|^2 + \tfrac{B}{2A}c^{-2}\psi_t^2\big). \tag{2.8}$$

With viscosity $b>0$ there is global well-posedness for small data [79]. In the inviscid case (2.8) is a quasilinear wave equation with local well-posedness and an expected gradient blowup (shocks) [24]. Two ODE-type mechanisms are relevant. (i) *Degeneracy*: for $b=0$ the Westervelt equation reads $\partial_t^2(p-kp^2)=c^2\Delta p$; spatially homogeneous solutions have $p-kp^2=A+Bt$, so for $B>0$ the branch with $p<1/(2k)$ satisfies $1-2kp=\sqrt{4kB(T-t)}$, which vanishes at a finite time T with $p_t\sim(T-t)^{-1/2}$: p stays bounded while the equation degenerates, a quasilinear analogue of quenching (our observation; we found no study of it). (ii) *Riccati-type nonlinearities*: the semilinear model $u_{tt}-\Delta u=(\partial_t u)^2$ has the ODE blowup $u_T=-\log(T-t)$. Speck [113] proves stable ODE-type blowup in $d=3$ for a class of quasilinear equations with such derivative-quadratic terms. In one dimension, Gough [65] and Liu–Raees [95] show that no smooth exact self-similar profiles exist, construct generalised self-similar solutions with logarithmic growth, and prove asymptotic stability of the relevant branches (including the ODE

branch). The spatial-derivative analogue is treated in [55]. Note that $u_{tt} - \Delta u = u_t^2 - |\nabla u|^2$ is linearised by $v = e^{-u}$ ($\Box v = 0$), so its blowup is the touchdown $v \to 0$ of a free wave.

2.5 Nematic liquid crystals, σ-models and wave maps

If the inertia of the director field $\boldsymbol{n}$ is retained and the elastic anisotropy and the coupling to flow are neglected, a nematic director $\boldsymbol{n} : \mathbb{R}^{1+3} \to S^2$ formally obeys the wave map equation $\boldsymbol{n}_{tt} - \Delta \boldsymbol{n} = (|\nabla \boldsymbol{n}|^2 - |\boldsymbol{n}_t|^2)\boldsymbol{n}$. With anisotropic (splay/bend) elasticity a planar director angle ψ satisfies the variational wave equation $\psi_{tt} - c(\psi)(c(\psi)\psi_x)_x = 0$, for which Glassey, Hunter and Zheng proved finite-time singularity formation (cusps; gradient blowup, not ODE-type) and nonlinear instability of constant states [58]. Self-similar blowup for co-rotational wave maps $\mathbb{R}^{1+3} \to S^3$ goes back to Shatah [111] and Turok–Spergel [116], with excited profiles in [7] and explicit profiles $\mathbb{R}^{1+d} \to S^d$, $d \geq 4$, in [5]. These are the founding examples of the Donninger programme (Section 2.8). The stability theory is developed for targets S^d with d equal to the space dimension; we are not aware of results specific to the S^2-valued director in three dimensions.

2.6 Relativistic strings, membranes and Born–Infeld

Timelike extremal (zero mean curvature) hypersurfaces in Minkowski space model relativistic strings and membranes and, in $1+1$ dimensions, give the Born–Infeld equation $(1 + u_x^2)u_{tt} - 2u_t u_x u_{tx} - (1 - u_t^2)u_{xx} = 0$. The origin is physical rather than industrial. Eggers and Hoppe describe singularity formation by self-similar solutions [46], extended to membranes in [47]. Yan constructs explicit self-similar solutions and states their stability inside a strict sub-cone of the backward light cone [123, 124]; by Proposition 7.5(c), the explicit solutions $\pm(T-t)\sqrt{1 - |x|^2/(T-t)^2}$ of [124] are null cones, on which the equation degenerates. These equations are quasilinear and lie outside the semilinear framework below.

2.7 Further models

MEMS plates, $u_{tt} + \Delta^2 u = -\lambda(1+u)^{-2}$ [90, 91], have no finite speed of propagation, and their natural scaling is $x \sim (T-t)^{1/2}$, so the light-cone method does not apply directly. Coupled MEMS models, in which the electric potential solves an elliptic problem in the moving gap, are analysed in [73, 91]. Cattaneo-type (hyperbolic) chemotaxis models are a further candidate. We have not reviewed their singularities.

2.8 The Donninger et al. method: literature

The method writes the equation as a first-order system in similarity coordinates on the backward light cone, so that the self-similar profile becomes a static solution. The linearised operator is treated as a compact perturbation of a generator $\mathbf{L}_0$ with explicit growth bound, obtained by Lumer–Phillips in $H^k \times H^{k-1}$. One proves mode stability, removes the symmetry-induced unstable eigenvalues by Riesz projections, modulates the blowup time (and the boost parameters), and closes a fixed-point argument. A survey is [29]. All entries below were checked against arXiv and journal or repository records.

- *Wave maps and Yang–Mills, conditional and mode stability.* [26]: nonlinear stability of the Shatah–Turok–Spergel co-rotational wave map $\mathbb{R}^{1+3} \to S^3$, conditional on mode stability. [42]: the linearised problem; it excludes unstable eigenvalues with large imaginary part or real part $> \frac{1}{2}$. [22]: rigorous mode stability of this wave map, completing the proof (note: wave maps, not Yang–Mills). [27]: conditional stability for $(1+5)$-dimensional equivariant Yang–Mills. [21]: mode stability for that Yang–Mills profile, via explicit approximate ("quasi-") solutions; this completes the stability proof. [59]: stable blowup for Yang–Mills

Model	Equation	Application	Self-similar profile	Symmetry modes	Stability status
Hyperbolic MEMS	$w_{tt}+\gamma w_t-\Delta w=-\lambda w^{-q}$, $q=2,3,4$	MEMS/NEMS touchdown	$\kappa_q(T-t)^{2/(q+1)}$ (quenching); formal non-trivial ones [57]	$\lambda=1$ (time); $\lambda=0$, $\ell=1$ (boosts)	Open in the literature (formal and numerical evidence only); target of this note
Nonlocal MEMS	as above, $\lambda\mapsto\lambda(1+\alpha\int w^{-1})^{-2}$	MEMS with series capacitor	as above (formally)	as above	Open
MEMS plate	$w_{tt}+\Delta^2 w=-\lambda w^{-2}$	MEMS plates	parabolic scaling	n/a	Open; light-cone method not applicable
Hyperbolic Gelfand	$u_{tt}+\gamma u_t-\Delta u=\delta e^u$	thermal explosion (semilinear model)	$\log\frac{2}{\delta}-2\log(T-t)$	$\lambda=1$; $\lambda=0$, $\ell=1$	Smooth data: [84]; 1D: [63, 1]; H^k light-cone stability open
Cattaneo system	(2.5)	laser heating, combustion	$-\log(T-t)$ (parabolic type)	$\lambda=1$; $\lambda=0$, $\ell=1$ (tilts, Section 5.6)	Open (not found)
Focusing KG	$u_{tt}-\Delta u+m^2u=\lvert u\rvert^{p-1}u$	softening foundations, ϕ^4	$\kappa(T-t)^{-2/(p-1)}$	$\lambda=1$; $\lambda=0$, $\ell=1$	Rates [85, 71]; $m=0$: [107]; $m>0$: expected [43]
Damped power	$u_{tt}+\gamma u_t-\Delta u=\lvert u\rvert^{p-1}u$	damped structures	as above	as above	Rates [69, 70]; 1D [68]; H^k, $d\geq 2$ open
Riccati derivative	$u_{tt}-\Delta u=(\partial_t u)^2$	nonlinear acoustics (model)	$-\log(T-t)$; generalised SS in 1D	$\lambda=1$ (time); $\lambda=0$ (shift $u\mapsto u+c$; tilts, $\ell=1$); no boosts	1D [65, 95]; 3D (other method) [113]
Westervelt, $b=0$	$\partial_t^2(p-kp^2)=c^2\Delta p$	HIFU, lithotripsy	$1-2kp\propto(T-t)^{1/2}$	formal: 1; $\frac{1}{4}$ (tilt); 0 (Section 8.7)	Open (quasilinear)
Wave maps	$\Box\boldsymbol{n}=-(\lvert\partial_t\boldsymbol{n}\rvert^2-\lvert\nabla\boldsymbol{n}\rvert^2)\boldsymbol{n}$	σ-models, nematics	Shatah–Turok–Spergel, Biernat–Bizoń	$\lambda=1$	Known (Section 2.8)
Variational wave	$\psi_{tt}-c(\psi)(c(\psi)\psi_x)_x=0$	nematic LC	cusp singularities	n/a	Instability of constants [58]
Timelike minimal surfaces	Born–Infeld	strings, membranes	[46, 47]	n/a	Partial (sub-cone, null-cone solutions) [124]

Table 1: Wave-type models with self-similar singularities. "Symmetry modes" are the non-negative eigenvalues of the formal linearisation around the ODE profile (Section 3); for co-rotational wave maps only time translation survives the symmetry reduction. Lower-order terms (γ, m) break Lorentz invariance but decay like $e^{-\tau}$, $e^{-2\tau}$.

in odd $d \geq 5$. [20]: mode stability of the Bizoń–Biernat wave maps for $d \geq 4$. [31]: existence and stability of self-similar blowup for wave maps into negatively curved targets, $d \geq 8$.

- *Power nonlinearities, ODE blowup.* [38]: $1 < p \leq 3$, $d = 3$, radial, energy topology. [39]: $p > 3$, $d = 3$, radial, stronger topology. [40]: $d = 3$ without symmetry, $H^2 \times H^1$. [41]: odd $d \geq 5$, radial. [28]: energy-critical $p = 5$, $d = 3$, radial; Strichartz estimates in similarity coordinates. [37]: the energy-critical case at optimal regularity (preprint). [118]: energy-critical, $3 \leq d \leq 6$ (preprint). [16]: cubic, odd $d \geq 5$, without symmetry. [107]: all d, all $p > 1$, without symmetry. [43]: strongly perturbed equations, $d = 3$. [94]: superconformal ODE family in all dimensions (preprint).
- *Non-trivial self-similar profiles, finite codimension.* [62]: explicit profile of the cubic equation for $d \geq 5$; codimension-one stability for $d = 7$. [23]: quadratic equation, explicit profile for $d \geq 7$, codimension-one stability for $d = 9$, non-radial ODE stability for $d = 7, 9$.
- *Beyond the light cone.* [6]: hyperboloidal similarity coordinates; global-in-space stability of the S^3 wave map. [18]: quadratic equation, $d = 7$, conditional codimension-one stability up to near the Cauchy horizon. [35]: global stability of the Yang–Mills profile, odd d. [60]: global-in-space stability for Yang–Mills in higher dimensions (preprint). [61]: globally stable wave-map profile in all $d \geq 3$. [36]: wave maps and Yang–Mills in regions approaching the future light cone of the singularity (preprint).
- *Optimal regularity and recent work.* [44]: wave maps $\mathbb{R}^{1+4} \to S^4$, small data in the critical Sobolev space. [45]: three-dimensional wave maps, fractional critical space. [34]: mode stability of self-similar wave maps without symmetry for $d \geq 4$ (preprint). [65, 95, 55]: one-dimensional derivative nonlinearities (Section 2.4).
- *Complementary approaches.* Merle–Zaag: blowup rate [98]; existence and universality of the profile in one dimension [99]; in higher dimensions, the stability of non-characteristic points and of the profile, and the dynamics near the explicit (Lorentz-transformed) stationary solutions in similarity variables [100, 101]. Bizoń et al.: numerics and heuristics [10], countable families of self-similar profiles [11]. Lyapunov functionals for perturbed and Klein–Gordon equations [69, 70, 71], and rates for NLKG [85]. Speck: quasilinear derivative-quadratic equations [113]. Blowup surfaces via Fuchsian/Nash–Moser methods [84], and scattering constructions of stable ODE-type blowup on spacelike hypersurfaces [77] (preprint).

2.9 Novelty assessment

We ran several targeted web searches per question (arXiv listings, journal and repository pages, and citing papers), current to 13 October 2026. Absence of evidence is not proof of absence; in particular the full texts of [82, 66, 84] were not accessible to us.

1. *Hyperbolic MEMS $w_{tt} - \Delta w = -\lambda w^{-q}$ (damped or not).* Existence, touchdown criteria and pull-in thresholds are rigorous [15, 51, 66, 81, 50, 82, 102]. The self-similar structure (ODE profile, non-trivial profiles, stability up to time translation) is established only formally and numerically [57, 73]. We found no rigorous proof of the quenching rate, of the profile, or of the stability of (2.4), in similarity coordinates or otherwise, for any $q > 1$ and any d. **Open**; a Donninger-type result would be new.

2. *Exponential equation $u_{tt} - \Delta u = e^u$.* For smooth data near constants, Kichenassamy's Nash–Moser construction [84] gives blowup on spacelike hypersurfaces in any dimension. In one dimension, blowup curves and rates are known [63, 1], and the Liouville equation is explicitly solvable. We found no Donninger-type theorem (stability of $\log 2 - 2\log(T - t)$ in $H^k \times H^{k-1}$ of the backward light cone with exponential decay in τ, $d \geq 2$) and no Merle–Zaag-type higher-dimensional profile result. **Partially known; the light-cone H^k statement appears open.**

3. *Focusing Klein–Gordon with mass.* Blowup rates are known [85, 71]. Stability of ODE blowup is known for $m = 0$ [107], and for strongly perturbed equations in $d = 3$ [43] (from the abstract we could not confirm whether a mass term is included). Since m^2u contributes an $O(e^{-2\tau})$ term, the extension is expected to be routine, but we found no explicit statement for general d and p. **Expected, not found explicitly.**

4. *Damped power equation* $u_{tt}+u_t-\Delta u = |u|^{p-1}u$. Blowup rates are known via Lyapunov functionals [69, 70], and the one-dimensional profile in [68]. We found no similarity-coordinate stability result for $d \geq 2$. The damping term is an $O(e^{-\tau})$ perturbation that is non-autonomous and breaks Lorentz invariance. **Open in the H^k light-cone sense, expected perturbative.**

5. $u_{tt} - \Delta u = (\partial_t u)^2$. In one dimension, stability of the (generalised) self-similar blowup is proved [65, 95]. In three dimensions, stable ODE-type blowup for a quasilinear class is proved by a different method [113]. A 2026 preprint (arXiv:2606.22282) treats radial blowup on spheres in higher dimensions for a quadratic-derivative nonlinearity; we could not verify its author list. **Largely known; a localised similarity-coordinate result for $d \geq 2$ was not found.**

For the models of Sections 2.2 (full Cattaneo system) and 2.4 (Westervelt degeneracy) we found no singularity analysis at all; both are quasilinear in the relevant regime and outside the scope of the semilinear method.

3 The Donninger et al. method for ODE-type blowup

This section fixes notation and isolates the part of the argument that is common to all models considered below. The method goes back to Donninger's work on wave maps [26] and was developed into a general scheme by Donninger and Schörkhuber [38, 39, 40, 41], with later refinements for mode stability [20], high Sobolev spaces in all dimensions [107, 61] and hyperboloidal similarity coordinates [6]; see [29] for a survey. We use it here as a black box with five steps, and we point out the (few) places where the industrial models need something extra.

3.1 Setting and self-similar variables

We consider

$$\partial_t^2 u - \Delta u = F(u) \qquad \text{in } [0,T) \times \mathbb{R}^d, \tag{3.1}$$

possibly with lower-order terms $\gamma\partial_t u + m^2 u$ that will be treated as perturbations (Section 3.4). By finite speed of propagation it suffices to study (3.1) in the backward light cone $\Gamma_T = \{(t,x) : 0 \leq t < T,\ |x| \leq T - t\}$ of the blowup point $(T, 0)$; in particular boundary conditions of the engineering problem (clamped edges of a MEMS plate, Dirichlet temperature, ...) are irrelevant as long as the cone base lies inside the physical domain.

Similarity coordinates are

$$\tau = -\log(T-t) + \log T, \qquad \xi = \frac{x}{T-t}, \qquad (\tau,\xi) \in [0,\infty) \times \overline{B^d},$$

so that $\partial_\tau = (T-t)\partial_t - x \cdot \nabla_x$ and $\nabla_\xi = (T-t)\nabla_x$. We write $\Lambda = \xi \cdot \nabla_\xi$. Suppose the nonlinearity has a scaling exponent $s \in \mathbb{R}$, i.e. $u \mapsto \mu^s u(\mu t, \mu x)$ maps solutions to solutions. This holds for $F(u) = |u|^{p-1}u$ with $s = 2/(p-1)$ and for the MEMS force $F(w) = -\lambda w^{-q}$ with $s = -2/(q+1)$ (a *negative* exponent: the gap w goes to zero). Setting

$$\psi_1(\tau,\xi) = (T-t)^s\, u(t,x), \qquad \psi_2(\tau,\xi) = (T-t)^{s+1}\, \partial_t u(t,x), \tag{3.2}$$

equation (3.1) becomes the autonomous first-order system

$$\partial_\tau \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = \underbrace{\begin{pmatrix} -\Lambda\psi_1 - s\psi_1 + \psi_2 \\ \Delta\psi_1 - \Lambda\psi_2 - (s+1)\psi_2 \end{pmatrix}}_{=:\, \mathbf{L}_0^{(s)}\Psi} + \begin{pmatrix} 0 \\ F(\psi_1) \end{pmatrix}. \tag{3.3}$$

For the exponential nonlinearity $F(u) = \delta e^u$ one instead sets $\psi_1 = u + 2\log(T-t)$, $\psi_2 = (T-t)\partial_t u$; the linear part is then $\mathbf{L}_0^{(0)}$ and an inhomogeneity -2 appears in the first component. The elementary but useful identity

$$\mathbf{L}_0^{(s)} = \mathbf{L}_0^{(0)} - s\,\mathbf{I} \tag{3.4}$$

shows that the scaling exponent only *shifts* the free spectrum; this is what makes the MEMS case ($s < 0$) structurally identical to the classical one.

3.2 ODE blowup and a universal spectrum

An *ODE blowup* solution is a spatially homogeneous solution $u_T(t) = \kappa(T-t)^{-s}$ (respectively $u_T = \log\kappa - 2\log(T-t)$), i.e. a constant fixed point $\Psi_* = (\kappa, s\kappa)$ of (3.3) (respectively $\Psi_* = (\log\kappa, 2)$). Its symmetry orbit consists of

- time translations $T \mapsto T'$ (in the fixed coordinates (τ, ξ) adapted to T, this is a genuine motion of the fixed point);
- Lorentz boosts: for $a \in \mathbb{R}^d$, $|a| < 1$,

$$u_{T,a}(t,x) = \kappa\,(1-|a|^2)^{s/2}\,(T - t + a\cdot x)^{-s}, \qquad \Psi_a(\xi) = \kappa(1-|a|^2)^{s/2}(1+a\cdot\xi)^{-s}, \tag{3.5}$$

  which is again an exact self-similar solution, smooth on $\overline{B^d}$ (direct substitution: with $f = T - t + a\cdot x$ one has $\partial_t^2 f^{-s} - \Delta f^{-s} = s(s+1)(1-|a|^2)f^{-s-2}$);
- space translations of the blowup point, which do not change u_T.

Linearising $F(\psi_1)$ at κ gives the constant potential $V_0 = F'(u_T)(T-t)^2$. The operator $\mathbf{L} = \mathbf{L}_0^{(s)} + \mathbf{L}'$, $\mathbf{L}'\Psi = (0, V_0\psi_1)$, has an eigenvalue λ with eigenfunction $(g, (\lambda + s + \Lambda)g)$ iff $v = (T-t)^{-s-\lambda} g(x/(T-t))$ solves $\partial_t^2 v - \Delta v = V_0(T-t)^{-2}v$, i.e. with $a := s + \lambda$,

$$(\delta_{ij} - \xi_i\xi_j)\partial_i\partial_j g - 2(a+1)\Lambda g - \big(a(a+1) - V_0\big)g = 0 \qquad \text{in } B^d. \tag{3.6}$$

Lemma 3.1 (V_0 is forced by the time-translation mode)**.** *For every ODE blowup with scaling exponent s one has $V_0 = (s+1)(s+2)$.*

Proof. $\partial_T u_T \propto (T-t)^{-s-1}$ solves the linearised equation, so $a = s+1$ (i.e. $\lambda = 1$, $g \equiv 1$) solves (3.6): $(s+1)(s+2) - V_0 = 0$. □

Thus $a^2 + a - V_0 = (a - a_1)(a - a_2)$ with $a_1 = s+1$, $a_2 = -(s+2)$, and for $F = |u|^{p-1}u$ one recovers $V_0 = p\kappa^{p-1} = ps(s+1)$, for MEMS $V_0 = q\lambda\kappa^{-q-1} = 2q(q-1)/(q+1)^2$ and for $F = \delta e^u$ $V_0 = 2$.

Writing $g(\xi) = |\xi|^\ell Y_\ell(\xi/|\xi|)\,h(|\xi|^2)$ with a spherical harmonic Y_ℓ, (3.6) becomes Gauss' hypergeometric equation

$$z(1-z)h'' + \big(c - (\alpha+\beta+1)z\big)h' - \alpha\beta h = 0, \quad c = \tfrac{d}{2} + \ell, \quad \alpha = \tfrac{a+\ell-a_1}{2}, \quad \beta = \tfrac{a+\ell-a_2}{2}, \tag{3.7}$$

(verified symbolically in `code/`), with Frobenius exponents $\{0, 1-c\}$ at $z = 0$ and $\{0,\, c-\alpha-\beta\}$ at $z = 1$, where $c - \alpha - \beta = \frac{d-1}{2} - s - \lambda$.

Theorem 3.2 (Spectrum of ODE blowup)**.** *Let $k \in \mathbb{N}$, $k > \frac{d}{2} - s$, and set $\omega_k := k - \frac{d}{2} + s > 0$. A number λ with $\operatorname{Re}\lambda > -\omega_k$ is an eigenvalue of $\mathbf{L}$ on $\mathcal{H}^k := H^k \times H^{k-1}(B^d)$ if and only if*

$$\lambda \in \{\, 1 - \ell - 2n \,\} \;\cup\; \{\, -2s - 2 - \ell - 2n \,\}, \qquad \ell, n \in \mathbb{N}_0,$$

and the eigenfunctions are $|\xi|^\ell Y_\ell \cdot$ (polynomial in $|\xi|^2$). In particular, for $s > -1$ the only eigenvalues with $\operatorname{Re}\lambda \geq 0$ are $\lambda = 1$ ($\ell = 0$, time translation, $g \equiv 1$) and $\lambda = 0$ ($\ell = 1$, d-fold, Lorentz boosts, $g = \xi_j$), and the spectral gap is

$$\omega_0(s) = \min\{1,\ 2s + 2\}. \tag{3.8}$$

Proof sketch. Since $\Gamma(c) \neq 0$ and the second Frobenius solution at $z = 0$ behaves like $|\xi|^{2-d-2\ell}$ (or $\log|\xi|$, or $|\xi|$ when $d = 1, \ell = 0$), an H^k eigenfunction ($k \geq 2$) must be a multiple of $h = {}_2F_1(\alpha, \beta; c; z)$. At $z = 1$ the second exponent produces $(1 - |\xi|^2)^\gamma$, $\gamma = \frac{d-1}{2} - s - \lambda$, possibly with a logarithm; such a function lies in $H^k(B^d)$ only if $\operatorname{Re}\gamma > k - \frac{1}{2}$ or $\gamma \in \mathbb{N}_0$ without log term. For $\operatorname{Re}\lambda > -\omega_k$ we have $\operatorname{Re}\gamma < k - \frac{1}{2}$, so h must be analytic at $z = 1$. By the $z \mapsto 1 - z$ connection formulae [25, §15.8] (including the degenerate cases $\gamma \in \mathbb{Z}$: for $\gamma = m \in \mathbb{N}_0$ the coefficient of $(1 - z)^m \log(1 - z)$, and for $\gamma = -m < 0$ the coefficient of the leading term $(1 - z)^{-m}$, is a nonzero multiple of $1/(\Gamma(\alpha)\Gamma(\beta))$) this happens iff $\alpha \in -\mathbb{N}_0$ or $\beta \in -\mathbb{N}_0$, in which case h is a polynomial. Solving $\alpha = -n$, $\beta = -n$ for $\lambda = a - s$ gives the two families. For $s > -1$ the second family is $\leq -2s - 2 < 0$, and $1 - \ell - 2n \geq 0$ only for $(\ell, n) \in \{(0, 0), (1, 0)\}$; the next eigenvalues are -1 ($\ell = 0, n = 1$ and $\ell = 2, n = 0$) and $-2s - 2$, whence (3.8). □

Remark 3.3. Theorem 3.2 contains, as special cases, the spectral statements of [38, 39, 41, 107] for $F = |u|^{p-1}u$, $p > 1$ ($s = 2/(p - 1)$, gap $\min\{1, \frac{2(p+1)}{p-1}\} = 1$), and it extends verbatim to the MEMS range $p = -q < -1$ ($s = -\frac{2}{q+1} \in (-1, 0)$, gap $\min\{1, \frac{2(q-1)}{q+1}\}$) and to the exponential case ($s = 0$, gap 1). The constraint $s > -1$ is exactly the condition $q > 1$ for MEMS touchdown to be of self-similar ODE type (Section 4). Semisimplicity of $\lambda = 1$ and $\lambda = 0$ (no Jordan chains) is checked by showing that $(\mathbf{L} - \lambda)\mathbf{u} = \mathbf{g}_\lambda$ has no $\mathcal{H}^k$ solution, an ODE computation of the same type as in [41, 107]; it is carried out model by model below.

3.3 The five steps

Step 1: free evolution. By (3.4) it suffices to understand $\mathbf{L}_0^{(0)}$. Heuristically, for a free wave $\|\nabla_\xi^k \psi_1(\tau)\|_{L^2(B^d)} = (T - t)^{k-d/2+s} \|\nabla_x^k u(t)\|_{L^2(B_{T-t})}$, and the last factor is bounded by local energy conservation, which suggests the decay rate $\omega_k = k - \frac{d}{2} + s$ on top-order seminorms. The low-order part is not dissipative: polynomial free waves of degree m give eigenvalues $-s - m$ of $\mathbf{L}_0^{(s)}$, and for the MEMS case $s < 0$ the constant wave $w \equiv 1$ gives the *positive* eigenvalue $-s = \sigma$. The rigorous version (inner products on $\mathcal{H}^k$ built from $\dot{H}^k$-type seminorms plus boundary/low-order terms, Lumer–Phillips) is modelled on [107] and [61]; a self-contained version for all s is Lemma 4.5 (we have not checked that the cited works state it in exactly this form): there is an equivalent inner product and a finite-rank operator $\mathbf{K}_0$ such that $\mathbf{L}_0^{(s)} - \mathbf{K}_0$ generates a semigroup of growth bound $\leq -\omega$ for any prescribed $\omega < \omega_k$ (after possibly increasing k). Since the finite-rank part is absorbed into the compact perturbation in Step 2, the positive free eigenvalue in the MEMS case is harmless.

Step 2: compact perturbation. $\mathbf{L}' + \mathbf{K}_0$ is compact on $\mathcal{H}^k$ (Rellich), hence $\mathbf{L}$ generates a semigroup $\mathbf{S}(\tau)$ with the same essential growth bound $\leq -\omega_k$, and $\sigma(\mathbf{L}) \cap \{\operatorname{Re}\lambda > -\omega_k\}$ consists of isolated eigenvalues of finite algebraic multiplicity [48, Ch. IV].

Step 3: mode stability. Determine all eigenvalues in $\operatorname{Re}\lambda > -\omega_0$; for the ODE profile this is Theorem 3.2. For non-explicit profiles (wave maps, Yang–Mills, non-trivial profiles of supercritical equations) this is the hard step and is done by supersymmetric removal of symmetry modes and the quasi-solution method of Costin, Donninger and Glogić [20].

Step 4: linear stability. With the Riesz projection $\mathbf{P}$ onto the symmetry eigenvalues ($\lambda = 1$, and $\lambda = 0$ in the nonradial case), $\|\mathbf{S}(\tau)(\mathbf{I}-\mathbf{P})\| \le M_\omega e^{-\omega\tau}$ for every $\omega < \min\{\omega_0, \omega_k\}$, because the growth bound of a semigroup equals the maximum of its spectral bound and its essential growth bound.

Step 5: nonlinear stability. Writing $\Psi = \Psi_* + \Phi$, the perturbation solves $\partial_\tau \Phi = \mathbf{L}\Phi + \mathbf{N}(\Phi)$ with $\mathbf{N}(\Phi) = (0, F(\kappa+\phi_1) - F(\kappa) - F'(\kappa)\phi_1)$. If $k > d/2$, $H^k(B^d)$ is an algebra embedded in L^∞, so for smooth F near κ (for MEMS: away from $w = 0$) one has $\|\mathbf{N}(\Phi) - \mathbf{N}(\tilde\Phi)\| \lesssim (\|\Phi\| + \|\tilde\Phi\|)\|\Phi - \tilde\Phi\|$. The Duhamel equation is solved in $\mathcal{X}_\omega = \{\Phi : \sup_\tau e^{\omega\tau}\|\Phi(\tau)\|_{\mathcal{H}^k} < \infty\}$ after subtracting the Lyapunov–Perron correction $\mathbf{C}(\Phi, \mathbf{u}) = \mathbf{P}\mathbf{u} + \int_0^\infty e^{-\tau'}\mathbf{P}_1\mathbf{N}(\Phi(\tau'))\,d\tau' + \int_0^\infty \mathbf{P}_0\mathbf{N}(\Phi(\tau'))\,d\tau'$. The correction is then removed by choosing the blowup time T^* (and, nonradially, the boost parameter a^*) via the Brouwer fixed point theorem: the derivative of the rescaled data with respect to (T, a) at the background is a non-degenerate map onto $\operatorname{ran}\mathbf{P}$, because the symmetry eigenfunctions are precisely the generators of these transformations. The outcome is a statement of the following form.

Theorem 3.4 (template)**.** *Let $k > \frac{d}{2} + \max\{0, -s\}$ be large enough and $0 < \omega < \omega_0$. There is $\varepsilon > 0$ such that for data (u_0, u_1) with $\|(u_0, u_1) - (u_{1,0}(0), \partial_t u_{1,0}(0))\|_{H^k \times H^{k-1}(B_{1+\varepsilon})} \le \varepsilon$ there exist $T^* \in (1-\varepsilon, 1+\varepsilon)$ and $a^* \in B_\varepsilon$ (with $a^* = 0$ for radial data) such that the solution blows up at $(T^*, 0)$ and*

$$\begin{aligned}\big\|(T^*-t)^s u(t, (T^*-t)\,\cdot) - \Psi_{a^*}\big\|_{H^k(B^d)} \\ + \big\|(T^*-t)^{s+1}\partial_t u(t, (T^*-t)\,\cdot) - (\Lambda + s)\Psi_{a^*}\big\|_{H^{k-1}(B^d)} \lesssim (T^*-t)^\omega.\end{aligned}$$

No translation parameter is needed: the ODE profile is translation invariant, and a space translation of a boosted profile is a time translation of it ($u_{T,a}(t, x - x_0) = u_{T - a\cdot x_0, a}(t, x)$). Applying the theorem at nearby base points shows that ODE blowup occurs along a spacelike hypersurface $t = T^*(x)$, with a^* playing the role of its gradient (cf. Remark 4.11).

3.4 Lower-order terms

Engineering models almost always contain damping or restoring forces. With $T - t = Te^{-\tau}$, a damping term $\gamma\partial_t u$ adds $-\gamma T e^{-\tau}\psi_2$ to the second line of (3.3), and a linear restoring force $m^2 u$ adds $-m^2T^2e^{-2\tau}\psi_1$. Two consequences: (i) the ODE blowup solution is no longer exactly self-similar, $u(t) = \kappa(T-t)^{-s}(1 + O(T-t))$ (for the member of the homogeneous family without the free mode $(T-t)^{s+2}$, and up to a factor $\log(T-t)$ in resonant cases such as MEMS with $q = 3$; see Proposition 6.8), so the background in similarity variables is $\Psi_* + O(e^{-\tau})$; (ii) the linearised flow is an *integrable* ($\int_0^\infty \|e^{-\tau}\mathbf{B}\|\,d\tau < \infty$) bounded perturbation of $\mathbf{S}(\tau)$. Both are compatible with Steps 4–5 as long as the decay rate of the forcing (1 for damping) is not below the target rate ω, which holds since $\omega < \omega_0 \le 1$. In the radial case only T must be modulated. In the nonradial case, the mass term preserves Lorentz invariance, whereas damping breaks it. Even then the neutral $\ell = 1$ modes are tangent to an exact family: $w = W(T - t + a\cdot x)$ solves the damped equation if W solves the spatially homogeneous equation with damping $\gamma/\sqrt{1-|a|^2}$ (after rescaling the variable by $\sqrt{1-|a|^2}$). Moreover, since there is no Jordan block at $\lambda = 0$, the neutral component obeys $\partial_\tau \mathbf{P}_0\Phi = \mathbf{P}_0\mathbf{G}(\Phi)$ with a forcing that is *integrable* in τ; hence $\mathbf{P}_0\Phi$ has a limit and a *static* choice of the boost parameter in the Lyapunov–Perron step suffices

(Proposition 4.12). A dynamical modulation is only needed when the neutral modes are forced non-integrably. The details are in Section 6 and in the model sections.

3.5 What the method gives, and what it does not

The output is *local*: stability of the blowup profile inside the backward light cone of the (moving) blowup point, for perturbations small in a high Sobolev norm, with an explicit exponential rate in τ, i.e. a power rate in $T^* - t$. It does not, by itself, determine the global blowup set or describe the solution outside the cone; for the latter see the hyperboloidal variant [6] and [18]. For the industrial models we treat, it nevertheless yields exactly the information an engineer would ask for: the *touchdown/runaway rate*, the *local profile*, and the *robustness* of both under perturbations of data and of lower-order model terms.

4 Hyperbolic MEMS: stability of self-similar touchdown

This section carries out the programme of Section 3 for the inertial MEMS model (2.3) in full, including the points where the singular nonlinearity and the *negative* scaling exponent $s = -\sigma$ require extra care. Throughout, the MEMS (voltage) parameter is called λ, as in the engineering literature, and the *spectral parameter* is called μ (it is called λ in Section 3).

Statement	Where	Status
ODE touchdown $w = c(T-t)^{2/(q+1)}$, Lorentz family, energy expansion	Proposition 4.1 and Lemma 4.2	elementary/known
$d = 1$: every smooth positive self-similar profile is a boosted ODE profile	Theorem 4.3	proved here
$d = 2, 3$: a sequence of non-trivial profiles $\psi_n \to A\lvert\xi\rvert^\sigma$	Table 3 and Conjecture 4.4	numerical; conjecture
Spectrum at the ODE profile, gap $\omega_0 = \min\{1, \frac{2(q-1)}{q+1}\}$	Proposition 4.6	proved here
No Jordan chains at $\mu = 1, 0$; Jordan block at $\mu = -1$ if $q = 3$	Lemma 4.7 and Remark 4.8	proved here ($d \le 3$)
Free operator: dissipative up to finite rank on $H^k \times H^{k-1}$	Lemma 4.5	proved here
Nonlinear stability modulo time translation and boosts	Theorem 4.10	proof sketch
Damping γw_t (radial *and* nonradial)	Proposition 4.12 and Corollary 4.13	proof sketch
Excited profiles have extra unstable modes	Section 4.9	numerical only

Table 2: Results of Section 4. "Proof sketch" means all estimates are indicated and the argument follows [40, 41, 107], but not every detail has been written out.

4.1 The model

In the small-aspect-ratio limit the deflection u of an electrostatically actuated membrane over a ground plate satisfies $\varepsilon^2 u_{tt} + \gamma u_t - \Delta u = -\lambda f(x)(1+u)^{-2}$ in Ω, $u = 0$ on $\partial\Omega$ [108, 51, 49]; see Section 2.1 for the variants. For the gap $w = 1 + u \in (0, 1]$, in the inertial time t/ε (and writing again t, γ for t/ε, γ/ε), and allowing the force laws w^{-q} ($q = 2$ Coulomb, $q = 3$ van der Waals, $q = 4$ Casimir), we study

$$w_{tt} + \gamma w_t - \Delta w = -\lambda\, w^{-q}, \qquad q > 1,\ \lambda > 0,\ \gamma \ge 0. \tag{4.1}$$

Touchdown (quenching) at (T, x_0) means $w(t,x) \to 0$ as $(t,x) \to (T, x_0)$; then $\partial_t w \to -\infty$ (infinite impact speed, see below). We put

$$\sigma := \frac{2}{q+1} \in (0,1), \qquad c^{q+1} := \frac{\lambda}{\sigma(1-\sigma)} = \frac{\lambda(q+1)^2}{2(q-1)} \qquad (q=2:\ \sigma=\tfrac{2}{3},\ c=(9\lambda/2)^{1/3}),$$

so $c = \kappa_q$ in (2.4). Two scalings will be used: if w solves (4.1) with damping γ, then $\mu_0^{-\sigma} w(\mu_0 t, \mu_0 x)$ solves it with damping $\mu_0\gamma$ (so only γT is scale-invariant), and $\lambda^{-1/(q+1)} w$ solves it with $\lambda = 1$ (used in the numerics; it changes the boundary value $w|_{\partial\Omega} = 1$, which is irrelevant for the local analysis).

4.2 ODE touchdown and its symmetry orbit

Proposition 4.1 (ODE touchdown)**.** *Let $\gamma = 0$ and let $W > 0$ solve $W'' = -\lambda W^{-q}$ on $[0,T)$ with $W(t) \to 0$ as $t \to T$. Let $E = \frac{1}{2}W'^2 + \lambda W^{1-q}/(1-q)$ (conserved; $E = \frac{1}{2}W'^2 + \lambda \log W$ if $q = 1$).*

(i) If $q > 1$, then with $\theta = T - t$, $W = c\,\theta^\sigma(1 + e\,\theta^{2-2\sigma} + O(\theta^{4-4\sigma}))$, $e = E/(c^2\sigma(3-2\sigma))$, and $W' = -c\sigma\theta^{\sigma-1}(1+o(1)) \to -\infty$. The exact power law $c(T-t)^\sigma$ is the zero-energy solution.

(ii) If $0 < q < 1$, touchdown occurs with finite speed, $W'(T) = -\sqrt{2E}$; if $q = 1$, $W = \theta\sqrt{2\lambda\log(1/\theta)}\,(1+o(1))$. In both cases there is no solution $c\,\theta^\sigma$ with $\sigma < 1$ (indeed $\sigma \geq 1$ forces $\sigma(1-\sigma) \leq 0$), i.e. $q > 1$ is exactly the condition $s = -\sigma > -1$ of Theorem 3.2.

(iii) Released from rest, $W(0) = 1$, $W'(0) = 0$, the touchdown time is $T_0 = \int_0^1 \big(\frac{2\lambda}{q-1}(W^{1-q} - 1)\big)^{-1/2} dW$, e.g. $T_0 = \pi/(2\sqrt{2\lambda})$ for $q = 2$ and $T_0 = \lambda^{-1/2}$ for $q = 3$.

Proof. (i) $W' < 0$ near T and $W'^2 = 2E + \frac{2\lambda}{q-1}W^{1-q}$, so $T - t = \int_0^W (2E + \frac{2\lambda}{q-1}V^{1-q})^{-1/2} dV$. Expanding the integrand for $V \to 0$ gives $\theta = \frac{2}{q+1}\sqrt{\frac{q-1}{2\lambda}}\,W^{\frac{q+1}{2}}(1 + O(W^{q-1}))$, i.e. $W^{q+1} = c^{q+1}\theta^2(1+O(\theta^{2-2\sigma}))$ since $\sigma(q-1) = 2-2\sigma$; the coefficient e follows by inserting the ansatz into E (the θ^0-terms give $E = c^2 e\,\sigma(3-2\sigma)$). (ii) For $q < 1$ the potential $\lambda W^{1-q}/(1-q)$ vanishes at $W = 0$; for $q = 1$ it diverges only logarithmically. (iii) Separation of variables. All formulas are checked in `mems_symbolic_checks.py` and, including e, numerically in `mems_damped_ode.py`. □

Lemma 4.2 (Symmetry orbit)**.** *For $T \in \mathbb{R}$ and $a \in \mathbb{R}^d$, $|a| < 1$,*

$$w^{T,a}(t,x) = c\,(1-|a|^2)^{-\sigma/2}\,(T - t + a\cdot x)^\sigma \tag{4.2}$$

solves $w_{tt} - \Delta w = -\lambda w^{-q}$ where $T - t + a\cdot x > 0$, in particular in the backward cone Γ_T, where $T - t + a\cdot x \geq (1-|a|)(T-t)$. In the similarity variables of $(T,0)$ it is the static profile

$$\psi_a(\xi) = c\,(1-|a|^2)^{-\sigma/2}(1 + a\cdot\xi)^\sigma, \qquad \psi_a \geq c\Big(\frac{1-|a|}{1+|a|}\Big)^{\sigma/2} > 0 \text{ on } \overline{B^d}. \tag{4.3}$$

Space translations act trivially on $w^{T,0}$ and as time translations on $w^{T,a}$: $w^{T,a}(t, x - x_0) = w^{T - a\cdot x_0, a}(t,x)$.

Proof. With $F = T - t + a\cdot x$, $\partial_t^2 F^\sigma - \Delta F^\sigma = -\sigma(1-\sigma)(1-|a|^2)F^{\sigma-2}$, and $\lambda c^{-q}(1 - |a|^2)^{q\sigma/2}F^{-q\sigma} = \sigma(1-\sigma)c\,(1-|a|^2)^{1-\sigma/2}F^{\sigma-2}$ because $q\sigma = 2 - \sigma$. (Also verified symbolically for $d = 3$ and general a.) □

The touchdown set of $w^{T,a}$ is the spacelike hyperplane $t = T + a\cdot x$; the boost parameter is the gradient of the touchdown-time function. Hence the orbit of the ODE solution under the symmetries is $(d+1)$-dimensional, which will match the $1 + d$ non-negative eigenvalues found in Proposition 4.6.

4.3 Self-similar profiles

Inserting $w = (T-t)^\sigma \psi(x/(T-t))$ into (4.1) with $\gamma = 0$ gives $w_{tt} - \Delta w + \lambda w^{-q} = -(T-t)^{\sigma-2}\mathcal{P}[\psi]$ with

$$\mathcal{P}[\psi] := (\delta_{ij} - \xi_i\xi_j)\partial_i\partial_j\psi - 2(1-\sigma)\Lambda\psi + \sigma(1-\sigma)\psi - \lambda\psi^{-q}, \tag{4.4}$$

and for radial $\psi(\rho)$, $\rho = |\xi|$,

$$(1-\rho^2)\psi'' + \Big(\frac{d-1}{\rho} - 2(1-\sigma)\rho\Big)\psi' + \sigma(1-\sigma)\psi - \lambda\psi^{-q} = 0. \tag{4.5}$$

At $\rho = 1$ the Frobenius exponents are 0 and $\frac{d-1}{2} + \sigma$. The solutions analytic at $\rho = 1$ form a one-parameter family $\psi(1) = b$, $\psi'(1) = (\lambda b^{-q} - \sigma(1-\sigma)b)/(d-3+2\sigma)$, except in the resonant case $d - 3 + 2\sigma = 0$, i.e. $(d,q) = (2,3)$, where analyticity *forces* $\psi(1) = c$ and $\psi'(1)$ is free. At $\rho = 0$ the regular solutions are parametrised by $\psi(0)$. Smooth radial profiles are therefore generically isolated. The constant $\psi \equiv c$ is always one of them; its boosts ψ_a, $a \neq 0$, are smooth but not radial.

Theorem 4.3 (Classification in $d = 1$)**.** *Let $d = 1$, $q > 1$, and let $\psi \in C^2(-1,1) \cap C^1[-1,1]$ be positive on $[-1,1]$ with $\mathcal{P}[\psi] = 0$. Then $\psi = \psi_a$ for a unique $a \in (-1,1)$. In particular $\psi \equiv c$ is the only even profile.*

Proof. Use rapidity coordinates in the cone, $T - t = R\cosh\eta$, $x = R\sinh\eta$, so $\xi = \tanh\eta$ and $w_{tt} - w_{xx} = R^{-1}\partial_R(R\partial_R w) - R^{-2}\partial_\eta^2 w$. For $w = (T-t)^\sigma\psi(\xi) = R^\sigma\Phi(\eta)$, $\Phi(\eta) := \cosh^\sigma\eta\,\psi(\tanh\eta)$, the profile equation becomes the *autonomous* ODE $\Phi'' = \sigma^2\Phi + \lambda\Phi^{-q}$ on $\mathbb{R}$, with conserved energy $E = \frac{1}{2}\Phi'^2 - \frac{1}{2}\sigma^2\Phi^2 + \frac{\lambda}{q-1}\Phi^{1-q}$. In terms of ψ,

$$2E = (1-\xi^2)^{1-\sigma}\Big[-\sigma^2\psi^2 + 2\sigma\xi\psi\psi' + (1-\xi^2)\psi'^2 + \tfrac{2\lambda}{q-1}\psi^{1-q}\Big].$$

The bracket is bounded on $[-1,1]$ because $\psi \in C^1$ and $\psi > 0$, and $1 - \sigma > 0$; letting $\xi \to 1$ gives $E = 0$. On $\{E = 0\}$, $\Phi'^2 = \sigma^2\Phi^{1-q}(\Phi^{q+1} - c^{q+1})$, so $\Phi \geq c$, and $\Phi'' > 0$. If Φ' had no zero, say $\Phi' > 0$, then $\Phi \downarrow L \geq c$ as $\eta \to -\infty$ with $\Phi' \to 0$, so $L = c$, but then $\Phi'' \to \sigma^2 c + \lambda c^{-q} > 0$ contradicts $\Phi' > 0$ near $-\infty$. Hence $\Phi'(\eta_0) = 0$, $\Phi(\eta_0) = c$ for some η_0, and by uniqueness for the initial value problem $\Phi = c\cosh^\sigma(\eta - \eta_0)$ (this solves the ODE, `mems_symbolic_checks.py`). Undoing the substitution gives $\psi = \psi_a$ with $a = -\tanh\eta_0$. □

This is the MEMS analogue of the one-dimensional classification of Merle and Zaag for power nonlinearities [99]. The argument uses the conserved energy of the rapidity ODE and has no direct analogue for $d \geq 2$: the radial equation on hyperbolic space, $\Phi'' + (d-1)\coth\eta\,\Phi' = \sigma(\sigma+d-1)\Phi + \lambda\Phi^{-q}$, only has a monotone energy, which tends to $-\infty$ for smooth profiles.

Non-trivial profiles for $d \geq 2$ (numerical). For $d \geq 2$, (4.5) has the singular solution

$$\psi_*(\rho) = A\rho^\sigma, \qquad A^{q+1} = \frac{\lambda}{\sigma(\sigma+d-2)}, \tag{4.6}$$

which is the *static* singular solution $w_*(x) = A|x|^\sigma$ of $\Delta w = \lambda w^{-q}$ (a membrane touching the plate at one point). It is analytic at $\rho = 1$ and singular only at $\rho = 0$. Near $\rho = 0$, the linearisation at ψ_* is $\Delta v + q\sigma(\sigma+d-2)|\xi|^{-2}v$, whose solutions $|\xi|^{-\frac{d-2}{2} \pm i\nu}$,

$$\nu^2 = q\sigma(\sigma+d-2) - \tfrac{(d-2)^2}{4}, \tag{4.7}$$

oscillate iff $\nu^2 > 0$. This Joseph–Lundgren-type condition holds for all $q > 1$ when $d = 2, 3$, and for $q = 2$ exactly when $2 \leq d \leq 7$. (To our knowledge this is the same range $2 \leq$

$d \le 7$ that produces the oscillating bifurcation diagram, with infinitely many fold points, of the stationary MEMS problem; see [49].) We solved (4.5) by two-sided shooting (regular series at $\rho = 0$ and $\rho = 1$, matched at $\rho = \frac{1}{2}$; robust under changes of the matching point; `mems_profiles_shooting.py`), scanning $\psi(0) \in [10^{-3}c, 50c]$ for $d = 2, 3$ and $\psi(0) \in [0.02c, 50c]$ for $d = 1$. Table 3 and Fig. 1 summarise the results.

- $d = 1$: only $\psi \equiv c$, as Theorem 4.3 requires.
- $d = 2, 3$, $q = 2, 3, 4$: besides $\psi_0 \equiv c$, a sequence of smooth positive profiles $\psi_1, \psi_2, \dots$ with $\psi_n(0) \downarrow 0$, each strictly increasing in ρ, converging to ψ_* away from $\rho = 0$ (e.g. $\sup_{[1/2,1]} |\psi_3 - \psi_*| < 10^{-4}c$ for $d = 3, q = 2$). The ratios $\psi_n(0)/\psi_{n+1}(0)$ converge to $e^{\sigma\pi/\nu}$. This is what the heuristic predicts: in the inner region the profile equation reduces to the scale-invariant static equation, so the inner solutions are rescalings $\ell^\sigma V(\rho/\ell)$ of a single function. Matching their oscillation $\cos(\nu \log \rho + \varphi)$ about ψ_* with the outer region quantises $\nu \log \ell$ modulo π.

(d, q)	$\psi_1(0)/c$	$\psi_2(0)/c$	$\psi_3(0)/c$	observed ratios	$e^{\sigma\pi/\nu}$
$(2, 2)$	0.14519	0.015139	0.0016489	9.59, 9.18	9.221
$(3, 2)$	0.24534	0.056547	0.012713	4.34, 4.45	4.443
$(2, 3)$	0.22125	0.033948	0.0055884	6.52, 6.07	6.134
$(3, 3)$	0.36986	0.12232	0.040285	3.024, 3.036, 3.0365	3.0365
$(2, 4)$	0.27636	0.053286	0.011243	5.19, 4.74, 4.83	4.811
$(3, 4)$	0.45746	0.18697	0.076792	2.447, 2.435, 2.437	2.437

Table 3: Non-trivial smooth radial self-similar profiles ($\psi_0 \equiv c$ omitted). Values of $\psi_n(0)/c$ are independent of λ. The limits $\psi_n(1) \to A$ are $A/c = 0.7937$ $(2, 2)$, 0.5848 $(3, 2)$, 1 $(2, 3)$, 0.7598 $(3, 3)$, 1.0845 $(2, 4)$, 0.8441 $(3, 4)$. Ratios are consecutive quotients $\psi_n(0)/\psi_{n+1}(0)$, including further members not listed.

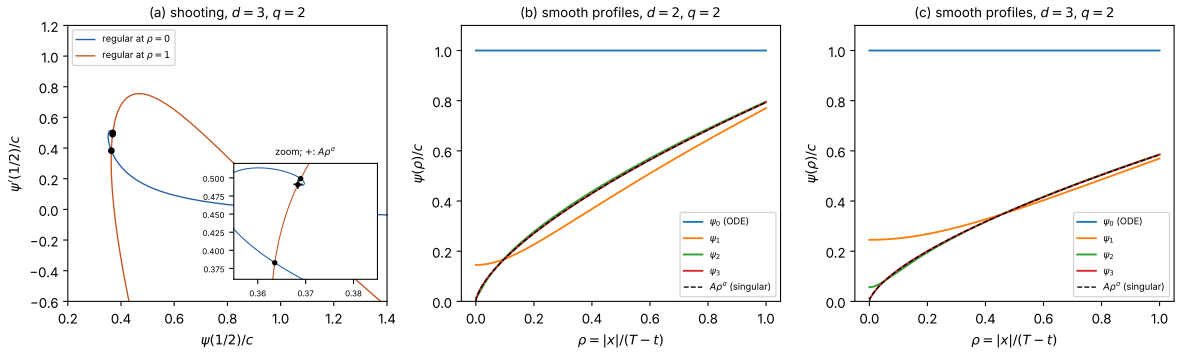


Figure 1: (a) Two-sided shooting for $d = 3$, $q = 2$: the curves of solutions regular at $\rho = 0$ (blue, parametrised by $\psi(0)$) and at $\rho = 1$ (orange, parametrised by $\psi(1)$) in the phase plane at $\rho = \frac{1}{2}$; intersections are smooth profiles. The inset shows the spiral of the blue curve around the singular solution $A\rho^\sigma$. (b), (c) The profiles $\psi_0 \equiv c, \psi_1, \psi_2, \psi_3$ for $d = 2, 3$, $q = 2$, and $A\rho^\sigma$.

Conjecture 4.4. *For $d \in \{2, 3\}$ and every $q > 1$ (more generally whenever $\nu^2 > 0$), (4.5) has infinitely many smooth positive solutions ψ_n. They are increasing in ρ, and $\psi_n \to \psi_*$ locally uniformly on $(0, 1]$ with $\psi_n(0)/\psi_{n+1}(0) \to e^{\sigma\pi/\nu}$. The profile ψ_n has exactly n unstable eigenvalues besides those generated by the symmetries, i.e. time and space translations and Lorentz boosts (Section 4.9). For $d = 1$ there are none (Theorem 4.3).*

For focusing power nonlinearities, countable families of smooth self-similar profiles are known [11], and a proof of Conjecture 4.4 would presumably follow the same shooting and

oscillation argument. Every ψ_n touches down at the tip with the *same rate* $(T-t)^\sigma$ but a *smaller constant*, $w(t,0)=\psi_n(0)(T-t)^\sigma$ with $\psi_n(0)<c$; the profile is focused at the centre instead of flat.

4.4 Similarity variables and the free operator

With $\psi_1=(T-t)^{-\sigma}w$, $\psi_2=(T-t)^{1-\sigma}\partial_t w$, $T-t=Te^{-\tau}$ (Section 3 with $s=-\sigma$), (4.1) becomes

$$\partial_\tau\psi_1=-\Lambda\psi_1+\sigma\psi_1+\psi_2,\qquad \partial_\tau\psi_2=\Delta\psi_1-\Lambda\psi_2-(1-\sigma)\psi_2-\lambda\psi_1^{-q}-\gamma Te^{-\tau}\psi_2. \tag{4.8}$$

For $\gamma=0$ the static solutions include $\Psi_a=(\psi_a,\Lambda\psi_a-\sigma\psi_a)$, and $\Psi_0=(c,-\sigma c)$. Writing $\Psi=\Psi_0+\Phi$,

$$\partial_\tau\Phi=\mathbf{L}\Phi+\mathbf{N}(\Phi),\quad \mathbf{L}=\mathbf{L}_0+\mathbf{L}',\quad \mathbf{L}'\Phi=(0,V_0\phi_1),\quad \mathbf{N}(\Phi)=(0,N(\phi_1)), \tag{4.9}$$

where $\mathbf{L}_0=\mathbf{L}_0^{(-\sigma)}$ is the free operator of (3.3) and

$$V_0=q\lambda c^{-q-1}=q\sigma(1-\sigma)=\frac{2q(q-1)}{(q+1)^2},\qquad N(\phi)=-\lambda\big[(c+\phi)^{-q}-c^{-q}+qc^{-q-1}\phi\big]. \tag{4.10}$$

We work in $\mathcal{H}^k:=H^k(B^d)\times H^{k-1}(B^d)$, $k\geq 2$, and set $\omega_k:=k-\frac{d}{2}-\sigma$. The free operator is *not* dissipative on low frequencies: polynomial free waves give eigenvalues $\sigma-m$, $m\in\mathbb{N}_0$, of $\mathbf{L}_0$ (e.g. $w\equiv 1$ gives $(1,0)$ with eigenvalue $\sigma>0$). The following lemma shows that this is a finite-rank effect. It is a soft version of the inner-product constructions of [107, 61].

Lemma 4.5 (Free operator)**.** *Let $k\geq 2$ with $\omega_k>0$. There is an inner product $\langle\cdot,\cdot\rangle$ on $\mathcal{H}^k$, equivalent to the standard one, such that for every $\varepsilon\in(0,\omega_k)$ there is a finite-rank operator $\mathbf{K}_\varepsilon$ for which $\mathbf{L}_0-\mathbf{K}_\varepsilon$ generates a C_0-semigroup with $\|e^{\tau(\mathbf{L}_0-\mathbf{K}_\varepsilon)}\|\leq e^{-(\omega_k-\varepsilon)\tau}$.*

Proof. Generation. For $f\in\mathcal{H}^k$ extend f to $\mathbb{R}^d$ (Sobolev extension), solve the free wave equation and transform back. By finite speed of propagation this defines a C_0-semigroup on $\mathcal{H}^k$, independent of the extension, whose generator is the closure of $\mathbf{L}_0$ on $C^\infty(\overline{B^d})^2$ (as in [41, 107]).

Top order. For $|\alpha|=k-1$ put $u=\partial^\alpha f$. Since $[\partial_i,\Lambda]=\partial_i$, $\partial^\alpha\mathbf{L}_0 f$ is the free operator with $s=-\sigma$ replaced by $s'=-\sigma+k-1$ applied to u. Integration by parts (using $\operatorname{Re}\int_B\nabla u_1\cdot\overline{\nabla\Lambda u_1}=(1-\frac{d}{2})\|\nabla u_1\|^2+\frac12\int_{\mathbb{S}^{d-1}}|\nabla u_1|^2$, the analogous identity for u_2, and $\int_B(\nabla u_1\cdot\nabla\bar u_2+\bar u_2\Delta u_1)=\int_{\mathbb{S}^{d-1}}\bar u_2\,\partial_r u_1$) gives, with $\|f\|_{\dot{\mathcal{H}}}^2:=\sum_{|\alpha|=k-1}(\|\nabla\partial^\alpha f_1\|_{L^2(B)}^2+\|\partial^\alpha f_2\|_{L^2(B)}^2)$,

$$\operatorname{Re}(\mathbf{L}_0 f,f)_{\dot{\mathcal{H}}}=-\omega_k\|f\|_{\dot{\mathcal{H}}}^2-\frac12\sum_{|\alpha|=k-1}\int_{\mathbb{S}^{d-1}}\big(|\nabla_T\partial^\alpha f_1|^2+|\partial_r\partial^\alpha f_1-\partial^\alpha f_2|^2\big)\,dS\leq-\omega_k\|f\|_{\dot{\mathcal{H}}}^2 \tag{4.11}$$

for $f\in C^\infty(\overline{B^d})^2$ (∇_T is the tangential gradient). This is the rigorous form of the heuristic decay rate of Section 3.3.

Low order. Let $\langle f,g\rangle:=(f,g)_{\dot{\mathcal{H}}}+(f,g)_{\mathcal{H}^{k-1}}$, with $\mathcal{H}^{k-1}=H^{k-1}\times H^{k-2}$; this is equivalent to the standard inner product. Since $\mathbf{L}_0:\mathcal{H}^k\to\mathcal{H}^{k-1}$ is bounded, (4.11) gives $\operatorname{Re}\langle\mathbf{L}_0 f,f\rangle\leq-\omega_k\|f\|^2+C\|f\|\,\|f\|_{\mathcal{H}^{k-1}}$. The embedding $J:\mathcal{H}^k\hookrightarrow\mathcal{H}^{k-1}$ is compact (Rellich). So for $\varepsilon>0$ there is a finite-rank orthogonal projection Q in $\mathcal{H}^k$ (spectral projection of J^*J) with $\|J(I-Q)\|\leq\varepsilon/(2C)$. Then $\|f\|_{\mathcal{H}^{k-1}}\leq\frac{\varepsilon}{2C}\|f\|+\|J\|\|Qf\|$, and $\operatorname{Re}\langle\mathbf{L}_0 f,f\rangle\leq-(\omega_k-\varepsilon)\|f\|^2+C_\varepsilon\|Qf\|^2$. With $\mathbf{K}_\varepsilon:=C_\varepsilon Q$, the operator $\mathbf{L}_0-\mathbf{K}_\varepsilon+(\omega_k-\varepsilon)$ is dissipative on a core. It is a bounded perturbation of a generator, so Lumer–Phillips gives the bound [48, Ch. II–III]. □

Since $\mathbf{L}'$ maps $\mathcal{H}^k$ into $\{0\}\times H^k\hookrightarrow\{0\}\times H^{k-1}$ compactly, $\mathbf{L}=(\mathbf{L}_0-\mathbf{K}_\varepsilon)+(\mathbf{K}_\varepsilon+\mathbf{L}')$ generates a semigroup $\mathbf{S}(\tau)$, and $\sigma(\mathbf{L})\cap\{\operatorname{Re}\mu>-\omega_k\}$ consists of isolated eigenvalues of finite algebraic multiplicity (analytic Fredholm theorem; [80, IV.5], [48]). In particular the positive free eigenvalue σ is harmless: it is not an eigenvalue of $\mathbf{L}$.

4.5 Spectral analysis

Proposition 4.6 (Spectrum)**.** *Let $k \geq 2$ with $\omega_k > 0$. A number μ with $\operatorname{Re}\mu > -\omega_k$ is an eigenvalue of $\mathbf{L}$ on $\mathcal{H}^k$ iff*

$$\mu \in \{\, 1-\ell-2n \,\} \;\cup\; \{\, -(2-2\sigma)-\ell-2n \,\}, \qquad \ell, n \in \mathbb{N}_0, \tag{4.12}$$

($\ell \in \{0,1\}$, the parity, if $d = 1$). The eigenfunctions are $f = (g, (\mu - \sigma + \Lambda)g)$ with $g = |\xi|^\ell Y_\ell(\xi/|\xi|)\, p(|\xi|^2)$, where Y_ℓ is a spherical harmonic and p is a polynomial. The non-negative eigenvalues are

- *$\mu = 1$ ($\ell = 0$): $f_T = (1, 1-\sigma)$, the time-translation mode, ∂_T of the rescaled $w^{T,0}$;*
- *$\mu = 0$ ($\ell = 1$, d-fold): $f_{L,j} = (\xi_j, (1-\sigma)\xi_j) = (c\sigma)^{-1}\partial_{a_j}\Psi_a|_{a=0}$, the Lorentz modes.*

The largest negative eigenvalues are $\mu = -(2-2\sigma)$ ($\ell = 0$, $f_E = (1, \sigma - 2)$, the energy mode*: the change of E in Proposition 4.1(i)) and $\mu = -1$ ($\ell = 0, n = 1$ and $\ell = 2, n = 0$). Hence the spectral gap is*

$$\omega_0 = \min\Big\{1,\ \frac{2(q-1)}{q+1}\Big\}: \qquad \omega_0 = \tfrac{2}{3}\ (q = 2), \qquad \omega_0 = 1\ (q \geq 3). \tag{4.13}$$

Proof. This is Theorem 3.2 with $s = -\sigma$; we record the MEMS-specific points (all identities verified in `mems_symbolic_checks.py`). The first component of $(\mu - \mathbf{L})f = 0$ gives $f_2 = (\mu - \sigma + \Lambda)f_1$. Then $f_1 = g$ solves (3.6) with $a = \mu - \sigma$, whose characteristic roots are $a_1 = 1-\sigma$, $a_2 = \sigma - 2$ (indeed $1 + 4V_0 = (\frac{3q-1}{q+1})^2$). By interior elliptic regularity $g \in C^\infty(B^d)$. Its projections $G_{\ell m}(r) = \int_{\mathbb{S}^{d-1}} g(r\omega)\overline{Y_{\ell m}(\omega)}d\omega$ lie in $H^k(\frac{1}{2}, 1)$ and are $O(r^\ell)$ at 0, hence $G_{\ell m} = \text{const} \cdot r^\ell {}_2F_1(\alpha, \beta; \ell + \frac{d}{2}; r^2)$ with $\alpha = \frac{\mu-1+\ell}{2}$, $\beta = \frac{\mu+2-2\sigma+\ell}{2}$. (For $d = 1$ the second solution at 0 has the opposite parity.) At $z = 1$ the exponents are 0 and $\delta(\mu) = \frac{d-1}{2} + \sigma - \mu$, and $\operatorname{Re}\mu > -\omega_k \iff \operatorname{Re}\delta < k - \frac{1}{2}$. Hence $(1-z)^\delta$ ($\delta \notin \mathbb{Z}$), $(1-z)^m \log(1-z)$ ($\delta = m \in \mathbb{N}_0$) and $(1-z)^{-m}$ ($\delta = -m < 0$) are not in H^k near $z = 1$. In all three cases the coefficient of the offending term in the connection formula [25, 15.8] is a nonzero multiple of $1/(\Gamma(\alpha)\Gamma(\beta))$. Therefore $\alpha \in -\mathbb{N}_0$ or $\beta \in -\mathbb{N}_0$, and conversely these give polynomial (hence admissible) eigenfunctions. The possibility that *both* Frobenius solutions at $z = 1$ are smooth arises only for $\delta = m \in \mathbb{N}$ with $(\alpha)_m(\beta)_m = 0$, which is again covered by α or $\beta \in -\mathbb{N}_0$. Since the solution regular at 0 is unique up to a factor, each harmonic $Y_{\ell m}$ contributes at most one eigenfunction. The two families overlap only if $\alpha - \beta = \sigma - \frac{3}{2} \in \mathbb{Z}$, i.e. for $q = 3$ (Remark 4.8). Solving $\alpha = -n$, $\beta = -n$ gives (4.12). The physical identification follows from Lemma 4.2, and from the spatially homogeneous perturbations $(T-t)^{\sigma-1}$ and $(T-t)^{2-\sigma}$ of the ODE (roots of $m(m-1) = V_0$), which correspond to $\mu = 1$ and $\mu = -(2-2\sigma)$. □

Lemma 4.7 (Semisimplicity)**.** *Let $d \leq 3$ and let $k \geq 2$ with $\omega_k > 0$. Then $(1 - \mathbf{L})g = f_T$ and $(-\mathbf{L})g = f_{L,j}$ have no solution $g \in \mathcal{D}(\mathbf{L})$. Consequently the Riesz projections $\mathbf{P}_1$ (at $\mu = 1$) and $\mathbf{P}_0$ (at $\mu = 0$) have ranks 1 and d, with $\operatorname{ran}\mathbf{P}_1 = \langle f_T \rangle$ and $\operatorname{ran}\mathbf{P}_0 = \langle f_{L,1}, \dots, f_{L,d} \rangle$.*

Proof. $(\mu - \mathbf{L})g = f$ is equivalent to $g_2 = (a+\Lambda)g_1 - f_1$ and $-\big[(\delta_{ij} - \xi_i\xi_j)\partial_{ij}g_1 - 2(a+1)\Lambda g_1 - (a(a+1) - V_0)g_1\big] = f_2 + (a+1)f_1 + \Lambda f_1$. For $f = f_T$ ($a = 1-\sigma$, so $a(a+1) = V_0$) the right side is $3 - 2\sigma$; for $f = f_{L,j}$ ($a = -\sigma$) it is $(3-2\sigma)\xi_j$. With $g_1 = G(r)$, resp. $g_1 = rK(r)\xi_j/|\xi|$, both reduce to the first-order equation

$$(1-r^2)P' + \Big(\frac{D-1}{r} - 2(2-\sigma)r\Big)P = -(3-2\sigma), \qquad P = G' \ (D = d), \quad P = K' \ (D = d+2).$$

The solution regular at $r = 0$ is $P = -(3-2\sigma)r^{1-D}(1-r^2)^{-(e+1)}\int_0^r s^{D-1}(1-s^2)^e ds$ with $e = \frac{3-D}{2} - \sigma$. Near $r = 1$, $\int_0^r s^{D-1}(1-s^2)^e\, ds = C_D + (1-r^2)^{e+1}A(r)$ with A analytic and

$$C_D = \frac{\Gamma(\frac{D}{2})\,\Gamma(e+1)}{2\,\Gamma(\frac{5}{2} - \sigma)} \neq 0$$

(analytic continuation of the Beta integral when $e < -1$). The only exception is $e + 1 \in -\mathbb{N}_0$, which for $d \le 3$ happens only for $D = 4$, $q = 3$, where a term $\log(1 - r^2)$ with coefficient $-\frac{1}{2}$ appears instead. Hence g_1 contains $C\,(1-r)^{\delta(\mu)}$ with $C \neq 0$ and $\delta(1) = \frac{d-3}{2} + \sigma$, $\delta(0) = \frac{d-1}{2} + \sigma$. Both are non-integers for $d \le 3$, except for $(d, q) = (2, 3)$, where $\delta(1) = 0$ and $\delta(0) = 1$ and one gets $\log(1 - r)$, resp. $(1 - r)\log(1 - r)$, instead. As $\delta(\mu) < k - \frac{1}{2}$, $g_1 \notin H^k$. (The constant C_D and the expansion were checked numerically with `mpmath`.) □

Remark 4.8 ($q = 3$: a Jordan block at the gap). For $q = 3$ the two families in (4.12) collide at $\mu = -1 = -\omega_0$ in the sector $\ell = 0$, with the single eigenfunction $f_E = (1, -\frac{3}{2})$. In this case $(-1 - \mathbf{L})g = f_E$ *is* solvable with $g = (|\xi|^2/d,\ |\xi|^2/(2d) - 1)$, so $\mu = -1$ carries a Jordan block (verified symbolically; in the collocation of Section 4.9 this double eigenvalue splits by $\approx 10^{-5}$, much more than the errors for simple eigenvalues). Hence for van der Waals forces the decay at the rate $\omega_0 = 1$ fails by a factor τ, which matches the $\theta \log \theta$ term in (4.14) below. This does not affect Theorem 4.10, where $\omega < \omega_0$.

Proposition 4.9 (Linear stability). *Let $d \le 3$, $k \ge 2$, $\omega_k > 0$, and $\mathbf{P} = \mathbf{P}_1 + \mathbf{P}_0$. Then $\mathbf{S}(\tau)\mathbf{P}_1 = e^{\tau}\mathbf{P}_1$, $\mathbf{S}(\tau)\mathbf{P}_0 = \mathbf{P}_0$, and for every $\omega < \min\{\omega_0, \omega_k\}$ there is M_ω with $\|\mathbf{S}(\tau)(\mathbf{I} - \mathbf{P})f\| \le M_\omega e^{-\omega\tau}\|(\mathbf{I} - \mathbf{P})f\|$.*

Proof. Fix $\omega < \omega' < \min\{\omega_0, \omega_k - \varepsilon\}$. On $\operatorname{Re}\mu \ge -\omega'$, $\mathbf{R}_{\mathbf{L}}(\mu) = \mathbf{R}_0(\mu)[\mathbf{I} - (\mathbf{K}_\varepsilon + \mathbf{L}')\mathbf{R}_0(\mu)]^{-1}$ with $\mathbf{R}_0$ the resolvent of $\mathbf{L}_0 - \mathbf{K}_\varepsilon$. Here $\|\mathbf{L}'\mathbf{R}_0(\mu)\| \lesssim |\mu|^{-1}$. Indeed, $u = \mathbf{R}_0(\mu)f$ satisfies $\mu u_1 = f_1 - \Lambda u_1 + \sigma u_1 + u_2 + (\mathbf{K}_\varepsilon u)_1$, so $\|u_1\|_{H^{k-1}} \lesssim |\mu|^{-1}\|f\|$. Also $\|\mathbf{K}_\varepsilon \mathbf{R}_0(\mu)\| \to 0$ as $|\mu| \to \infty$: write $\mathbf{K}_\varepsilon$ via vectors in the dense set $\mathcal{D}(\mathbf{L}_0^*)$, up to a small error. Hence $\mathbf{R}_{\mathbf{L}}$ is uniformly bounded for $|\mu|$ large. On the remaining compact set it is analytic except at $\mu = 1, 0$ (Proposition 4.6). The Gearhart–Prüss theorem [48, V.1.11] applied to $\mathbf{L}|_{\operatorname{ran}(\mathbf{I}-\mathbf{P})}$ gives the bound. Lemma 4.7 gives the action on $\operatorname{ran}\mathbf{P}$. □

Required regularity. To see the full gap one needs $\omega_k \ge \omega_0$, i.e. $k \ge \frac{d}{2} + \sigma + \omega_0$, together with $k > \frac{d}{2}$ (for the nonlinearity) and $k \ge 2$. Since $\sigma + \omega_0 = \min\{1 + \sigma, 2 - \sigma\} \le \frac{3}{2}$,

$$k = 2 \ (d = 1), \qquad k = 3 \ (d = 2, 3) \qquad \text{suffice for every } q > 1.$$

For $(d, q) = (1, 3), (3, 3)$ the value $\omega_k = \omega_0 = 1$ is attained exactly, which is harmless since $\omega < \omega_0$. For smaller admissible k (e.g. $k = 2$, $d = 2$) everything holds with $\omega < \min\{\omega_0, \omega_k\}$.

4.6 Nonlinear stability of ODE touchdown

Normalise $T_0 = 1$ by scaling. The reference solution $w^{1,0} = c(1-t)^\sigma$ has data $(c, -\sigma c)$ at $t = 0$.

Theorem 4.10 (Stable self-similar touchdown). *Let $\gamma = 0$, $q > 1$, $\lambda > 0$, $d \in \{1, 2, 3\}$, $k = 2$ if $d = 1$ and $k = 3$ if $d = 2, 3$, and $0 < \omega < \omega_0 = \min\{1, \frac{2(q-1)}{q+1}\}$. There are $\delta > 0$, $M \ge 1$ such that the following holds. Let real data $(w_0, w_1) = (c, -\sigma c) + (v_0, v_1)$ satisfy $\|(v_0, v_1)\|_{H^k \times H^{k-1}(B_{1+\delta})} \le \delta/M$. Then there exist $T^* \in [1 - \delta, 1 + \delta]$ and $a^* \in \mathbb{R}^d$, $|a^*| \le \delta$, such that the solution w of (4.1) with these data exists in the backward cone Γ_{T^*}, is positive there, touches down at $(T^*, 0)$, and satisfies, for $0 \le t < T^*$,*

$$(T^* - t)^{|\alpha| - \sigma - \frac{d}{2}} \big\|\partial_x^\alpha \big(w(t) - w^{T^*, a^*}(t)\big)\big\|_{L^2(B_{T^*-t})} \lesssim \delta\,(T^* - t)^\omega, \quad |\alpha| \le k,$$

$$(T^* - t)^{|\alpha| + 1 - \sigma - \frac{d}{2}} \big\|\partial_x^\alpha \partial_t \big(w(t) - w^{T^*, a^*}(t)\big)\big\|_{L^2(B_{T^*-t})} \lesssim \delta\,(T^* - t)^\omega, \quad |\alpha| \le k - 1.$$

In particular, uniformly for $|x| \le T^* - t$,

$$w(t, x) = (T^* - t)^\sigma \Big[\psi_{a^*}\Big(\frac{x}{T^* - t}\Big) + O\big(\delta (T^* - t)^\omega\big)\Big],$$

$$\partial_t w(t, 0) = -\sigma c\,(1 - |a^*|^2)^{-\frac{\sigma}{2}} (T^* - t)^{\sigma - 1}\big(1 + O((T^* - t)^\omega)\big).$$

If the data are radial (even, if $d = 1$), then $a^* = 0$.

Proof sketch. 1. Data and unknowns. For $T \in [1-\delta, 1+\delta]$ and $|a| \le \delta$, the rescaled data are $\mathbf{U}(v,T) = (T^{-\sigma} w_0(T\cdot), T^{1-\sigma} w_1(T\cdot))$ on B^d ($B_T \subset B_{1+\delta}$). We write $\Psi(\tau) = \Psi_a + \Phi(\tau)$, so that $\Phi(0) = \mathbf{u}(T,a) := \mathbf{U}(v,T) - \Psi_a$ and

$$\partial_\tau \Phi = \mathbf{L}\Phi + \mathbf{G}_a(\Phi), \qquad \mathbf{G}_a(\Phi) = \big(0,\ (V_a - V_0)\phi_1 + N_a(\phi_1)\big),$$

with $V_a = q\lambda\psi_a^{-q-1}$, $\|V_a - V_0\|_{C^k(\overline{B})} \lesssim |a|$, and $N_a(\phi) = -\lambda[(\psi_a+\phi)^{-q} - \psi_a^{-q} + q\psi_a^{-q-1}\phi]$.

2. Nonlinearity. As $k > \frac{d}{2}$, $H^k(B^d)$ is an algebra and $\|\phi\|_{L^\infty} \le C_k \|\phi\|_{H^k}$. For $\|\phi\|_{H^k} \le \delta$ with $C_k\delta \le c/4$ we have $\psi_a + \phi \ge c/4$ on $\overline{B^d}$, i.e. the solution stays away from the singularity of w^{-q}, and $N_a(\phi) = \phi^2 \int_0^1 (1-\theta) F''(\psi_a + \theta\phi)\, d\theta$ with $F(w) = -\lambda w^{-q}$ smooth on $[c/4, \infty)$. Moser estimates give $\|N_a(\phi) - N_a(\tilde\phi)\|_{H^{k-1}} \lesssim (\|\phi\|_{H^k} + \|\tilde\phi\|_{H^k})\|\phi - \tilde\phi\|_{H^k}$.

3. Modified Duhamel equation. In $\mathcal{X} = \{\Phi \in C([0,\infty), \mathcal{H}^k) : \|\Phi\|_{\mathcal{X}} := \sup_\tau e^{\omega\tau}\|\Phi(\tau)\| \le \delta\}$ solve

$$\Phi(\tau) = \mathbf{S}(\tau)\big[\mathbf{u} - \mathbf{C}(\Phi, \mathbf{u})\big] + \int_0^\tau \mathbf{S}(\tau - \tau')\mathbf{G}_a(\Phi(\tau'))\, d\tau',$$
$$\mathbf{C}(\Phi, \mathbf{u}) = \mathbf{P}\mathbf{u} + \int_0^\infty (e^{-\tau'}\mathbf{P}_1 + \mathbf{P}_0)\mathbf{G}_a(\Phi(\tau'))\, d\tau'.$$

By Step 2, $\|\mathbf{G}_a(\Phi)(\tau)\| \lesssim (\delta^2 + |a|\delta)e^{-\omega\tau}$, with the corresponding Lipschitz bound. By Proposition 4.9, the right-hand side maps $\mathcal{X}$ into itself and is a contraction, provided $\|(\mathbf{I}-\mathbf{P})\mathbf{u}\| \le \delta/(2M_\omega)$. This holds because $\mathbf{u}(T,a) = [\mathbf{U}(0,T) - \Psi_0] + [\Psi_0 - \Psi_a] + [\mathbf{U}(v,T) - \mathbf{U}(0,T)]$, with

$$\mathbf{U}(0,T) - \Psi_0 = -c\sigma(T-1)f_T + O((T-1)^2), \qquad \Psi_0 - \Psi_a = -c\sigma\, a\cdot f_L + O(|a|^2),$$

so $\|(\mathbf{I}-\mathbf{P})\mathbf{u}\| \lesssim \delta/M + \delta^2$. The fixed point $\Phi_{T,a}$ is continuous in (T,a).

4. Choice of (T^*, a^*). In the basis $f_T, f_{L,1}, \dots, f_{L,d}$ of $\operatorname{ran}\mathbf{P}$, $\mathbf{C}(\Phi_{T,a}, \mathbf{u}(T,a)) = -c\sigma\big((T-1)f_T + a\cdot f_L\big) + \mathbf{R}(T,a)$ with $\mathbf{R}$ continuous and $|\mathbf{R}| \lesssim \delta/M + \delta^2$. (The terms are exactly the symmetry generators, Proposition 4.6.) By Brouwer's fixed point theorem there is $(T^*, a^*) \in [1-\delta, 1+\delta] \times \overline{B_\delta}$ with $\mathbf{C} = 0$, for M large and δ small. Then Φ solves the true Duhamel equation. For radial data everything can be done in the radial subspace, where $\mathbf{P}_0 = 0$, so $a^* = 0$.

5. Back to w. The mild solution coincides with the (strong) solution of (4.1) in Γ_{T^*} (approximation by smooth data and finite speed of propagation). Since $\psi_1 = \psi_{a^*} + \phi_1$ and $\|\phi_1(\tau)\|_{L^\infty} \lesssim \delta e^{-\omega\tau}$, $w > 0$ in the cone and $w(t,0) \to 0$. The estimates are $\|\Phi(\tau)\|_{\mathcal{H}^k} \le \delta e^{-\omega\tau}$ rewritten with $e^{-\tau} = (T^* - t)/T^*$. The formula for $\partial_t w(t,0)$ uses $H^{k-1} \hookrightarrow C^0$, which holds since $k - 1 > \frac{d}{2}$ for our choices of k. □

Remark 4.11. (a) *No translation parameter is needed.* The ODE profile is translation invariant, and $w^{T,a}(\cdot, \cdot - x_0) = w^{T - a\cdot x_0, a}$. Applying the theorem at every nearby base point x_0 (translated data) shows that, near $x_0 = 0$, touchdown happens along a hypersurface $t = T^*(x_0)$. For the exact solutions $a = \nabla T^*$. We expect $a^* = \nabla T^*(0)$ and $T^* \in C^1$ in general, as for the 1D power case [99]; we have not proved it. (b) The rate $\omega_0 = \frac{2}{3}$ for $q = 2$ is dictated by the ODE energy mode $\mu = -(2 - 2\sigma)$. For $q \ge 3$ it is the PDE mode $\mu = -1$, which by Remark 4.8 is not semisimple when $q = 3$. (c) The theorem is in the style of [40, 41, 107]. The new features are $s < 0$ (Lemma 4.5) and the need to keep the solution away from $w = 0$, which the L^∞ control from $k > \frac{d}{2}$ provides.

4.7 Damping and other integrable perturbations

In (4.8), damping contributes $-\gamma T e^{-\tau}\psi_2$, a perturbation decaying at rate $1 \ge \omega_0$. Exact self-similar solutions no longer exist (there is no scaling symmetry), but the touchdown remains asymptotically self-similar.

Damped ODE. For $W''+\gamma W'=-\lambda W^{-q}$, inserting $W=c\theta^\sigma(1+c_1\theta+\dots)$, $\theta=T-t$, the $\theta^{\sigma-1}$-terms give $c_1\sigma(3-q)=\gamma\sigma$. Including the free energy mode,

$$W=c\,\theta^\sigma\Big(1+e\,\theta^{2-2\sigma}+\frac{\gamma}{3-q}\,\theta+\dots\Big)\ (q\neq 3),\qquad W=c\,\theta^{1/2}\Big(1+\frac{\gamma}{4}\,\theta\log\theta+e\,\theta+\dots\Big)\ (q=3). \tag{4.14}$$

For $q=2$, $c_1=\gamma$. For $q<3$ the damping correction is positive (larger gap at the same remaining time and the same e); for $q>3$ it is negative. For $q=3$ the exponents 1 and $2-2\sigma$ resonate. Both formulas are confirmed numerically (fitted coefficients within 0.4% of the predicted ones; `mems_damped_ode.py`). In similarity variables the background is $\Psi_0+O(\gamma Te^{-\tau})$ (or $O(\gamma T\tau e^{-\tau})$ if $q=3$).

Proposition 4.12 (Damped touchdown, sketch)**.** *Assume the hypotheses of Theorem 4.10, except that the damping satisfies $0\le\gamma\le\delta/M$, i.e. $\gamma T_0\le\delta/M$ in general units. Then the conclusions of Theorem 4.10 hold verbatim, for radial and nonradial data, with the same comparison profiles w^{T^*,a^*}.*

Sketch. Add $-\gamma Te^{-\tau}(0,\psi_{a,2}+\phi_2)$ to $\mathbf{G}_a$. Its size is $\lesssim\gamma e^{-\tau}\le\gamma e^{-\omega\tau}$, its Lipschitz constant is $\lesssim\gamma$, and $\int_0^\tau e^{-\omega'(\tau-\tau')}e^{-\tau'}d\tau'\lesssim e^{-\omega\tau}$ since $\omega<\omega'<\omega_0\le 1$. Steps 3–5 go through with $\delta^2+|a|\delta$ replaced by $\delta^2+|a|\delta+\gamma$. The key point for the nonradial case is that the neutral component obeys $\frac{d}{d\tau}\mathbf{P}_0\Phi=\mathbf{P}_0\mathbf{G}_a(\Phi)$ (no Jordan block). The right side is *integrable* in τ, so $\mathbf{P}_0\Phi(\tau)$ has a limit, and the static choice of a by Brouwer makes it zero. A dynamical modulation of the boost (cf. Section 3.4) would be needed only for a non-integrable forcing of the neutral modes. □

Corollary 4.13 (Arbitrary damping; sketch)**.** *Let $\gamma\ge 0$ and let W be any solution of the damped ODE that touches down at $T_0>0$. Then data close enough to $(W(0),W'(0))$ in $H^k\times H^{k-1}(B_{T_0+\eta})$ lead to touchdown at $(T^*,0)$, $T^*\to T_0$ as the data converge, with the asymptotics of Theorem 4.10.*

Sketch. By (4.14), at $t_1=T_0-\theta_1$ the rescaled data of W differ from $(c,-\sigma c)$ by $O(\theta_1^{\min\{1,2-2\sigma\}})$ (times log if $q=3$), and the rescaled damping is $\gamma\theta_1$. Choose θ_1 so small that both are $\le\delta/(2M)$. By local well-posedness and continuous dependence on $[0,t_1]$ (the solution stays near $W(t)\ge W(t_1)>0$), data close enough to those of W produce data at t_1 satisfying the hypotheses of Proposition 4.12 after rescaling by θ_1. □

The same argument covers any perturbation that is $O(e^{-\tau})$ in similarity variables. For instance a smooth dielectric profile $f(x)>0$ in $-\lambda f(x)w^{-q}$ gives $\lambda f(Te^{-\tau}\xi)\psi_1^{-q}=\lambda f(0)\psi_1^{-q}+O(e^{-\tau})$. Touchdown at a point x_0 is then ODE-like with $c=c(\lambda f(x_0))$. This is not written out.

4.8 Physical interpretation

- *Rate and impact speed.* For the Coulomb force ($q=2$) the gap closes like $w\approx(9\lambda/2)^{1/3}(T-t)^{2/3}$, i.e. $1+u\approx(9\lambda/2)^{1/3}\big((T-t)/\varepsilon\big)^{2/3}$ in the original time. The impact speed $u_t\sim-\frac{2}{3}(9\lambda/2)^{1/3}\varepsilon^{-2/3}(T-t)^{-1/3}$ is infinite. In the parabolic model ($\varepsilon=0$) the ODE $W'=-\lambda W^{-2}$ gives instead $w\approx(3\lambda(T-t))^{1/3}$; for the rigorous parabolic theory see [49] and references therein. For $\varepsilon>0$, inertia dominates damping once $T-t\ll\varepsilon^2/\gamma$ (the scale-invariant damping is $\gamma(T-t)/\varepsilon^2$). Hence the *final* touchdown asymptotics is the hyperbolic one, Corollary 4.13, and the parabolic rate is at most an intermediate regime for $\varepsilon^2\ll T-t\ll 1$.

- *Flat profile, touchdown on a hypersurface.* The stable profile is constant in the light cone. Near touchdown the gap closes uniformly over a region of size $\sim T-t$ (in units of the wave

speed), and the touchdown set is locally a spacelike hypersurface $t = T^*(x)$ (Remark 4.11(a)). This differs sharply from parabolic MEMS, where touchdown is typically at isolated points. The focused profiles ψ_n of Conjecture 4.4 touch down at a point but are unstable.

- *Locality.* Only data in the base of the cone, $B_{T^*}(x_0)$, matter. The clamped boundary, the shape of Ω and the far-away dielectric profile are irrelevant as long as this ball lies inside Ω.

- *An explicit scenario.* Release the undamped device ($q = 2$) from rest, $w \equiv 1$, $w_t \equiv 0$, and let $B_R(x_0) \subset \Omega$. By finite speed of propagation $w(t,x) = W(t)$ for $|x - x_0| < R - t$. If $T_0 = \pi/(2\sqrt{2\lambda}) < R$, i.e. $\lambda > \pi^2/(8R^2)$, the membrane therefore touches down *simultaneously on the ball* $|x - x_0| < R - T_0$ (unless it has quenched elsewhere earlier; we have not excluded this). By Theorem 4.10 and Proposition 4.12 this flat touchdown is stable under small perturbations of the data, of the damping and of the permittivity. It then occurs on a nearby spacelike hypersurface with a boosted ODE profile. For $R = 1$ the threshold $\pi^2/8 \approx 1.234$ should be compared with the static pull-in values $\lambda^* \approx 0.350$ (interval $(-1, 1)$), 0.789 (unit disk) and 1.299 (unit ball in $\mathbb{R}^3$) (`mems_damped_ode.py`).

- *Dimensions.* $d = 1$ describes strings and membrane strips (beams with bending stiffness, Δ^2, are *not* covered: there is no light cone); $d = 2$ describes the membrane (e.g. a disk); $d = 3$ is included for comparison with the wave-equation literature.

4.9 Numerical checks

The eigenvalue problem in the sector ℓ is the quadratic pencil $z(1-z)h'' + (\ell + \frac{d}{2} - (b + \frac{3}{2})z)h' - \frac{1}{4}(b^2 + b - V)h = 0$, $b = \mu - \sigma + \ell$, $g = |\xi|^\ell Y_\ell h(|\xi|^2)$. We solved it by Chebyshev collocation in $z \in [0,1]$ without boundary conditions, keeping eigenvalues that are stable under refinement ($N = 20, 28$; larger N starts to resolve the non-smooth generalised eigenfunctions of $\operatorname{Re}\mu \le -\omega_k$ and produces spurious eigenvalues). For $V \equiv V_0$, $q = 2, 3, 4$, $d = 1, 2, 3$ and $\ell \le 2$, all eigenvalues with $\operatorname{Re}\mu > -1.9$ agree with (4.12) to $\le 10^{-5}$. No others appear. For $q = 3$ the double eigenvalue -1 splits as expected for a Jordan block. An independent discretisation of the full first-order operator $\mathbf{L}$ for $d = 1$ gives $\{1, 0, -\frac{2}{3}, -1, -\frac{5}{3}\}$ for $q = 2$ (both parities), as predicted. For the excited profiles ($d = 3$, $q = 2$, $V = q\lambda\psi_n^{-q-1}$) we find:

- ψ_1: $\{3.4955, 1\}$ for $\ell = 0$; $\{1, 0\}$ for $\ell = 1$ (space translation, which is non-trivial for a non-constant profile, and boost); none for $\ell = 2$.
- ψ_2: $\{20.35, 3.598, 1\}$ for $\ell = 0$ (this needs $N \ge 120$).

Thus ψ_1 has one and ψ_2 two genuinely unstable directions, as in Conjecture 4.4. See Fig. 2 (`mems_spectrum_numerics.py`). The independent numerics section of this note provides a further check.

4.10 Relation to Gimperlein–He–Lacey

The closest prior work is [57] (see also [73]). According to the summary in Section 2.1, it studies the self-similar quenching of (2.3) formally and numerically:

- it treats the ODE profile $\kappa_q(T-t)^{2/(q+1)}$;
- it obtains formally a sequence of self-similar solutions;
- it gives evidence that radial solutions increasing in $|x|$ quench at the origin, that the trivial profile is stable up to time translation, and that the non-trivial ones are unstable.

Caveat: the text of [57] could not be consulted while this section was written (no network access). The points below compare with that summary only, and should be checked against the paper.

(a) *Profile and constant.* $\kappa_q = c$, with $c^{q+1} = \lambda(q+1)^2/(2(q-1))$, (2.4).

(b) *Non-trivial profiles.* Our ψ_n (Table 3) are increasing in $|x|$ and accumulate at the *static* singular solution $A|x|^{2/(q+1)}$. The ratio law $\psi_n(0)/\psi_{n+1}(0) \to e^{\sigma\pi/\nu}$ gives a sharp quan-

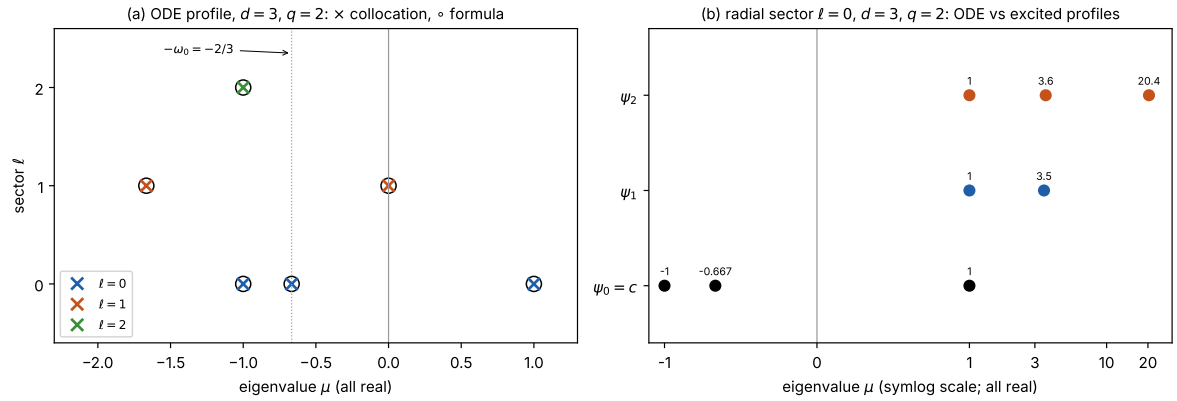


Figure 2: (a) Eigenvalues of the linearisation at the ODE profile, $d = 3$, $q = 2$, sectors $\ell = 0, 1, 2$: collocation ($\times$) versus (4.12) ($\circ$); the dotted line is $-\omega_0 = -\frac{2}{3}$ (energy mode). (b) Radial eigenvalues ($\mathrm{Re}\,\mu > -0.3$ for $\psi_{1,2}$) of the ODE profile $\psi_0 = c$ and of the first two excited profiles: ψ_n has n unstable eigenvalues in addition to the time-translation eigenvalue 1.

titative test for identifying them with the sequence of [57]. Theorem 4.3 shows that the sequence has no counterpart in $d = 1$.

(c) *Stability.* We give a Donninger-type proof (complete at the spectral level, a proof sketch for the nonlinear step, $d \leq 3$) of the stability of the ODE profile: modulo time translation for radial data and modulo time translation plus Lorentz boosts in general (Theorem 4.10), with damping (Proposition 4.12 and Corollary 4.13). We also give the decay rate $(T-t)^{\omega}$, $\omega < \min\{1, \frac{2(q-1)}{q+1}\}$, and identify the slowest stable mode as the ODE energy mode for $q < 3$. The numerically computed extra unstable eigenvalues of ψ_1, ψ_2 support the instability of the non-trivial profiles.

4.11 Caveats and open points

1. Theorem 4.10, Proposition 4.12, and Corollary 4.13 are proof sketches. The standard parts are referenced, not reproduced: generation of $\mathbf{L}_0$ on $\mathcal{H}^k$, equivalence of mild and strong solutions, continuity of the fixed point in (T, a), and Moser estimates. Lemmas 4.5 and 4.7 and Proposition 4.6 are complete modulo routine details. Lemma 4.7 is stated for $d \leq 3$; the general case needs the same Beta-function check.
2. The statements are local in the backward cone. Which points touch down first, the regularity of the touchdown hypersurface, $a^* = \nabla T^*$, and the behaviour outside the cone are open.
3. The existence of the non-trivial profiles for $d \geq 2$ is numerical (Conjecture 4.4); their spectra are low-precision numerics. Nonradial non-trivial profiles were not searched for.
4. For $q = 3$ the eigenvalue $-\omega_0$ is a Jordan block (Remark 4.8), so the rate $\omega_0 = 1$ itself is not attained.
5. Not covered: plate models with Δ^2, and nonlocal (series capacitor) models. In the latter the coefficient $\lambda(1 + \alpha \int w^{-1})^{-2}$ depends on the whole solution, and a flat touchdown on a set of positive measure would drive it to 0 (heuristic, not analysed).

5 Hyperbolic thermal explosion: exponential nonlinearities

In fast thermal processes (pulsed laser heating and processing, ignition of solid propellants and energetic materials, rapid combustion) the relevant time scales can approach the thermal relaxation time of the medium, and Fourier's law is replaced by the Maxwell–Cattaneo–Vernotte

(MCV) law [13, 117, 76, 114]. Combined with the Frank-Kamenetskii (FK) exothermic source [52, 4], this produces damped wave equations with exponential nonlinearities. The main points of this section are:

(1) There are two inequivalent hyperbolic models ("reduced" and "full"); their blowup is *qualitatively different*.

(2) Reduced model: the ODE blowup $\theta \simeq \log(2\tau_*/\delta) - 2\log(T-t)$ is stable modulo time translation and Lorentz boosts; the linearised spectrum is explicit, the spectral gap is 1 (Theorem 5.8).

(3) In $d = 1$ the reduced model is Liouville's equation; the explicit general solution confirms the theorem, including the sharpness of the gap (Section 5.5).

(4) Full (MCV-consistent) model: the blowup is $\theta \simeq -\log(\delta(T-t))$ (the *parabolic* rate) with a *hyperbolic* spatial structure; there is no Lorentz invariance, the neutral modes come from a family of tilted plane waves, and $\lambda = -1$ carries a Jordan block (Section 5.6).

(5) With Arrhenius kinetics there is no blowup at all; FK blowup is an intermediate asymptotic regime (Section 5.7).

Throughout, *known* results are attributed, statements labelled "proof sketch" are derived here with the Donninger machinery but not written out in full detail, and purely numerical observations are labelled as such. All algebraic identities below are verified symbolically in `code/therm_symbolic.py`; spectra in `code/therm_spectrum.py`; simulations in `code/therm_sim1d.py`.

5.1 Modelling: Cattaneo flux and Frank-Kamenetskii source

Let Θ be the absolute temperature, Θ_0 the ambient temperature, ρc the volumetric heat capacity, k the conductivity, τ_r the thermal relaxation time, and $QAe^{-E/(R\Theta)}$ the volumetric heat release (zeroth-order Arrhenius kinetics, reactant consumption neglected). Energy balance and MCV law read

$$\rho c\,\partial_t\Theta + \nabla\cdot q = QA\,e^{-E/(R\Theta)}, \qquad \tau_r\,\partial_t q + q = -k\nabla\Theta. \tag{5.1}$$

With the FK variable $\theta = E(\Theta-\Theta_0)/(R\Theta_0^2)$ and $\varepsilon = R\Theta_0/E$ one has the *exact* identity $e^{-E/(R\Theta)} = e^{-E/(R\Theta_0)}\exp\!\big(\theta/(1+\varepsilon\theta)\big)$. Scaling $x = Lx'$, $t = t_0 t'$ with the diffusion time $t_0 = \rho cL^2/k$, and $q = (kR\Theta_0^2/(EL))\,q'$, gives (primes dropped)

$$\theta_t + \nabla\cdot q = f_\varepsilon(\theta) := \delta\, e^{\theta/(1+\varepsilon\theta)}, \qquad \tau_*\, q_t + q = -\nabla\theta, \tag{5.2}$$

with the FK parameter $\delta = QAEL^2e^{-E/(R\Theta_0)}/(kR\Theta_0^2)$ and the dimensionless relaxation time $\tau_* = \tau_r/t_0 = \tau_r k/(\rho cL^2)$; the dimensionless thermal wave speed is $\tau_*^{-1/2}$. Applying $(1+\tau_*\partial_t)$ to the balance law and subtracting the divergence of the flux law eliminates q:

$$\boxed{\tau_*\theta_{tt} + \theta_t - \Delta\theta = f_\varepsilon(\theta) + \tau_*\,\partial_t f_\varepsilon(\theta) = f_\varepsilon(\theta) + \tau_* f_\varepsilon'(\theta)\,\theta_t} \qquad \text{(full model)}. \tag{5.3}$$

Equation (5.3) with data $(\theta,\theta_t)|_{t=0} = (\theta_0,\theta_1)$ is equivalent to Eq. (5.2) with any q_0 satisfying $-\nabla\cdot q_0 = \theta_1 - f_\varepsilon(\theta_0)$. The *reduced* model

$$\tau_*\theta_{tt} + \theta_t - \Delta\theta = f_\varepsilon(\theta) \qquad \text{(reduced model)} \tag{5.4}$$

drops the term $\tau_*\partial_t f_\varepsilon(\theta)$. It is obtained if the relaxation is applied to the *whole* energy balance (a single phase lag $\rho c\,\Theta_t(t+\tau_r) = -\nabla\cdot q_{\text{Fourier}}(t) + Q(t)$, expanded to first order), or by discarding

$\tau_*\partial_t f$ as small. Both forms occur in the literature on hyperbolic reaction–diffusion equations; e.g. the full form $\tau_* n_{tt} + n_t = Dn_{xx} + F(n) + \tau_* F'(n) n_t$ is the starting point of [96], while the reduced form is what one obtains by simply adding $\tau_*\theta_{tt}$ to the parabolic Gelfand problem. We could not verify, within our (limited) literature search, which of the two forms is used in the thermal-explosion-specific literature; the distinction is essential near blowup. We show below that the dropped term is *dominant* at blowup, so the two models are not asymptotically equivalent near thermal runaway.

Normal forms. In the FK limit $\varepsilon \to 0$, $f_0(\theta) = \delta e^\theta$. Rescale time $t \mapsto \sqrt{\tau_*}\, t$ (wave speed 1) and put $\gamma := \tau_*^{-1/2}$. The reduced model becomes (keeping δ, as in Section 3)

$$u_{tt} + \gamma u_t - \Delta u = \delta e^u, \qquad u(t,x) = \theta(\sqrt{\tau_*}\, t, x), \tag{5.5}$$

and the full model becomes, with $u(t,x) = \theta(\sqrt{\tau_*}\, t, x) + \log(\sqrt{\tau_*}\, \delta)$,

$$(\partial_t + \gamma)(u_t - e^u) = \Delta u \quad \Longleftrightarrow \quad u_{tt} - \Delta u = e^u u_t + \gamma\,(e^u - u_t). \tag{5.6}$$

Both are invariant under $u \mapsto u(\mu t, \mu x) + \kappa \log \mu$ ($\kappa = 2$ reduced, $\kappa = 1$ full) *provided* $\gamma \mapsto \mu\gamma$. Hence the damping enters only through the dimensionless product γT (T = blowup time); near the singularity it is a perturbation of relative size $\gamma(T-t) \to 0$.

Remark 5.1 (Quenching form). For readers used to MEMS models: $w = e^{-u/2}$ turns Eq. (5.5) with $\gamma = 0$ into the quenching problem $w_{tt} - \Delta w = (w_t^2 - |\nabla w|^2)/w - \delta/(2w)$, and the ODE blowup below becomes the linear touchdown $w = \sqrt{\delta/2}\,(T-t)$.

5.2 Reduced model: ODE blowup, symmetries, similarity variables

Let first $\gamma = 0$: $u_{tt} - \Delta u = \delta e^u$. The spatially homogeneous solutions satisfy $u'' = \delta e^u$; those that blow up are $e^u = 2a^2/(\delta \sinh^2(a(T-t)))$, $2/(\delta(T-t)^2)$, $2a^2/(\delta \sin^2(a(T-t)))$, all of the form $u = u_T + O((T-t)^2)$ with the *ODE blowup*

$$u_T(t) = \log \frac{2}{\delta} - 2\log(T-t). \tag{5.7}$$

Symmetries. Time and space translations, rotations, the scaling $u \mapsto u(\mu t, \mu x) + 2\log\mu$, the shift $u \mapsto u + c_0$, $\delta \mapsto \delta e^{-c_0}$ (so δ can be normalised on $\mathbb{R}^d$), and Lorentz boosts. Boosting Eq. (5.7) and re-centring the singular hyperplane at $(T, 0)$ gives the family (checked symbolically for $d = 3$)

$$u_{T,a}(t,x) = \log \frac{2}{\delta} - 2\log(A_0(T-t) + a\cdot x), \qquad A_0 = A_0(a) = \sqrt{1+|a|^2}, \quad a \in \mathbb{R}^d, \tag{5.8}$$

singular on the spacelike hyperplane $t = T + a\cdot x/A_0$ (velocity a/A_0, $|a/A_0| < 1$). Space translations and scalings map $u_{T,a}$ to $u_{T',a}$, so $\{u_{T,a}\}$ is the full $(d+1)$-parameter symmetry orbit of the ODE blowup for a fixed blowup point $x_0 = 0$.

Similarity variables. With the conventions of Section 3, $\tau = \log T - \log(T-t)$, $\xi = x/(T-t)$, $\Lambda = \xi\cdot\nabla_\xi$, $\psi_1 = u + 2\log(T-t)$, $\psi_2 = (T-t)u_t$, Eq. (5.5) becomes

$$\partial_\tau \psi_1 = -\Lambda\psi_1 + \psi_2 - 2, \qquad \partial_\tau \psi_2 = \Delta\psi_1 - \Lambda\psi_2 - \psi_2 + \delta e^{\psi_1} - \gamma T e^{-\tau}\psi_2. \tag{5.9}$$

The boosted ODE blowups are *static* solutions (for $\gamma = 0$):

$$\Psi_a(\xi) = \Big(\log\tfrac{2}{\delta} - 2\log(A_0 + a\cdot\xi),\ \tfrac{2A_0}{A_0 + a\cdot\xi}\Big), \qquad \Psi_0 = (\log\tfrac{2}{\delta},\, 2), \tag{5.10}$$

smooth on $\overline{B^d}$ for every a. Writing $\psi = \Psi_0 + \varphi$,

$$\begin{aligned} \partial_\tau \varphi &= \mathbf{L}\varphi + \mathbf{N}(\varphi) - \gamma T e^{-\tau}(0, 2+\varphi_2), \\ \mathbf{L}\varphi = \begin{pmatrix} -\Lambda\varphi_1 + \varphi_2 \\ \Delta\varphi_1 - \Lambda\varphi_2 - \varphi_2 + 2\varphi_1 \end{pmatrix}, &\qquad \mathbf{N}(\varphi) = \begin{pmatrix} 0 \\ 2(e^{\varphi_1} - 1 - \varphi_1) \end{pmatrix}, \end{aligned} \tag{5.11}$$

since $\delta e^c = 2$ for $c = \log(2/\delta)$; thus $V_0 = 2$, $s = 0$ in the notation of the note.

Damped homogeneous blowup. For $\gamma > 0$ the profile Eq. (5.7) is not exact. A formal Fuchsian expansion of $u'' + \gamma u' = \delta e^u$ gives

$$u(t) = u_T(t) + \gamma(T-t) + \tfrac{2}{3}\gamma^2 (T-t)^2 \log(T-t) + b\,(T-t)^2 + \dots, \tag{5.12}$$

with a free constant b: the damping produces an $e^{-\tau}$ correction in similarity variables and a resonant logarithm at order $e^{-2\tau}$ (the $(T-t)^2$ mode is the eigenvalue -2 below).

5.3 Linearised operator and its spectrum

Let $\mathcal{H}^k := H^k(B^d) \times H^{k-1}(B^d)$, $k \in \mathbb{N}$, $k > d/2$. Write $\mathbf{L} = \mathbf{L}_0 + \mathbf{L}'$ with the free part $\mathbf{L}_0\varphi = (-\Lambda\varphi_1 + \varphi_2,\ \Delta\varphi_1 - \Lambda\varphi_2 - \varphi_2)$ and $\mathbf{L}'\varphi = (0, 2\varphi_1)$, which is compact on $\mathcal{H}^k$ (Rellich).

Lemma 5.2 (Free evolution)**.** *Let φ be a smooth solution of $\partial_\tau\varphi = \mathbf{L}_0\varphi$ and $E_k(\varphi) = \|D^k\varphi_1\|^2_{L^2(B^d)} + \|D^{k-1}\varphi_2\|^2_{L^2(B^d)}$, where D^j is the tensor of all j-th derivatives. Then $\frac{d}{d\tau}E_k \le -2(k - \frac{d}{2})E_k$. Consequently the semigroup generated by $\mathbf{L}_0$ on $\mathcal{H}^k$ has essential growth bound $\le -(k - \frac{d}{2})$; since $\{\operatorname{Re}\lambda < \frac{d}{2} - k\} \subset \sigma_p(\mathbf{L}_0)$, the free decay rate on $\mathcal{H}^k$ is exactly $k - \frac{d}{2}$. (The finitely many eigenvalues of $\mathbf{L}_0$ with $\operatorname{Re}\lambda > \frac{d}{2} - k$, e.g. $\lambda = 0$ with eigenfunction $(1, 0)$, live on a finite-dimensional invariant space of polynomials.)*

Proof. Differentiating, $\partial_\tau D^k\varphi_1 = -(\Lambda + k)D^k\varphi_1 + D^k\varphi_2$ and $\partial_\tau D^{k-1}\varphi_2 = \Delta D^{k-1}\varphi_1 - (\Lambda + k)D^{k-1}\varphi_2$. Use $-(\Lambda f, f) = \frac{d}{2}\|f\|^2 - \frac{1}{2}\int_{\mathbb{S}^{d-1}} |f|^2$ and integrate the cross term by parts: $(D^k\varphi_2, D^k\varphi_1) + (\Delta D^{k-1}\varphi_1, D^{k-1}\varphi_2) = \int_{\mathbb{S}^{d-1}} \partial_r D^{k-1}\varphi_1 \cdot D^{k-1}\varphi_2$. Hence

$$\tfrac{1}{2}\tfrac{d}{d\tau}E_k = -(k - \tfrac{d}{2})E_k - \tfrac{1}{2}\int_{\mathbb{S}^{d-1}} (|D^k\varphi_1|^2 + |D^{k-1}\varphi_2|^2 - 2\,\partial_r D^{k-1}\varphi_1 \cdot D^{k-1}\varphi_2) \le -(k - \tfrac{d}{2})E_k,$$

because $|\partial_r D^{k-1}\varphi_1| \le |D^k\varphi_1|$ on $|\xi| = 1$. The kernel of E_k is $Y = \mathcal{P}_{k-1} \times \mathcal{P}_{k-2}$ (polynomials), which is finite dimensional and $\mathbf{L}_0$-invariant; on $\mathcal{H}^k/Y$ the seminorm $E_k^{1/2}$ is an equivalent norm (Bramble–Hilbert), and the induced quotient semigroup decays like $e^{-(k-d/2)\tau}$. This gives the bound on the essential growth bound. Conversely, for $\operatorname{Re}\lambda < \frac{d}{2} - k$ both Frobenius solutions at $|\xi| = 1$ of the eigenvalue equation (see the proof of Proposition 5.3) lie in H^k, so every such λ is an eigenvalue. □

In the power case the same energy structure (with top-order rate $s + k - \frac{d}{2}$) underlies the semigroup theory of [41, 107], which works with suitable equivalent inner products; we have not checked that those constructions carry over verbatim to $s = 0$, but the quotient argument above is all that is used below. Since $\mathbf{L}'$ is compact, $\sigma(\mathbf{L}) \cap \{\operatorname{Re}\lambda > \frac{d}{2} - k\}$ consists of isolated eigenvalues of finite algebraic multiplicity.

Proposition 5.3 (Point spectrum, reduced model)**.** *Let $k > d/2$ and $\operatorname{Re}\lambda > \frac{d}{2} - k$. Then $\lambda \in \sigma_p(\mathbf{L})$ if and only if*

$$\lambda \in \{1 - \ell - 2n\} \cup \{-2 - \ell - 2n\}, \qquad \ell, n \in \mathbb{N}_0 \quad (\ell \in \{0, 1\} \text{ if } d = 1). \tag{5.13}$$

The eigenfunctions are $\varphi = (g, (\lambda + \Lambda)g)$ with $g = |\xi|^\ell Y_\ell(\xi/|\xi|)\,{}_2F_1(\alpha, \beta; \ell + \frac{d}{2}; |\xi|^2)$, a polynomial, where $\alpha = \frac{\lambda+\ell-1}{2}$, $\beta = \frac{\lambda+\ell+2}{2}$ and Y_ℓ is a spherical harmonic of degree ℓ.

λ	sector	eigenfunction g	mult.	origin
1	$\ell=0$	1	1	time translation $\partial_T u_{T,0}$
0	$\ell=1$	ξ_j	d	boosts, $\partial_{a_j}\Psi_a\vert_{a=0}=-2(\xi_j,\xi_j)$
-1	$\ell=0,2$	$1-\vert\xi\vert^2/d$; harm. quadratics	$1+\dim\mathcal{Y}_2$	curvature of blowup surface
-2	$\ell=0,1,3$	1; …		energy of the ODE; …

Table 4: Leading eigenvalues of $\mathbf{L}$ (reduced model, $V_0=2$, $s=0$). The interpretation of $\lambda=-1$ follows from the exact 1D solution, Proposition 5.11.

Proof. Eliminating $\varphi_2=(\lambda+\Lambda)\varphi_1$ gives, for $g=\varphi_1$,

$$(\delta_{ij}-\xi_i\xi_j)\partial_i\partial_j g-2(\lambda+1)\Lambda g-\big(\lambda(\lambda+1)-2\big)g=0\quad\text{in } B^d. \tag{5.14}$$

The operator is elliptic in the open ball, so $g\in C^\infty(B^d)$, and the projection of g onto spherical harmonics of degree ℓ has the form $|\xi|^\ell Y_\ell\,h(\rho)$, $\rho=|\xi|^2$, with h smooth on $[0,1)$. A direct computation (verified for general d,ℓ) turns Eq. (5.14) into $4|\xi|^\ell Y_\ell$ times the hypergeometric equation $\rho(1-\rho)h''+[c-(\alpha+\beta+1)\rho]h'-\alpha\beta h=0$ with $c=\ell+\frac{d}{2}$ and α,β as stated. Regularity at $\rho=0$ selects $h={}_2F_1(\alpha,\beta;c;\rho)$. At $\rho=1$ the Frobenius exponents are 0 and $c-\alpha-\beta=\frac{d-1}{2}-\lambda$; the corresponding branch $(1-\rho)^{\frac{d-1}{2}-\lambda}$ (multiplied by $\log(1-\rho)$ in the integer case) is *not* in H^k near $\rho=1$, because $\frac{d-1}{2}-\operatorname{Re}\lambda<k-\frac12$. By the connection formulae [106, §15.8], the coefficient of this branch in ${}_2F_1(\alpha,\beta;c;\rho)$ is a non-vanishing multiple of $1/(\Gamma(\alpha)\Gamma(\beta))$ in all three cases $c-\alpha-\beta\notin\mathbb{Z}$, $\in\mathbb{N}_0$ (logarithmic term), $\in-\mathbb{N}$ (pole-type leading term $\Gamma(\alpha+\beta-c)\Gamma(c)/(\Gamma(\alpha)\Gamma(\beta))$). Hence λ is an eigenvalue iff $\alpha\in-\mathbb{N}_0$ or $\beta\in-\mathbb{N}_0$, i.e. Eq. (5.13), and then ${}_2F_1$ is a polynomial. (For $d=1$ use the even/odd decomposition, $\ell\in\{0,1\}$.) □

Lemma 5.4 (Semisimplicity)**.** *The eigenvalues $\lambda=1$ and $\lambda=0$ of $\mathbf{L}$ on $\mathcal{H}^k$, $k>d/2$, are semisimple.*

Proof. Let $P(\lambda)g:=(\Lambda+\lambda)(\Lambda+\lambda+1)g-\Delta g-2g$. A Jordan chain $(\mathbf{L}-\lambda_0)w=\varphi_0$ with $\varphi_0=(g_0,(\lambda_0+\Lambda)g_0)$ forces $P(\lambda_0)w_1=-P'(\lambda_0)g_0=-(2\Lambda+2\lambda_0+1)g_0$. For $(\lambda_0,\ell)\in\{(1,0),(0,1)\}$ we have $g_0=|\xi|^\ell Y_\ell$, $\alpha=0$, $\beta=\frac32$, and $P'(\lambda_0)g_0=3g_0$; in the ℓ-sector this becomes $\rho(1-\rho)h''+(c-\frac52\rho)h'=\frac34$. Since $\frac{d}{d\rho}\big[\rho^c(1-\rho)^{\beta+1-c}h'\big]=\rho^{c-1}(1-\rho)^{\beta-c}\big[\rho(1-\rho)h''+(c-(\beta+1)\rho)h'\big]$, regularity at 0 gives $\rho^c(1-\rho)^{\frac52-c}h'(\rho)=\frac34\int_0^\rho s^{c-1}(1-s)^{\frac32-c}\,ds$. Write $q=\frac52-c$. If $q>0$ the right side tends to $\frac34B(c,q)\neq0$, so $h'\sim\frac34B(c,q)(1-\rho)^{-q}$; if $q=0$ it diverges like $-\frac34\log(1-\rho)$; if $q<0$, $q\notin\mathbb{Z}$, it equals $\frac34B(c,q)+(1-\rho)^q\times$(analytic) by analytic continuation in q, with $B(c,q)=\Gamma(c)\Gamma(q)/\Gamma(\frac52)\neq0$, so again $h'\ni\frac34B(c,q)(1-\rho)^{-q}$; if $q\in-\mathbb{N}$ the integrand contains $(1-s)^{-1}$ with coefficient $(-1)^{c-5/2}\binom{c-1}{c-5/2}\neq0$, producing $(1-\rho)^{-q}\log(1-\rho)$ in h'. In every case $h\ni(1-\rho)^{c-\frac32}$ (times $\log(1-\rho)$ in the integer cases) with non-zero coefficient, which is not in H^k near $\rho=1$ since $c-\frac32=\frac{d-1}{2}-\lambda_0<k-\frac12$. □

Corollary 5.5 (Spectral gap)**.** *For $k>d/2$ set $\omega_0:=\min\{1,k-\frac{d}{2}\}$. Then $\sigma(\mathbf{L})\cap\{\operatorname{Re}\lambda>-\omega_0\}=\{0,1\}$; $\lambda=1$ is simple with eigenfunction $(1,1)$, $\lambda=0$ is semisimple of multiplicity d with eigenfunctions (ξ_j,ξ_j). With $\mathbf{P}$ the Riesz projection onto these $d+1$ modes, $\|e^{\tau\mathbf{L}}(1-\mathbf{P})\|\le C_\omega e^{-\omega\tau}$ for every $\omega<\omega_0$. If $k\ge\frac{d}{2}+1$, then $\omega_0=1$.*

Remark 5.6 (Boosted profiles; polynomial structure; numerics). (i) A mode $e^{\lambda\tau}g(\xi)$ corresponds to the solution $v=(T-t)^{-\lambda}g(x/(T-t))$ of the linearised equation, homogeneous of degree $-\lambda$ in $(t-T,x)$; Lorentz transformations fixing $(T,0)$ preserve the backward cone, homogeneity and smoothness and conjugate the linearisations at $u_{T,0}$ and $u_{T,a}$. Hence $\mathbf{L}_a$ (linearisation at Ψ_a) has the same point spectrum Eq. (5.13). (ii) $\mathbf{L}$ maps $\mathcal{P}_N\times\mathcal{P}_N$ (polynomials in ρ times

$|\xi|^\ell Y_\ell$) into itself and is block upper-triangular in the monomial basis, with diagonal blocks $\begin{pmatrix} -(\ell+2m) & 1 \\ 2 & -(\ell+2m)-1 \end{pmatrix}$, whose eigenvalues are exactly $1-\ell-2m$ and $-2-\ell-2m$ – a two-line proof that Eq. (5.13) are eigenvalues (completeness needs the connection argument). Consequently Chebyshev collocation in ρ is exact in exact arithmetic; in floating point ($N = 32$) it reproduces Eq. (5.13) to within $3 \cdot 10^{-5}$ for $\operatorname{Re}\lambda > -1.6$ ($d \le 5$, $\ell \le 4$), while non-normality produces spurious complex pairs for $\operatorname{Re}\lambda \lesssim -2.5$ (for the full model of Section 5.6 the double eigenvalue -1 is split by up to 0.07, the typical $\sqrt{\text{rounding}}$-sensitivity of a Jordan block). Independently, the connection coefficient computed by direct numerical integration of the radial ODE agrees with the closed form to 10^{-20}, and the argument principle applied to it counts exactly the zeros $\{1\}$ ($\ell = 0$), $\{0\}$ ($\ell = 1$), $\emptyset$ ($\ell = 2$) in $[-0.45, 2.45] \times [-3, 3]$, for this model with $d = 2$ and for the full model of Section 5.6 with $d = 3$ (`code/therm_spectrum.py`).

5.4 Nonlinear stability of the ODE blowup (reduced model)

Lemma 5.7 (Nonlinearity)**.** *Let $k > d/2$ and C_k the algebra constant of $H^k(B^d)$. For $\|f\|_{H^k}, \|g\|_{H^k} \le R$,*

$$\|e^f - 1 - f - (e^g - 1 - g)\|_{H^k} \le (e^{C_k R} - 1)\|f - g\|_{H^k} \le C_k R e^{C_k R}\|f - g\|_{H^k}.$$

Hence $\mathbf{N} : \mathcal{H}^k \to \mathcal{H}^k$ is locally Lipschitz with $\|\mathbf{N}(\varphi) - \mathbf{N}(\tilde\varphi)\| \lesssim (\|\varphi\| + \|\tilde\varphi\|)\|\varphi - \tilde\varphi\|$ on bounded sets, and the same holds for $\mathbf{N}_a(\varphi) = (0, V_a(e^{\varphi_1} - 1 - \varphi_1))$, $V_a = 2/(A_0 + a \cdot \xi)^2$.

Proof. $\|f^n - g^n\|_{H^k} \le n C_k^{n-1} R^{n-1}\|f - g\|_{H^k}$; sum $\sum_{n\ge 2} \frac{1}{n!}(\cdot)$. □

Theorem 5.8 (Stable ODE blowup for $u_{tt} - \Delta u = \delta e^u$; proof sketch)**.** *Let $d \ge 1$, $\delta > 0$, $k \in \mathbb{N}$ with $k > d/2$, and $0 < \omega < \min\{1, k - \frac{d}{2}\}$. There are $\varepsilon_0, M > 0$ such that the following holds. Let $(f, g) \in H^k \times H^{k-1}(B_2^d)$ with $\|(f, g)\| \le \varepsilon \le \varepsilon_0$ and consider Eq. (5.5) with $\gamma = 0$ and data $u(0) = u_1(0) + f$, $u_t(0) = \partial_t u_1(0) + g$, where $u_1 = u_{T=1}$ is Eq. (5.7). Then there exist $T \in [1 - M\varepsilon, 1 + M\varepsilon]$ and $a \in \mathbb{R}^d$, $|a| \le M\varepsilon$, such that the solution exists in the backward light cone $\Gamma_T = \{(t, x) : 0 \le t < T,\ |x| \le T - t\}$ and, for $t \in [0, T)$,*

$$(T-t)^{j-\frac{d}{2}}\,\big\|u(t) - u_{T,a}(t)\big\|_{\dot H^j(B_{T-t})} \le M\varepsilon\,(T-t)^\omega, \qquad j = 0, \dots, k,$$
$$(T-t)^{j+1-\frac{d}{2}}\,\big\|\partial_t u(t) - \partial_t u_{T,a}(t)\big\|_{\dot H^j(B_{T-t})} \le M\varepsilon\,(T-t)^\omega, \qquad j = 0, \dots, k-1.$$

In particular $\sup_{|x|\le T-t} |u(t,x) - u_{T,a}(t,x)| \le M\varepsilon(T-t)^\omega$, i.e. $e^u = \frac{2}{\delta(A_0(T-t)+a\cdot x)^2}(1 + O(\varepsilon(T-t)^\omega))$ in Γ_T. For radial data, $a = 0$.

Proof sketch. We follow [38, 39] (modulation of T) and [41, 107] (non-radial data and Lorentz modulation). *Step 1.* For parameters (T, a) near $(1, 0)$ write $\psi = \Psi_a + \varphi$ in the similarity variables of $(T, 0)$: $\partial_\tau \varphi = \mathbf{L}_a \varphi + \mathbf{N}_a(\varphi)$, with $\mathbf{L}_a = \mathbf{L}_0 + (0, V_a \cdot)$ and data $\varphi(0) = \mathbf{U}(f, g; T, a)$, which depends smoothly on (T, a) and satisfies $\mathbf{U}(0, 0; 1, 0) = 0$. *Step 2 (linear).* By Lemma 5.2, compactness of $\mathbf{L}_a - \mathbf{L}_0$, Proposition 5.3 and Lemma 5.4 and Remark 5.6(i), $\mathbf{L}_a$ has, for $|a|$ small, exactly the unstable eigenvalue 1 (eigenfunction from ∂_T) and the semisimple eigenvalue 0 (eigenfunctions $\partial_{a_j}\Psi_a$), the rest of the spectrum in $\{\operatorname{Re}\lambda \le -\omega_0\}$, and $\|e^{\tau \mathbf{L}_a}(1 - \mathbf{P}_a)\| \le Ce^{-\omega\tau}$ uniformly in a (upper semicontinuity of the spectrum under the small bounded perturbation $\mathbf{L}_a - \mathbf{L}$). *Step 3 (nonlinear, fixed T, a).* Solve the modified Duhamel equation $\varphi(\tau) = e^{\tau \mathbf{L}_a}(\mathbf{U} - \mathbf{C}(\varphi, \mathbf{U})) + \int_0^\tau e^{(\tau-\sigma)\mathbf{L}_a}\mathbf{N}_a(\varphi(\sigma))\,d\sigma$, $\mathbf{C}(\varphi, \mathbf{U}) = \mathbf{P}_a\mathbf{U} + \int_0^\infty e^{-\sigma \mathbf{L}_a}\mathbf{P}_a\mathbf{N}_a(\varphi(\sigma))d\sigma$ (Lyapunov–Perron correction on the $(d+1)$-dimensional range of $\mathbf{P}_a$), by contraction in $\{\sup_\tau e^{\omega\tau}\|\varphi(\tau)\|_{\mathcal{H}^k} \le C\varepsilon\}$, using Lemma 5.7. *Step 4 (choice of T, a).* Since $\partial_T \mathbf{U}$ and $\partial_a \mathbf{U}$ at $(1, 0)$ span $\operatorname{ran}\mathbf{P}_0$ (they are the symmetry modes), the finite-dimensional map $(T, a) \mapsto \mathbf{C}(\varphi_{T,a}, \mathbf{U}(f, g; T, a)) \in \operatorname{ran}\mathbf{P}_a$ has a zero with $|T - 1| + |a| \lesssim \varepsilon$ (Brouwer); for these parameters φ solves the original Duhamel equation.

Step 5. Local well-posedness of Eq. (5.5) in $H^k \times H^{k-1}$ ($k > d/2$, e^u smooth) and finite speed of propagation identify φ with the solution in Γ_T; undoing the similarity transformation, $\|\varphi(\tau)\|_{\mathcal{H}^k}$ is equivalent to the weighted norms in the statement. □

Remark 5.9 (Damping). For $\gamma > 0$, Eq. (5.11) contains the extra terms $-\gamma T e^{-\tau}(0, \Psi_{a,2}+\varphi_2)$: a linear perturbation with integrable norm $\gamma T e^{-\tau}$ and a forcing of size $\gamma T e^{-\tau}$. Both are harmless in Steps 2–4 for $\omega < 1$, provided γT is small, and the conclusion of Theorem 5.8 holds with $u_{T,a}$ still the *undamped* boosted profile and $M\varepsilon$ replaced by $M(\varepsilon+\gamma)$ (the boosted profiles are only approximate solutions since damping breaks Lorentz invariance; the defect $O(\gamma(T-t))$ matches Eq. (5.12)). Smallness of γT is no restriction: by the scaling symmetry it means $T - t_0 \ll \tau_r$ at the starting time t_0, which can always be reached by Cauchy stability on $[0, t_0]$ for data close to the damped ODE blowup. The forcing is spatially constant and decays like $e^{-\tau}$; it does not resonate with $\lambda = -1$ (whose $\ell = 0$ eigenfunction $1 - |\xi|^2/d$ is not constant), but its second-order effect at $e^{-2\tau}$ resonates with the constant mode $\lambda = -2$, which produces the logarithm in Eq. (5.12).

5.5 An exact check in one dimension: Liouville's equation

For $d = 1$, $\gamma = 0$, Eq. (5.5) is the hyperbolic Liouville equation [93]. In null coordinates $\sigma = t + x$, $\eta = t - x$ it reads $4u_{\sigma\eta} = \delta e^u$.

Proposition 5.10 (Liouville's formula). *Let F, G be smooth with $F'(\sigma)G'(\eta) > 0$. Then*

$$u(t,x) = \log \frac{8\,F'(t+x)\,G'(t-x)}{\delta\,\big(F(t+x)+G(t-x)\big)^2} \tag{5.15}$$

solves $u_{tt} - u_{xx} = \delta e^u$ wherever $F + G \neq 0$. Conversely, every C^2 solution is locally of this form: $u_{\sigma\sigma} - \frac{1}{2}u_\sigma^2$ depends on σ only and equals the Schwarzian $\mathcal{S}(F) = F'''/F' - \frac{3}{2}(F''/F')^2$, so $F = y_1/y_2$ for two solutions of the Hill equation $y'' + \frac{1}{2}(u_{\sigma\sigma} - \frac{1}{2}u_\sigma^2)y = 0$ (similarly G), computable from the Cauchy data; (F, G) is unique up to the diagonal action of $PSL(2,\mathbb{R})$ that leaves Eq. (5.15) invariant.

For smooth Cauchy data, F and G are local diffeomorphisms into $\mathbb{R} \cup \{\infty\}$ (F', G' never vanish since $u(0,\cdot)$ is finite; a Möbius normalisation makes F, G finite near any given point), and the solution blows up exactly on $\{F(t+x)+G(t-x) = 0\}$. The ODE blowup corresponds to $F = (T-\sigma)/2$, $G = (T-\eta)/2$ (then $F + G = T - t$).

Proposition 5.11 (Local structure of 1D blowup). *Let u be given by Eq. (5.15) and let $F(T + x_0) + G(T - x_0) = 0$. Then near (T, x_0) the blowup set is a smooth spacelike curve $t = T(x)$, $T(x_0) = T$, $T'(x_0) = \frac{q-p}{p+q} \in (-1, 1)$ with $p = F'(T+x_0)$, $q = G'(T-x_0)$. In similarity variables centred at (T, x_0), in every $C^m([-1,1])$,*

$$\begin{aligned} \psi(\tau,\cdot) &= \Psi_a + Te^{-\tau}(g_1, g_2) + O(e^{-2\tau}), \qquad \frac{a}{A_0} = T'(x_0), \\ g_1(\xi) &= T''(x_0)\,A_0^2\,\frac{1-\xi^2}{1+T'(x_0)\xi}, \qquad g_2 = (\Lambda - 1)g_1, \end{aligned} \tag{5.16}$$

and (g_1, g_2) is an eigenfunction of $\mathbf{L}_a$ with eigenvalue -1.

Proof. The implicit function theorem applies since $\partial_t(F + G) = p + q \neq 0$, and $pq > 0$ gives $|T'| < 1$. Expanding F, G to third order at the blowup point and inserting $t = T - s$, $x = x_0 + s\xi$, $s = Te^{-\tau}$, into Eq. (5.15) gives the expansion (verified symbolically); the leading term reproduces Ψ_a exactly, with $1 - a^2/A_0^2 = 4pq/(p+q)^2$, and g_1 is identified with the curvature by differentiating $F(T(x)+x)+G(T(x)-x) = 0$ twice. Because the expansion is in powers of $s = Te^{-\tau}$ and solves Eq. (5.9), its $O(s)$ coefficient solves $(\mathbf{L}_a + 1)(g_1, g_2) = 0$; this is also checked directly. □

Corollary 5.12 (What the exact solution tells us)**.** *For $d = 1$: (i) every blowup point of a smooth solution is non-characteristic, the blowup curve is smooth and spacelike, and near each blowup point the solution converges to the boosted ODE profile in every C^m norm on the light cone at the rate $T - t$; (ii) the modulation parameters of Theorem 5.8 are $T = T(x_0)$ and $a/A_0 = T'(x_0)$; (iii) the rate $e^{-\tau}$ is attained iff $T''(x_0) \neq 0$, i.e. generically, so the gap $\omega_0 = 1$ of Corollary 5.5 is* sharp*, and the first stable eigenvalue -1 is excited precisely by the curvature of the blowup curve (cf. Table 4); (iv) perturbations of the data act on (F, G) and hence move $T(x_0), T'(x_0)$: the $d + 1 = 2$ unstable/neutral directions are exactly the symmetry modes.*

Item (i) contrasts with the power nonlinearity in 1D, where characteristic blowup points exist [99]. Statements of this type are presumably classical; related 1D results are the C^1 spacelike blowup curve for mixed problems [63, 64] and the blowup rate near non-characteristic points for rough data [1].

Numerical cross-check. We solved Eq. (5.9) ($d = 1$, Chebyshev collocation in ξ, no boundary conditions since the cone boundary is outflow, RK4 in τ) with data taken from an explicit Liouville solution, and located T by bisection using the instability e^{τ}. The bisection reproduces the exact T to 10^{-14}, the boost to six digits, and the error $\|\Psi(\tau) - \Psi_a\|_{H^1 \times H^1}$ decays with fitted rate -1.0016 (exact solution: -1.0016); with damping $\gamma = 1$ (same data) the fitted rate is -0.986, see Fig. 3(b,c).

5.6 The full Cattaneo model: a different blowup

Proposition 5.13 (Homogeneous dynamics of the full model)**.** *For spatially homogeneous solutions, Eq. (5.3) is $(1 + \tau_* \partial_t)(\theta' - f_\varepsilon(\theta)) = 0$, i.e. $\theta' = f_\varepsilon(\theta) + Ce^{-t/\tau_*}$, $C = \theta'(0) - f_\varepsilon(\theta(0))$. For $\varepsilon = 0$ every blowing-up solution satisfies $\theta(t) = -\log(\delta(T - t)) + O(T - t)$, and $\theta = -\log(\delta(T - t))$ is an exact solution. There is no solution with $\theta \sim c - 2\log(T - t)$: inserting this ansatz leaves the unbalanced term $-2\tau_* \delta e^c (T - t)^{-3}$.*

Thus the term $\tau_* \partial_t f(\theta)$ dropped in Eq. (5.4) *dominates* at blowup: the MCV-consistent model blows up at the *parabolic* rate $e^\theta \sim (\delta(T - t))^{-1}$ (and, if $C = 0$, at the classical FK ignition time), whereas the reduced model predicts the faster $e^\theta \sim 2\tau_*/(\delta(T - t)^2)$. Physically, in a homogeneous hot spot there is no flux, and relaxing the flux cannot change the local runaway. The *spatial* structure is nevertheless hyperbolic, as we now show.

Similarity variables and profiles. For Eq. (5.6) the scaling has $\kappa = 1$; set $\psi_1 = u + \log(T - t)$, $\psi_2 = (T - t)u_t$:

$$\partial_\tau \psi_1 = -\Lambda \psi_1 + \psi_2 - 1, \qquad \partial_\tau \psi_2 = \Delta \psi_1 - \Lambda \psi_2 - \psi_2 + e^{\psi_1} \psi_2 - \gamma T e^{-\tau} (\psi_2 - e^{\psi_1}). \tag{5.17}$$

The ODE profile $\Psi_0^{\mathrm{F}} = (0, 1)$ ($u = -\log(T - t)$) is exact for *every* γ. Equation (5.6) is not Lorentz invariant ($e^u u_t$ is not a scalar), but for $\gamma = 0$ it has the exact *tilted* plane-wave family

$$u_{T,v}(t, x) = \log(1 - |v|^2) - \log(T - t + v \cdot x), \qquad \Psi_v^{\mathrm{F}} = \Big(\log(1 - |v|^2) - \log(1 + v \cdot \xi),\ \tfrac{1}{1 + v \cdot \xi}\Big), \quad |v| < 1, \tag{5.18}$$

singular on the hyperplane $t = T + v \cdot x$; for $\gamma > 0$ it has the defect $\gamma T e^{-\tau} |v|^2 / (1 + v \cdot \xi)$. In the original variables of Eq. (5.3), $\theta \simeq \log(1 - |v|^2) - \log(\delta(T - t + \sqrt{\tau_*}\, v \cdot x))$.

Linearisation. At Ψ_0^{F},

$$\mathbf{L}_{\mathrm{F}} \varphi = \begin{pmatrix} -\Lambda \varphi_1 + \varphi_2 \\ \Delta \varphi_1 - \Lambda \varphi_2 + \varphi_1 \end{pmatrix}, \qquad \mathbf{N}_{\mathrm{F}}(\varphi) = \begin{pmatrix} 0 \\ (e^{\varphi_1} - 1 - \varphi_1) + (e^{\varphi_1} - 1)\varphi_2 \end{pmatrix}. \tag{5.19}$$

The derivative nonlinearity cancels the term $-\varphi_2$ of the free operator: $\mathbf{L}_{\mathrm{F}} = \mathbf{L}_0 + \mathbf{B} + \mathbf{K}$ with $\mathbf{B}\varphi = (0, \varphi_2)$ bounded but *not compact* and $\mathbf{K}\varphi = (0, \varphi_1)$ compact. In physical variables the linearisation is $v_{tt} - \Delta v - \frac{v_t}{T-t} - \frac{v}{(T-t)^2} = 0$: a singular *anti-damping*. Repeating the proof of Lemma 5.2, $\frac{d}{d\tau} E_k \le -2(k - \frac{d}{2} - 1)E_k$ for $\mathbf{L}_0 + \mathbf{B}$, so the essential growth bound of $\mathbf{L}_{\mathrm{F}}$ is at most $\frac{d}{2} + 1 - k$; on the other hand every λ with $\operatorname{Re}\lambda < \frac{d+1}{2} - k$ is an eigenvalue. (We expect the latter bound to be the true one; the cruder bound suffices for k large.)

Proposition 5.14 (Spectrum, full model)**.** *Let $k > \frac{d}{2} + 1$ and $\operatorname{Re}\lambda > \frac{d}{2} + 1 - k$. Then $\lambda \in \sigma_p(\mathbf{L}_{\mathrm{F}})$ iff*

$$\lambda \in \{1 - \ell - 2n\} \cup \{-1 - \ell - 2n\}, \qquad \ell, n \in \mathbb{N}_0, \tag{5.20}$$

with $\alpha = \frac{\lambda+\ell-1}{2}$, $\beta = \frac{\lambda+\ell+1}{2}$, $c - \alpha - \beta = \frac{d}{2} - \lambda$ in the proof of Proposition 5.3. The eigenvalue 1 (time translation, $g = 1$) is simple and 0 (tilts, $g = \xi_j = -\partial_{v_j} \Psi^{\mathrm{F}}_{v,1}|_{v=0}$) is semisimple of multiplicity d; the proof of Lemma 5.4 applies with $\beta = 1$. In the $\ell = 0$ sector, $\lambda = -1$ is a double root ($\alpha = -1$, $\beta = 0$) and carries a Jordan block:

$$v = T - t \quad \text{and} \quad v = -(T-t)\log(T-t) - \frac{|x|^2}{d\,(T-t)} \tag{5.21}$$

both solve the linearised equation (the second is a generalised eigenfunction, $\varphi \propto e^{-\tau}(\tau \cdot 1 + \text{const} - |\xi|^2/d)$). For $k > \frac{d}{2} + 2$ the spectral gap is 1, but the semigroup on the stable subspace decays only like $(1+\tau)e^{-\tau}$.

The Jordan block has a transparent origin: a formal Fuchsian expansion $u = -\log\phi + \log(1 - |\nabla T|^2) + a(x)\,\phi\log\phi + \ldots$, $\phi = T(x) - t$, of Eq. (5.6) with $\gamma = 0$ forces $a = -\Delta T/(2(1 - |\nabla T|^2))$: a *curved* blowup surface produces $(T-t)\log(T-t)$ terms. (For the reduced model the analogous order is log-free, cf. Eq. (5.16).) Exact polynomial computations confirm: in the full model $\lambda = -1$ ($\ell = 0$) and $\lambda = -2$ ($\ell = 1$) have algebraic multiplicity 2 and geometric multiplicity 1, while all eigenvalues of the reduced model are semisimple in each sector.

Theorem 5.15 (Full model; proof sketch, less checked than Theorem 5.8)**.** *Let $d \ge 1$, $k \in \mathbb{N}$, $k > \frac{d}{2} + 2$, $0 < \omega < 1$. There are $\varepsilon_0, M > 0$ such that for data ε-close in $H^k \times H^{k-1}(B_2^d)$ to those of $u_1 = -\log(1-t)$, and $\gamma \le \varepsilon_0$, the solution of Eq. (5.6) blows up at some T with $|T - 1| \le M\varepsilon$, and with some $|v| \le M\varepsilon$ the bounds of Theorem 5.8 hold with $u_{T,a}$ replaced by $u_{T,v}$ from Eq. (5.18) and $M\varepsilon$ by $M(\varepsilon + \gamma)$.*

Why the Donninger argument goes through, and what changes. (i) The derivative nonlinearity costs no regularity: it sits in the second component, and $H^k \cdot H^{k-1} \subset H^{k-1}$ for $k > d/2$, so Lemma 5.7 holds for $\mathbf{N}_{\mathrm{F}}$. (ii) The non-compact term $\mathbf{B}$ shifts the free decay rate down by (at most) one; this only costs regularity ($k > \frac{d}{2} + 2$ instead of $k \ge \frac{d}{2} + 1$ for a gap 1). (iii) There is no Lorentz symmetry; the d neutral modes are produced by the tilted family Eq. (5.18), which is all that the modulation in Step 4 needs (for $\gamma > 0$ its defect is $O(\gamma|v|^2 e^{-\tau})$, handled as in Remark 5.9). (iv) The Jordan block at -1 forces $\omega < 1$, which is anyway required by the nonlinear argument. The parts we have not checked in detail are the uniform-in-v spectral perturbation for $\mathbf{L}_{\mathrm{F},v}$ (which is now a non-compact perturbation of $\mathbf{L}_{\mathrm{F}}$, but small in operator norm) and the precise semigroup bound. For the closest rigorous analogues with time-derivative nonlinearities see [95] (quadratic u_t^2-type nonlinearities) and [65] ($d = 1$).

Numerical evidence (Fig. 3(c)). For $d = 1$, $\gamma \in \{0, 1\}$ and generic smooth data, the similarity-variable evolution with bisection in T converges to a tilted profile Ψ_v^{F}; the tilt read off from the limit profile agrees with the slope $T'(0)$ of the blowup curve (obtained independently by bisection at $x_0 = \pm 0.02$) to 3–4 digits ($v = -0.05710$ vs. $T'(0) = -0.05714$ for $\gamma = 0$; -0.09340 vs. -0.09342 for $\gamma = 1$), and $e^{\tau}\|\Psi(\tau) - \Psi_v^{\mathrm{F}}\|$ grows linearly in τ (Jordan block), whereas for the reduced model it is constant.

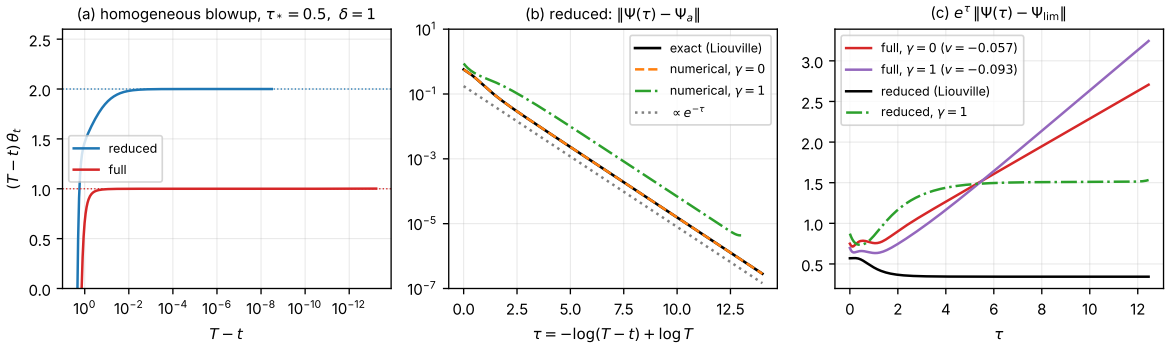


Figure 3: (a) Homogeneous blowup of Eqs. (5.3) and (5.4) with $\varepsilon = 0$, $\tau_* = 0.5$, $\delta = 1$, $\theta(0) = \theta'(0) = 0$: $(T-t)\theta_t \to 2$ for the reduced and $\to 1$ for the full model. (b) Reduced model, $d = 1$: distance to the boosted profile in similarity variables, exact Liouville solution vs. numerics ($\gamma = 0$), and the damped case $\gamma = 1$; all decay like $e^{-\tau}$ (gap 1, sharp). (c) $e^{\tau}\times$distance to the limiting profile: constant for the reduced model ($\gamma = 0, 1$), linear growth ($\tau e^{-\tau}$ decay, Jordan block at $\lambda = -1$) for the full model Eq. (5.6) ($\gamma = 0, 1$; v = fitted tilt). Code: `code/therm_sim1d.py`.

Known	Derived here (proof sketch)	Numerical / conjectural
Liouville's formula [93]; MCV law and both model forms [13, 96]; for $\Box u = e^u$: blowup on a surface with logarithmic behaviour for data near constants, any d [84]; 1D blowup curves and rates [63, 64, 1]; Donninger machinery for powers [38, 39, 41, 107].	Spectra Eqs. (5.13) and (5.20) with all integer/log cases; semisimplicity; Jordan block (full); free rates; Theorem 5.8 and Remark 5.9; exact 1D structure (Proposition 5.11 and Corollary 5.12); homogeneous dynamics of the full model; tilted family; Theorem 5.15 (less checked); Arrhenius global existence.	1D simulations of both models (rates, tilt = slope); exact essential growth bound $\frac{d+1}{2} - k$ for the full model; non-ODE (non-homogeneous) self-similar profiles were not searched for.

Table 5: Status of the results of Section 5.

5.7 Arrhenius kinetics: no blowup

For $\varepsilon > 0$, $f_\varepsilon(\theta) = \delta \exp(\theta/(1+\varepsilon\theta))$ on $\theta > -1/\varepsilon$ (absolute temperature positive) satisfies $0 < f_\varepsilon \le \delta e^{1/\varepsilon}$, and f_ε together with all derivatives tends to 0 as $\theta \downarrow -1/\varepsilon$; hence f_ε extends by 0 to a C^∞, bounded, globally Lipschitz function on $\mathbb{R}$ (hyperbolic models have no maximum principle, so this extension matters). Then Eq. (5.4) is a semilinear wave equation with globally Lipschitz nonlinearity, and Eq. (5.3) has in addition the term $\tau_* f_\varepsilon'(\theta)\theta_t$, linear in θ_t with a bounded (possibly anti-damping) coefficient. Standard energy estimates give global smooth solutions growing at most exponentially: *there is no finite-time blowup.* Thermal "explosion" is a rapid transition to $\theta = O(1/\varepsilon)$ (temperature rise of order Θ_0; reactant consumption, neglected here, limits it further). The FK blowup is an *intermediate asymptotic* regime, valid while $\varepsilon\theta \ll 1$: in similarity time up to $\tau_{\max} \approx 1/(2\varepsilon)$ (reduced) or $\tau_{\max} \approx 1/\varepsilon$ (full). For typical activation energies $E/(R\Theta_0) = 1/\varepsilon \approx 20$–$40$ this window spans many e-folds, so the convergence rates above are physically meaningful.

5.8 Status, literature and caveats

Novelty. For the method in general see the survey [29]. We did not find a Donninger-type (similarity variables, H^k, spectral) stability theorem for $u_{tt} - \Delta u = e^u$, nor any analysis of

the MCV-consistent equation with the $\partial_t(e^u)$ nonlinearity. However, Kichenassamy's result [84] (Fuchsian reduction and Nash–Moser, data near constants, i.e. near the ODE blowup) appears, judging from its abstract, to imply already stability of the ODE blowup for $\Box u = e^u$ in a high-regularity sense, including the logarithmic terms (absent iff the blowup surface has vanishing scalar curvature); we have only seen the abstract of [84]. Theorem 5.8 should therefore be viewed as a routine but quantitative adaptation (H^k, $k > d/2$, explicit rate $(T-t)^\omega$, data only on a ball), while the full Cattaneo model (Section 5.6) appears to be new. The literature search was limited (search budget), so this assessment is provisional.

Caveats. (1) Theorems 5.8 and 5.15 are proof sketches; the uniform semigroup bounds on the stable subspace (beyond the quasi-compactness of Lemma 5.2) and the modulation in (T, a) are adapted from the power case without writing out all details. (2) For the full model only the crude essential bound $\frac{d}{2} + 1 - k$ is proved. (3) Damping requires γT small, reached by Cauchy stability. (4) The analysis is local in a light cone; boundary conditions, the Gelfand threshold δ_{crit} of bounded domains, and whether blowup occurs at all for given data are not addressed. (5) Results concern the FK limit $\varepsilon = 0$; with Arrhenius kinetics there is no blowup (Section 5.7).

6 The power family, Klein–Gordon and lower-order perturbations

This section (i) puts softening "Duffing continua", the focusing Klein–Gordon (ϕ^4) equation and MEMS touchdown into one family $\partial_t^2 u - \Delta u = F_p(u)$, $p \notin [-1, 1]$, and proves the spectral theorem for its ODE blowup in all angular sectors (a detailed version of Theorem 3.2, including the integer, logarithmic and Jordan cases); (ii) compares it with the published results; (iii) treats damping and restoring forces $\gamma\partial_t u + m^2 u$, which destroy exact self-similarity, via an "asymptotically autonomous" lemma (Lemma 6.10) that the model sections can cite. Labels: **(K)** known (with reference), **(P)** proved here, **(S)** sketched here, **(N)** numerical or conjectural. Symbolic claims are checked in `code/power_symbolic.py`, numerical ones in `code/power_spectrum.py`.

6.1 Industrial models and the unified family

Duffing continua. A string ($d = 1$) or membrane ($d = 2$) with density ϱ, tension $\mathcal{T}$, viscous damping c_v, on a nonlinear Winkler foundation with restoring force $k_1 u - k_3 u^3$ per unit area obeys

$$\varrho\, \partial_t^2 u + c_v\, \partial_t u - \mathcal{T}\Delta u + k_1 u = k_3 u^3. \tag{6.1}$$

For a *softening* foundation ($k_3 > 0$; continuum version of the softening Duffing oscillator [103, 87]) the potential $\frac{1}{2}k_1 u^2 - \frac{1}{4}k_3 u^4$ has a barrier at $u_b = (k_1/k_3)^{1/2}$, and homogeneous motions above it escape in finite time. With $t = t_* t'$, $x = (\mathcal{T}/\varrho)^{1/2} t_* x'$, $u = (\varrho/(k_3 t_*^2))^{1/2} u'$ (primes dropped),

$$\partial_t^2 u + \gamma\, \partial_t u + m^2 u - \Delta u = u^3, \qquad \gamma = c_v t_*/\varrho, \quad m^2 = k_1 t_*^2/\varrho. \tag{6.2}$$

Taking t_* equal to the blowup time shows that γ, m measure the damping time and foundation period *relative to the blowup time*. For $\gamma = 0$ this is the focusing cubic Klein–Gordon (ϕ^4) equation. Physical caveats: the cubic law is a truncation, so "blowup" means dynamic escape (snap-through) and the self-similar profile describes the escaping region only while the foundation law is valid; Euler–Bernoulli beams (∂_x^4 instead of $-\Delta$) are not covered.

The unified family. For $p \in \mathbb{R} \setminus [-1,1]$ let

$$\partial_t^2 u - \Delta u = F_p(u) := \operatorname{sgn}(p-1)\,|u|^{p-1}u, \qquad s := \tfrac{2}{p-1} \in (-1,0) \cup (0,\infty). \tag{6.3}$$

For $p > 1$ this is the focusing power equation; for $p = -q < -1$ and $u = w > 0$ it is the MEMS equation $\partial_t^2 w - \Delta w = -w^{-q}$ for the gap w (λ in $-\lambda w^{-q}$ scales out), with $s = -\sigma = -\frac{2}{q+1}$ as in Section 4. The sign makes the force point *towards* the singularity ($u \to \infty$, resp. $w \to 0$). The critical Sobolev exponent is $s_c = \frac{d}{2} - s$. See Table 6.

model	p	s	$V_0 = (s+1)(s+2)$	$\omega_0 = \min\{1, 2s+2\}$
softening string/membrane, ϕ^4	3	1	6	1
focusing NLKG/NLW	$p > 1$	$\frac{2}{p-1}$	$\frac{2p(p+1)}{(p-1)^2}$	1
MEMS, general	$-q < -1$	$-\frac{2}{q+1}$	$\frac{2q(q-1)}{(q+1)^2}$	$\min\{1, \frac{2(q-1)}{q+1}\}$
MEMS, parallel plate	-2	$-\frac{2}{3}$	$\frac{4}{9}$	$\frac{2}{3}$

Table 6: The unified family (6.3); ω_0 is the spectral gap of Corollary 6.7.

6.2 ODE blowup and the excluded range $-1 \le p \le 1$

Proposition 6.1 (ODE profile; **(P)**). *Let $p \neq \pm 1$, $s = \frac{2}{p-1}$. Then $u'' = F_p(u)$ has a solution $u_T = \kappa(T-t)^{-s}$, $\kappa > 0$, iff $p \notin [-1,1]$, and then*

$$\kappa^{p-1} = \operatorname{sgn}(p-1)\,s(s+1) = |s(s+1)|, \qquad V_0 := F_p'(u_T)(T-t)^2 = p\,s(s+1) = (s+1)(s+2).$$

For MEMS with general λ: $\kappa^{q+1} = \lambda/(\sigma(1-\sigma))$, $V_0 = \lambda q \kappa^{-q-1} = q\sigma(1-\sigma) = (1-\sigma)(2-\sigma) > 0$ (note $F'(w) = +\lambda q w^{-q-1}$).

Proof. $u_T'' = s(s+1)\kappa(T-t)^{-s-2}$, $F_p(u_T) = \operatorname{sgn}(p-1)\kappa^p (T-t)^{-sp}$ and $sp = s+2$. For $s > 0$ and $s \in (-1,0)$ the product $\operatorname{sgn}(p-1)s(s+1)$ is positive; for $p \in (-1,1)$, $s < -1$ and it is negative. Euler's relation $uF_p'(u) = pF_p(u)$ gives $V_0 = p\,s(s+1)$, and $ps = s+2$. □

Remark 6.2 (Why $-1 \le p \le 1$ is excluded; **(P)**). For $p \in (-1,1)$ a self-similar solution would have $s < -1$, so u_T *and* $\partial_t u_T$ tend to 0: no singularity of $(u, \partial_t u)$; even with the opposite sign (where κ exists) it is non-generic, the second eigenvalue family below starting at $-2s-2 > 0$. For the MEMS sign and $0 < q < 1$ the gap closes with finite speed, $w \approx \sqrt{2E}(T-t)$, and for $q = 1$ with a logarithmic correction, $w \approx (T-t)\sqrt{2|\log(T-t)|}$ (from $\frac{1}{2}w'^2 + \log w = E$). Self-similar ODE blowup thus exists exactly for $s \in (-1,0) \cup (0,\infty)$, which is also where $\omega_0 > 0$.

Symmetries. Besides T, the boosted profiles (3.5) $u_{T,a} = \kappa(1-|a|^2)^{s/2}(T - t + a\cdot x)^{-s}$, $|a| < 1$, blow up on the spacelike hyperplane $\{t = T + a\cdot x\}$ and are stationary in similarity variables, $\psi_a(\xi) = \kappa(1-|a|^2)^{s/2}(1 + a\cdot\xi)^{-s}$; this holds verbatim in the MEMS range. Space translations act trivially on u_T.

6.3 The linearised operator and the free flow

With $\Psi = \Psi_* + \Phi$, $\Psi_* = (\kappa, s\kappa)$ in (3.3), $\partial_\tau \Phi = \mathbf{L}\Phi + \mathbf{N}(\Phi)$ on $\mathcal{H}^k = H^k(B^d) \times H^{k-1}(B^d)$, where

$$\mathbf{L}\Phi = \begin{pmatrix} -\Lambda\phi_1 - s\phi_1 + \phi_2 \\ \Delta\phi_1 - \Lambda\phi_2 - (s+1)\phi_2 + V_0\phi_1 \end{pmatrix}, \quad \mathbf{N}(\Phi) = \begin{pmatrix} 0 \\ F_p(\kappa+\phi_1) - F_p(\kappa) - V_0\phi_1 \end{pmatrix}.$$

Since $\kappa > 0$, F_p is real analytic near κ for every real p (non-integer p and the singular MEMS force included); for $k > \frac{d}{2}$ and $\|\phi_1\|_{L^\infty} \le \kappa/2$, $\mathbf{N}$ is locally Lipschitz with a quadratic bound. No regularity restriction on p arises.

Lemma 6.3 (Polynomial invariance; **(P)**). *Let $\mathcal{P}_N$ be the polynomials of degree $\le N$ in ξ. Then $\mathbf{L}$ maps $\mathcal{P}_N^2$ into itself and acts on homogeneous parts of degree j (modulo $\mathcal{P}_{j-1}^2$) as $M_j = \left(\begin{smallmatrix} -s-j & 1 \\ V_0 & -s-1-j \end{smallmatrix}\right)$, with eigenvalues $1-j$ and $-2s-2-j$. For $V_0 = 0$ (the free operator $\mathbf{L}_0$) they are $-s-j$, $-s-1-j$; in particular $\mathbf{L}_0(1,0) = -s(1,0)$. On constants $\mathbf{L}$ has the eigenvalues 1 (time translation) and $-2s-2$.*

Proof. Λ multiplies homogeneous polynomials of degree j by j, Δ lowers the degree by two, and $\det(M_j - \mu) = (\mu + j - 1)(\mu + j + 2s + 2)$ since $V_0 = (s+1)(s+2)$. □

So every eigenvalue of Theorem 6.6 with a polynomial eigenfunction is visible algebraically; the content of the theorem is that there are no others. Note also that polynomial (collocation, Galerkin) discretisations of $\mathbf{L}$ are *exact* on $\mathcal{P}_N^2$ and cannot detect non-polynomial eigenfunctions (Section 6.8).

Lemma 6.4 (Free flow; (a) **(P)**, (b) **(S)**). *Let $\omega_k := k - s_c = k - \frac{d}{2} + s$. (a) $-s$ is an eigenvalue of $\mathbf{L}_0$ on every $\mathcal{H}^k$ (eigenfunction $(1,0)$, i.e. $u \equiv$ const), so the growth bound of $e^{\tau \mathbf{L}_0}$ is $\ge -s$; in the MEMS range $-s = \sigma > 0$ and the free flow grows. (b) $\mathbf{L}_0 = \tilde{\mathbf{L}}_0 + \mathbf{K}$ with $\mathbf{K}$ of finite rank and $\|e^{\tau \tilde{\mathbf{L}}_0}\| \le M e^{-\omega_k \tau}$. Hence, if $\omega_k > 0$, $\sigma(\mathbf{L}) \cap \{\operatorname{Re}\lambda > -\omega_k\}$ consists of isolated eigenvalues of finite algebraic multiplicity.*

Proof of (b), sketch. For $|\beta| = k-1$ let $f = \partial^\beta \phi_1$, $g = \partial^\beta \phi_2$. Since $\partial^\beta \Lambda = (\Lambda + k - 1)\partial^\beta$, the free flow gives $\partial_\tau f = -(\Lambda + s + k - 1)f + g$, $\partial_\tau g = \Delta f - (\Lambda + s + k)g$, and with $\int_B h \Lambda h = -\frac{d}{2}\int_B h^2 + \frac{1}{2}\int_{\partial B} h^2$,

$$\tfrac{1}{2}\tfrac{d}{d\tau}(\|\nabla f\|^2 + \|g\|^2) = -\omega_k(\|\nabla f\|^2 + \|g\|^2) \\ - \tfrac{1}{2}\int_{\partial B}(|\nabla f|^2 - (\partial_r f)^2 + (\partial_r f - g)^2) \le -\omega_k(\|\nabla f\|^2 + \|g\|^2),$$

the boundary term having a sign because ∂B is the light cone. So the seminorm

$$E_k(\Phi) = \textstyle\sum_{|\beta|=k-1}(\|\nabla \partial^\beta \phi_1\|^2 + \|\partial^\beta \phi_2\|^2)$$

decays at rate $2\omega_k$. Its kernel $\mathcal{P}_{k-1} \times \mathcal{P}_{k-2}$ is finite-dimensional and $\mathbf{L}_0$-invariant (Lemma 6.3), and $E_k^{1/2}$ is equivalent to the quotient norm (Bramble–Hilbert), so $\mathbf{L}_0$ is block-triangular with a quotient block of growth bound $\le -\omega_k$ and a finite-rank remainder. $\mathbf{L}' = (0, V_0\phi_1)$ is compact, and the rest is [48, Ch. IV]. Rigorous inner products of this type are constructed in [107, 61]. □

Remark 6.5 (Regularity threshold; the Lumer–Phillips step). The free decay rate is $\omega_k = k - s_c$, so exponential decay needs regularity strictly above scaling, $k > s_c$; at $k = s_c$ one needs Strichartz estimates in similarity coordinates [28, 37, 118]. To see the full gap one needs $k > s_c + \omega_0$; for MEMS $k > \frac{d}{2} + \sigma + \omega_0$ (e.g. $q = 2$: $k \ge 2$ for $d = 1$, $k \ge 3$ for $d = 2$). The simplest Lumer–Phillips argument, an inner product in which $\mathbf{L}_0$ itself is dissipative with bound $-s$, is sharp by (a) and yields the rate $\min\{1, s\}$; we believe this is the origin of the exponent $\min\{1, s_p\}$ in [107]. In the MEMS range it yields nothing ($-s = \sigma > 0$; no inner product makes $\mathbf{L}_0$ dissipative with negative bound), and the splitting (b) is necessary. The finite-rank part is absorbed into the compact perturbation, so the positive free eigenvalue σ is harmless: it is not an eigenvalue of $\mathbf{L}$.

6.4 The spectral theorem in all angular sectors

Let $\mathcal{Y}_\ell$ be the spherical harmonics of degree ℓ ($d \ge 2$); for $d = 1$, $\mathcal{Y}_0 = \{1\}$, $\mathcal{Y}_1 = \{\operatorname{sgn}\xi\}$, $\mathcal{Y}_{\ell \ge 2} = \{0\}$.

Theorem 6.6 (Spectrum of the ODE blowup; **(P)**). *Let $d \geq 1$, $p \in \mathbb{R} \setminus [-1,1]$, $k \in \mathbb{N}$, $\omega_k > 0$, and $\operatorname{Re}\lambda > -\omega_k$. Then λ is an eigenvalue of $\mathbf{L}$ on $\mathcal{H}^k$ iff*

$$\lambda \in \Sigma_s := \{1-\ell-2n\} \cup \{-2s-2-\ell-2n\}, \qquad n \in \mathbb{N}_0,\ \mathcal{Y}_\ell \neq \{0\}.$$

In the sector ℓ the eigenfunctions are $(g, (\lambda+s+\Lambda)g)$, $g = |\xi|^\ell Y_\ell\, {}_2F_1(\alpha,\beta;\frac{d}{2}+\ell;|\xi|^2)$, $Y_\ell \in \mathcal{Y}_\ell$, $\alpha = \frac{\lambda+\ell-1}{2}$, $\beta = \frac{\lambda+\ell+2s+2}{2}$ (a polynomial of degree n in $|\xi|^2$). All eigenvalues are real, and:

(a) $\lambda = 1$ occurs only for $\ell = 0$, with $\Phi = (1, s+1)$ = time-translation mode;

(b) $\lambda = 0$ occurs only for $\ell = 1$, with $\Phi = (\xi_j, (s+1)\xi_j)$, $j = 1,\dots,d$, the boost modes $\propto \partial_{a_j}\Psi_a|_{a=0}$;

(c) both are semisimple. The two families collide iff $s \in \frac{1}{2}+\mathbb{Z}$ ($p \in \{5, \frac{7}{3}, \frac{9}{5}, \dots\}$ or $q = 3$); at the lowest collisions ($s = -\frac{1}{2}, \lambda = -1$; $s = \frac{1}{2}, \lambda = -3$; $\ell = 0$) there is a Jordan block of length two.

Proof. Reduction. An eigenfunction has $\phi_2 = (\lambda+s+\Lambda)\phi_1$, and $g = \phi_1$ solves (3.6) with $a = s+\lambda$, which is elliptic in the open ball; so $g \in C^\infty(B^d)$. Each spherical-harmonic component $|\xi|^\ell Y_\ell h(|\xi|^2)$ is again an H^k solution, and h solves (3.7) with $c = \frac{d}{2}+\ell$, $\alpha+\beta+1 = a+\ell+\frac{3}{2}$, $\alpha\beta = \frac{1}{4}[(a+\ell)(a+\ell+1)-V_0]$; with $a_1 = s+1$, $a_2 = -(s+2)$ this gives the stated α,β. (The sector ℓ in dimension d is the radial problem in dimension $d+2\ell$ with $\lambda \mapsto \lambda+\ell$.) *At $z = 0$* the second Frobenius solution gives $|\xi|^{2-d-\ell}Y_\ell$ (or $\log|\xi|$, or $|\xi|$, $\operatorname{sgn}\xi$ for $d \leq 2$), which is not smooth; so $h \propto {}_2F_1(\alpha,\beta;c;z)$. *At $z = 1$* the exponents are 0 and $\mu = \frac{d-1}{2} - s - \lambda$; the non-analytic local solutions behave like $(1-|\xi|^2)^\mu$, $(1-|\xi|^2)^m\log(1-|\xi|^2)$ ($\mu = m \in \mathbb{N}_0$) or $(1-|\xi|^2)^{-m}$ ($\mu = -m < 0$), and lie in H^k near ∂B only if $\operatorname{Re}\mu > k - \frac{1}{2}$, i.e. $\operatorname{Re}\lambda < -\omega_k$. Hence, for $\operatorname{Re}\lambda > -\omega_k$, h is an eigenfunction iff it is analytic at $z = 1$. *Monodromy.* The hypergeometric equation is Fuchsian with singular points $0, 1, \infty$. A solution analytic at 0 and at 1 has trivial monodromy and extends to an entire function of polynomial growth (as ∞ is regular singular), i.e. a polynomial; and ${}_2F_1(\alpha,\beta;c;z)$ is a polynomial iff $\alpha \in -\mathbb{N}_0$ or $\beta \in -\mathbb{N}_0$. This handles the integer and logarithmic cases uniformly (alternatively use the connection formulae [106, §15.8]). $\alpha = -n$ gives $\lambda = 1-\ell-2n$, $\beta = -n$ gives $\lambda = -2s-2-\ell-2n$. *(a), (b)*: $1-\ell-2n \geq 0$ iff $(\ell,n) \in \{(0,0),(1,0)\}$, and $-2s-2-\ell-2n < 0$ for $s > -1$. *(c)*: a generalised eigenvector solves $(\mathbf{L}-\lambda)\Phi = \Phi_\lambda$; eliminating ϕ_2 gives an inhomogeneous hypergeometric equation with polynomial right-hand side, so by the same monodromy argument an $\mathcal{H}^k$ solution is a polynomial. By Lemma 6.3 its degree is at most $\max\{\deg\Phi_\lambda, j_*\}$, where $M_{j_*}-\lambda$ is singular, and the degree-j_* block requires the top part of Φ_λ to lie in $\operatorname{ran}(M_{j_*}-\lambda)$. For $\lambda = 1$ ($j_* = 0$) and $\lambda = 0$ ($j_* = 1$) this fails, as λ is a simple eigenvalue of M_{j_*} for $s > -1$. At a collision both $\alpha,\beta \in -\mathbb{N}_0$ and the eigenfunction is shared by the two families (e.g. $s = -\frac{1}{2}$, $\lambda = -1$: $g \equiv 1$), and a generalised eigenvector exists: $(\mathbf{L}+1)\Phi = (1, -\frac{3}{2})$ with $\phi_1 = -|\xi|^2/3$ ($d = 3$), and for $s = \frac{1}{2}$, $\lambda = -3$, $(\mathbf{L}+3)\Phi = (1, -\frac{5}{2})$ with $\phi_1 = -\frac{2}{3}|\xi|^2 + \frac{2}{15}|\xi|^4$ ($d = 3$). That all non-colliding eigenvalues are semisimple, and all collisions give Jordan blocks (a double zero of $1/(\Gamma(\alpha)\Gamma(\beta))$), is expected but only checked in these cases **(S)**. □

Corollary 6.7 (Spectral gap; **(P)**). *If $\omega_k > \omega_0$, then $\sup\{\operatorname{Re}\lambda : \lambda \in \sigma(\mathbf{L}) \setminus \{0,1\}\} = -\omega_0$,*

$$\omega_0 = \min\{1, 2s+2\} = \begin{cases} 1, & p > 1 \text{ or } p = -q,\ q \geq 3, \\ \frac{2(q-1)}{q+1}, & p = -q,\ 1 < q \leq 3. \end{cases}$$

The gap eigenvalue is -1 ($\ell = 0, n = 1$ and $\ell = 2, n = 0$) for $p > 1$, $q \geq 3$, and the spatially homogeneous mode $-2s-2 = -2(1-\sigma)$ for $q < 3$; at $q = 3$ they form a Jordan block. For the parallel-plate force $q = 2$: $\omega_0 = \frac{2}{3}$.

6.5 What is proved in the literature

All results below concern $p > 1$; for the MEMS range see Section 4. **(K)**: [38]: $d = 3$, radial, $1 < p \leq 3$, stability of u_T in the energy topology (complementing [98, 99]); here $k = 1$

and $\omega_1 = s - \frac{1}{2} \geq \frac{1}{2}$, consistent with Remark 6.5. [39]: $d = 3$, radial, $p > 3$, an open set of radial data above energy regularity blows up via u_T. [40]: $d = 3$, energy supercritical, *no symmetry assumption.* [41]: radial data in odd $d \geq 5$, energy-supercritical powers, data in $H^{(d+1)/2} \times H^{(d-1)/2}$. [107]: all $d \geq 1$, all $p > 1$, no symmetry; data in $H^k \times H^{k-1}$, $k > \frac{d}{2}$; convergence to a Lorentz-boosted ODE profile with parameters (T^*, β^*), with decay exponent governed by $\min\{1, s_p\}$; dissipative inner products on H^k for all d and integer k. [94]: superconformal powers, all dimensions, mode stability via Lorentz transformations and asymptotic stability of the ODE blowup family, extending [101] (subconformal, energy space, explicit Lorentz family, no symmetry) and [107]. Critical case $d = 3$, $p = 5$: [28, 37]; cubic in higher dimensions: [16].

Cross-check. In all these works the only non-negative eigenvalues are $\lambda = 1$ and, nonradially, $\lambda = 0$, as in Theorem 6.6. The exponent $\min\{1, s_p\}$ of [107] equals the gap $\omega_0 = 1$ for $p \leq 3$ but is smaller for $p > 3$; it reflects the free eigenvalue $-s$ of constants (Remark 6.5), not an eigenvalue of $\mathbf{L}$ (the second family starts at $-2s-2 < -s$). With Lemma 6.4(b) the rate should improve to $\min\{1, \omega_k\} - \varepsilon$, i.e. $1 - \varepsilon$ for k large **(S)**. *Source check:* the full texts could not be accessed here; the statements are taken from abstracts/indexing data and `bib/lit.bib`, and the details (ranges of p, spaces, rates) should be checked against the papers.

6.6 Lower-order perturbations: damping and restoring forces

Consider, with $\gamma, m^2 \in \mathbb{R}$,

$$\partial_t^2 u + \gamma\, \partial_t u + m^2 u - \Delta u = F_p(u). \tag{6.4}$$

With $T - t = Te^{-\tau}$ and (3.2) one obtains *exactly* (**(P)**, symbolic identity)

$$\partial_\tau \Psi = \mathbf{L}_0 \Psi + \begin{pmatrix} 0 \\ F_p(\psi_1) \end{pmatrix} - \begin{pmatrix} 0 \\ \gamma T e^{-\tau} \psi_2 + m^2 T^2 e^{-2\tau} \psi_1 \end{pmatrix}: \tag{6.5}$$

non-autonomous terms of size $e^{-\tau}$ (damping) and $e^{-2\tau}$ (mass), with T in the coefficients.

Proposition 6.8 (Perturbed ODE blowup; expansion **(P)**, existence **(S)**)**.** *As $\rho = T - t \downarrow 0$, spatially homogeneous solutions of (6.4) blowing up at T form a one-parameter family,*

$$U_T = \kappa \rho^{-s}(1 + a_1 \rho + a_2 \rho^2 + \dots) + c\, \rho^{s+2}(1 + \dots),$$
$$a_1 = \frac{\gamma s}{2(2s+1)}, \qquad a_2 = \frac{\gamma^2(3s^2 - 5s - 4)}{24(2s+1)^2} + \frac{m^2}{6s}.$$

The order-j coefficient obeys $-(j+1)(2s+2-j)a_j =$ (lower orders); the indicial roots $j = -1$, $j = 2s + 2$ are the eigenvalues 1, $-2s - 2$ of Lemma 6.3. If $2s + 2 \in \mathbb{N}$ and compatibility fails, $\rho^{2s+2} \log \rho$ terms appear: for $p = 3$ iff $\gamma^2(9m^2 - 2\gamma^2) \neq 0$ (cf. the Painlevé condition $m^2 = 2\gamma^2/9$ for the damped Duffing equation), for $p = 5$ iff $\gamma(16m^2 - 3\gamma^2) \neq 0$, and for MEMS $q = 3$ at first order whenever $\gamma \neq 0$: $U_T = \sqrt{2}\rho^{1/2}(1 + \frac{\gamma}{4}\rho \log \rho + c\rho + \dots)$. Hence $\Psi_T(\tau) - \Psi_* = O(e^{-\tau})$ if $\gamma \neq 0$ ($O(\tau e^{-\tau})$ for $q = 3$) and $O(e^{-2\tau})$ if $\gamma = 0$.

Lemma 6.9 (The tilted family survives damping; **(P)**)**.** *For $|a| < 1$, $u = W(T - t + a \cdot x)$ solves (6.4) iff $(1-|a|^2)W'' - \gamma W' + m^2 W = F_p(W)$. Hence $u_{T,a}(t, x) := U^{\gamma_a}\big((T - t + a \cdot x)/\sqrt{1 - |a|^2}\big)$, with $U^{\gamma_a}(\rho)$ the perturbed ODE blowup of Proposition 6.8 for damping $\gamma_a = \gamma/\sqrt{1 - |a|^2}$, is an exact solution blowing up on $\{t = T + a \cdot x\}$, and $\Psi_{T,a}(\tau) = \Psi_a + O(e^{-\tau})$.*

Proof. $\partial_t^2 u = W''$, $\Delta u = |a|^2 W''$, $\partial_t u = -W'$; then rescale the argument of W by $\sqrt{1 - |a|^2}$. □

Thus the mass preserves Lorentz invariance (for $\gamma = 0$ the $u_{T,a}$ are boosts of U_T) and damping breaks it, but the d-parameter tilted family remains an exact family of solutions for

every γ; only translation invariance in (t,x) is used. The neutral modes $\lambda=0$ are therefore still generated by exact solutions, and no dynamical modulation of the boost parameter (as anticipated in Section 3.4) is needed.

Lemma 6.10 (Asymptotically autonomous Lyapunov–Perron lemma; **(S)**)**.** *Let* $\mathbf{L}$ *generate a* C_0*-semigroup* $\mathbf{S}$ *on a Hilbert space* $\mathcal{H}$*,* $\mathbf{P}=\mathbf{P}_1+\mathbf{P}_0$ *a finite-rank projection commuting with* $\mathbf{S}$*,* $\mathbf{L}\mathbf{P}_1=\mathbf{P}_1$*,* $\mathbf{L}\mathbf{P}_0=0$*, and* $\|\mathbf{S}(\tau)(\mathbf{I}-\mathbf{P})\|\le Me^{-\bar\omega\tau}$ *for some* $\bar\omega\in(0,1]$*. For* θ *in a closed ball* $\overline{B}\subset\mathbb{R}^{\operatorname{rank}\mathbf{P}}$ *let* $\mathbf{B}_\theta(\tau)\in\mathcal{B}(\mathcal{H})$ *be strongly continuous with* $\sup_\tau\|\mathbf{B}_\theta(\tau)\|\le\beta$ *(in applications* $\mathbf{B}_\theta(\tau)=O(|a|)+O(e^{-\tau})$*), and* $\mathbf{N}_\theta(\tau,\cdot)$ *with* $\mathbf{N}_\theta(\tau,0)=0$*,* $\|\mathbf{N}_\theta(\tau,\Phi)-\mathbf{N}_\theta(\tau,\tilde\Phi)\|\le C_N(\|\Phi\|+\|\tilde\Phi\|)\|\Phi-\tilde\Phi\|$ *for* $\|\Phi\|,\|\tilde\Phi\|\le1$*, all continuous in* θ*. Consider*

$$\partial_\tau\Phi=\mathbf{L}\Phi+\mathbf{B}_\theta(\tau)\Phi+\mathbf{N}_\theta(\tau,\Phi),\qquad\Phi(0)=\mathbf{u}_\theta.\tag{6.6}$$

For every $\omega\in(0,\bar\omega)$ *there are* $\beta_0,\delta_0,C>0$ *such that if* $\beta\le\beta_0$ *and* $\|\mathbf{u}_\theta\|\le\delta\le\delta_0$*:*

(i) there is a unique Φ_θ *with* $\sup_\tau e^{\omega\tau}\|\Phi_\theta(\tau)\|\le C\delta$ *solving* $\Phi(\tau)=\mathbf{S}(\tau)[\mathbf{u}_\theta-\mathbf{C}_\theta(\Phi)]+\int_0^\tau\mathbf{S}(\tau-\sigma)\mathbf{G}_\theta(\sigma)\,d\sigma$*, where* $\mathbf{G}_\theta=\mathbf{B}_\theta\Phi+\mathbf{N}_\theta(\sigma,\Phi)$ *and* $\mathbf{C}_\theta(\Phi)=\mathbf{P}\mathbf{u}_\theta+\int_0^\infty(e^{-\sigma}\mathbf{P}_1+\mathbf{P}_0)\mathbf{G}_\theta(\sigma)\,d\sigma$*;*

(ii) Φ_θ *solves* (6.6) *iff* $\mathbf{C}_\theta(\Phi_\theta)=0$*;*

(iii) if $\theta\mapsto\mathbf{P}\mathbf{u}_\theta$ *is a homeomorphism onto a neighbourhood of* 0 *in* $\operatorname{ran}\mathbf{P}$ *with* δ*-independent Lipschitz inverse (transversality), then for small* δ *some* θ^* *has* $\mathbf{C}_{\theta^*}(\Phi_{\theta^*})=0$*.*

An inhomogeneity $e^{-\tau}\mathbf{g}_\theta(\tau)$*,* $\|\mathbf{g}_\theta\|\le\delta$*, may be added; then (i)–(iii) hold for* $\omega<\min\{\bar\omega,1\}$*.*

Sketch. In $\mathcal{X}=\{\Phi:\|\Phi\|_{\mathcal{X}}=\sup e^{\omega\tau}\|\Phi(\tau)\|<\infty\}$,

$$\int_0^\tau Me^{-\bar\omega(\tau-\sigma)}(\beta e^{-\omega\sigma}\|\Phi\|_{\mathcal{X}}+C_Ne^{-2\omega\sigma}\|\Phi\|_{\mathcal{X}}^2)d\sigma\le Me^{-\omega\tau}\Big(\tfrac{\beta}{\bar\omega-\omega}\|\Phi\|_{\mathcal{X}}+\tfrac{C_N}{\omega}\|\Phi\|_{\mathcal{X}}^2\Big),$$

while the corrected unstable and neutral parts $-\int_\tau^\infty(e^{\tau-\sigma}\mathbf{P}_1+\mathbf{P}_0)\mathbf{G}_\theta\,d\sigma$ are $O(\beta\omega^{-1}e^{-\omega\tau}\|\Phi\|_{\mathcal{X}}+C_Ne^{-2\omega\tau}\|\Phi\|_{\mathcal{X}}^2)$. So the map is a contraction on a small ball if $\beta\le\beta_0$; a decay factor $e^{-\sigma}$ in $\mathbf{B}_\theta$ makes the term integrable but not small, which is why the smallness of β (Remark 6.11) is needed. A forcing is handled by $\int_0^\tau e^{-\bar\omega(\tau-\sigma)}e^{-\sigma}d\sigma\le(1+\tau)e^{-\min\{\bar\omega,1\}\tau}$. (ii), (iii) are as in [38, 107] (Brouwer). □

Remark 6.11 (Making β small). Linearising (6.5) at the exact family $\Psi_{T,a}(\tau)$ of Lemma 6.9 gives (6.6) with $\theta=(T,a)$, no inhomogeneity, and $\mathbf{B}_\theta(\tau)\Phi=\big(0,[F_p'(\psi_{T,a,1}(\tau))-V_0]\phi_1-\gamma Te^{-\tau}\phi_2-m^2T^2e^{-2\tau}\phi_1\big)$, so $\beta\lesssim|a|+|\gamma|T+m^2T^2$. Here $|a|=O(\delta)$; to make the rest small, evolve the data to $t_1=1-\eta$ by local well-posedness on balls (finite speed of propagation keeps the solution $C(\eta)\varepsilon$-close to the reference on $B_{\eta+\delta}$) and rescale by η: $(\gamma,m^2)\mapsto(\gamma\eta,m^2\eta^2)$.

Theorem 6.12 (ODE blowup under lower-order terms; **(S)**, conditional)**.** *Let* $d\ge1$*,* $p\in\mathbb{R}\setminus[-1,1]$*,* $\gamma,m^2\in\mathbb{R}$*,* $k\in\mathbb{N}$*,* $k>\max\{\frac{d}{2},s_c\}$*, and let* $\bar\omega\in(0,1]$ *be such that the autonomous estimate* $\|\mathbf{S}(\tau)(\mathbf{I}-\mathbf{P})\|_{\mathcal{H}^k}\lesssim e^{-\bar\omega\tau}$ *holds (by Theorem 6.6 and Lemma 6.4 any* $\bar\omega<\min\{\omega_0,\omega_k\}$*; for* $p>1$ *and* $\bar\omega<\min\{1,s\}$ *this is contained in [107], as we read it), and* $0<\omega<\bar\omega$*. Let* U *be a spatially homogeneous solution of* (6.4) *on* $[0,1)$ *with ODE blowup at* 1*. There is* $\varepsilon>0$ *such that for data with* $\|(u_0,u_1)-(U(0),U'(0))\|_{H^k\times H^{k-1}(B_{1+\varepsilon})}\le\varepsilon$ *there exist* T^* *near* 1 *and small* $a^*\in\mathbb{R}^d$ *(*$a^*=0$ *for radial data) such that* u *blows up at* $(T^*,0)$ *and*

$$\|\Psi(\tau)-\Psi_{T^*,a^*}(\tau)\|_{\mathcal{H}^k}\lesssim e^{-\omega\tau},\qquad\|\Psi(\tau)-\Psi_{a^*}\|_{\mathcal{H}^k}\lesssim e^{-\omega\tau}+e^{-\tau}$$

($e^{-2\tau}$ *instead of* $e^{-\tau}$ *if* $\gamma=0$*;* $\tau e^{-\tau}$ *for* $q=3$*,* $\gamma\neq0$*). The blowup point is not a modulation parameter: any* x_0 *near* 0 *may replace* 0*, and then* (T^*,a^*) *depend on* x_0 *(*a^* *is the slope of the blowup surface for the exact tilted solutions, and presumably to leading order in general).*

Sketch. Reduce to small (γ,m^2) (Remark 6.11); $\mathbf{N}_\theta$ is quadratic uniformly in τ because $\psi_{T,a,1}(\tau)$ stays within $\kappa/4$ of ψ_a. Transversality: by Proposition 6.8 and Lemma 6.9, $\partial_T\Psi_{T,a}(0)$ and $\partial_{a_j}\Psi_{T,a}(0)$ are, up to nonzero factors and errors $O(|\gamma|+m^2+|a|)$, the eigenfunctions of Theorem 6.6(a),(b). Apply Lemma 6.10; the second estimate follows from Lemma 6.9. □

Remark 6.13 (Rates and resonances). (1) Since $\omega_0 \le 1$ in the whole family, damping and mass never decay *slower* than the gap. If one linearises at Ψ_a instead of $\Psi_{T,a}(\tau)$, the forcing has rate 1 (damping) or 2 (mass); for $\omega_0 = 1$ rate-1 forcing can resonate with the eigenvalue -1 and produce $\tau e^{-\tau}$ (for $a \neq 0$ the forcing has $\ell = 2$ components), and at $q = 3$ the Jordan block at -1 does the same. Only the exponent is affected. (2) Lower-order nonlinear terms f with $|f^{(j)}(u)| \lesssim |u|^{p'-j}$ contribute $(T-t)^{s+2} f((T-t)^{-s}\psi_1) = O(e^{-s(p-p')\tau})$; they are perturbative iff $s(p-p') > 0$ ($p' < p$ for powers, weaker singularity $q' < q$ for MEMS; a constant load has rate $s+2$). If $s(p-p') < \omega_0$ the convergence rate drops to $s(p-p')$ but Lemma 6.10 still applies; this is the only way a perturbation in this family decays more slowly than the gap. (3) The reduction, the tilted family and the abstract lemma are elementary; the deep input is the autonomous linear estimate, the same as in [107]. Local well-posedness on balls for (6.4) and the Lipschitz dependence on θ are routine and not written out.

Literature on the perturbed equations (K). Hamza and Zaag constructed Lyapunov functionals in similarity variables for $\partial_t^2 u = \Delta u + |u|^{p-1}u + f(u) + g(\partial_t u)$ with lower-order f, g (including the Klein–Gordon mass and linear damping) and proved ODE-type blowup rates: subconformal [69], conformal [70], Klein–Gordon and other perturbations [71], superconformal [72]. Killip, Stovall and Visan [85] obtained ODE-type blowup behaviour for focusing NLKG, so the mass is a lower-order effect at blowup. The closest semigroup result we know is [43] on strongly perturbed wave equations, where the blowup solution is no longer self-similar; we could not check whether it covers $\gamma\partial_t u$. To our knowledge (search not exhaustive) no Donninger-type stability theorem for the damped NLW or NLKG has been published; Theorem 6.12 suggests that little beyond the autonomous theory is needed.

6.7 Non-ODE self-similar profiles

In the energy-supercritical range there are self-similar solutions $u = (T-t)^{-s}U(|x|/(T-t))$ with $U \not\equiv \kappa$. **(K)**; the explicit formulas below were re-verified symbolically **(P)**.

- Bizoń–Maison–Wasserman [11]: $d = 3$, odd $p \ge 7$, a countable family U_n ($U_0 = \kappa$); numerically U_n has n unstable modes besides time translation, and U_1 is conjectured to be the co-dimension-one critical solution between blowup and dispersal **(N)**.
- Glogić–Schörkhuber [62]: cubic, explicit profile in every $d \ge 5$, co-dimension-one stable without symmetry assumptions in $d = 7$. One checks that ($\rho = |x|/(T-t)$)

$$u^* = \frac{1}{T-t}\,\frac{2\sqrt{2}\sqrt{(d-1)(d-4)}}{d-4+3\rho^2} \quad \text{solves } \partial_t^2 u - \Delta u = u^3 \qquad \Big(d=7:\ u^* = \frac{4(T-t)}{(T-t)^2+|x|^2}\Big).$$

- Csobo–Glogić–Schörkhuber [23]: quadratic, explicit profile for $d \ge 7$, co-dimension-one stable in $d = 9$; extended beyond the light cone in [18]. One checks that in $d = 9$

$$u^* = \frac{1}{(T-t)^2}\,\frac{240\,(7-\rho^2)}{(7+5\rho^2)^2} \quad \text{solves } \partial_t^2 u - \Delta u = u^2,$$

and is positive in the cone; we expect, but have not confirmed, that this is their profile.

These live in high dimensions and are irrelevant for strings and membranes ($d \le 2$), where the ODE profile is the generic one. MEMS profiles: Section 4.

6.8 Numerical confirmation

By Lemma 6.3, Chebyshev collocation reproduces the polynomial eigenvalues exactly and cannot see others; in double precision the lower ones are also badly conditioned (spurious "continuum" for $\operatorname{Re}\lambda < \frac{d-1}{2} - s$, where $(1-|\xi|^2)^\mu$ is continuous). We use collocation ($N = 24$,

$V_0 = F_p'(\kappa)$ computed numerically from F_p, so the MEMS signs are tested) only for the well-conditioned eigenvalues $1, 0, -1, -2s-2$, and test completeness by a *monodromy test* mirroring the proof: the Frobenius solution regular at $z=0$ is continued numerically (DOP853, tolerance 10^{-12}) around $|z-1| = \frac{1}{2}$; it returns to itself iff it is single-valued at $z=1$, and single-valued solutions with a pole are rejected via their Laurent coefficients. Table 7 lists the zeros on a fine real grid ($d=3$): all lie within 10^{-6} of Σ_s and there are no others; on the complex box $\operatorname{Re}\lambda \in [-4.5, 1.3]$, $\operatorname{Im}\lambda \in [0.2, 3]$ the defect stays $\geq 4 \cdot 10^{-4}$. At $q=3$ the defect vanishes quadratically at $\lambda = -1$ ($10^{-4}, 10^{-6}, 10^{-8}$ at distances $10^{-2}, 10^{-3}, 10^{-4}$: Jordan block), at $q=2$ linearly. **(N)**

case ($d=3$)	$\ell=0$	$\ell=1$	$\ell=2$
$p=3$ ($s=1$)	$1, -1, -3, -4$	$0, -2, -4$	$-1, -3$
$p=7$ ($s=\frac{1}{3}$)	$1, -1, -\frac{8}{3}, -3$	$0, -2, -\frac{11}{3}, -4$	$-1, -3$
$p=-2$ ($q=2$)	$1, -\frac{2}{3}, -1, -\frac{8}{3}, -3$	$0, -\frac{5}{3}, -2, -\frac{11}{3}, -4$	$-1, -\frac{8}{3}, -3$
$p=-3$ ($q=3$)	$1, -1^{(2)}, -3^{(2)}$	$0, -2^{(2)}, -4^{(2)}$	$-1, -3^{(2)}$

Table 7: Eigenvalues in $[-4.6, 1.3]$ from the monodromy test; $^{(2)}$: collision of the two families (double zero). The single-valued pole solution ${}_2F_1 = (1-z)^{-1}$ at $p=3$, $\ell=2$, $\lambda=1$ is correctly rejected.

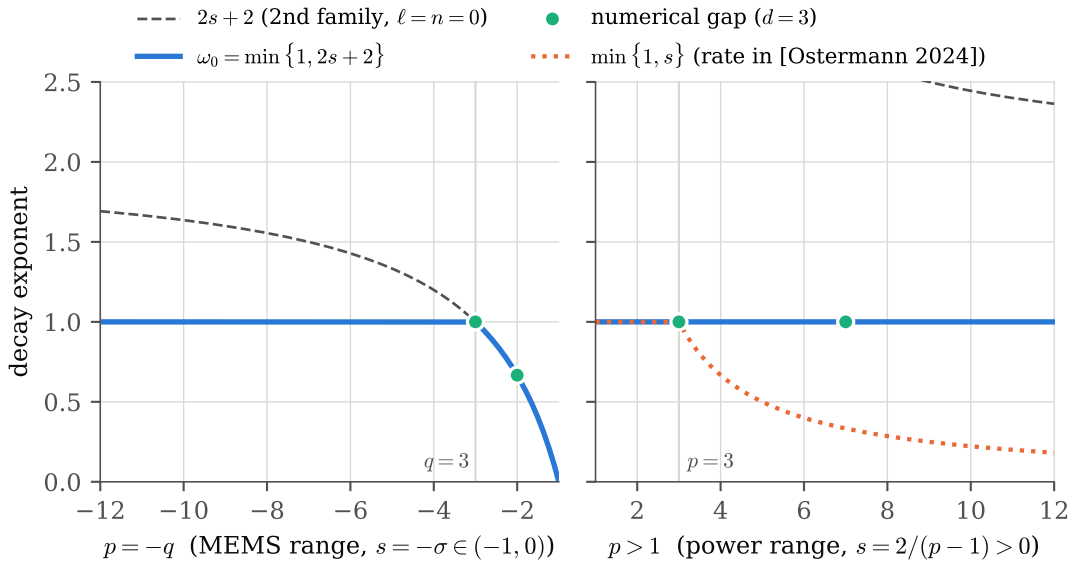


Figure 4: Spectral gap $\omega_0 = \min\{1, 2s+2\}$ over the MEMS range $p=-q<-1$ (left) and $p>1$ (right). Dashed: top of the second family, $2s+2$; dotted: the exponent $\min\{1,s\}$ of [107], reflecting the free eigenvalue $-s$; dots: numerically confirmed gaps ($d=3$).

6.9 Summary and caveats

(P): for all $p \notin [-1,1]$ (powers and MEMS), $V_0 = (s+1)(s+2)$; the spectrum in $\operatorname{Re}\lambda > -\omega_k$, $\omega_k = k - s_c$, is $\{1-\ell-2n\} \cup \{-2s-2-\ell-2n\}$ with polynomial eigenfunctions; $\lambda=1$ and the d-fold $\lambda=0$ are semisimple symmetry modes; the gap is $\min\{1, 2s+2\}$ ($=\frac{2}{3}$ for $q=2$); Jordan blocks at $s=\pm\frac{1}{2}$; no self-similar ODE blowup for $-1 \leq p \leq 1$; the tilted family survives damping. **(S)**: the free-flow splitting, the improved rate $1-\varepsilon$ for $p>1$, Lemma 6.10 and Theorem 6.12, conditional on the autonomous linear estimate (for MEMS, Section 4). **(N)**: the monodromy confirmation; Jordan blocks at all collisions; the co-dimension claims of [11].

Caveats: exponential decay needs $k > s_c$ but the nonlinear argument uses $k > \frac{d}{2}$; the models are only valid up to amplitudes where the cubic foundation law (or parallel-plate force) holds; the literature in Section 6.5 was checked against abstracts, not full texts.

7 Geometric models: wave maps and relativistic membranes

This section treats two *geometric* wave equations: the wave map (σ-model) equation, which is semilinear and for which the Donninger programme is essentially complete, and the timelike minimal-surface (Nambu–Goto, scalar Born–Infeld) equation, which is quasilinear and for which it is not. Throughout we separate **(K)** known results (with references), **(D)** what is derived or computed here (scripts `code/geom_*.py`), and **(C)** conjectural statements.

7.1 Wave maps: origin, scaling, and what is industrially relevant

A wave map $\boldsymbol{u} : \mathbb{R}^{1+d} \to S^m \subset \mathbb{R}^{m+1}$ is a critical point of $\frac{1}{2}\int(|\partial_t \boldsymbol{u}|^2 - |\nabla \boldsymbol{u}|^2)$ under the constraint $|\boldsymbol{u}| = 1$:

$$\partial_t^2 \boldsymbol{u} - \Delta \boldsymbol{u} = (|\nabla \boldsymbol{u}|^2 - |\partial_t \boldsymbol{u}|^2)\boldsymbol{u}. \tag{7.1}$$

Where it comes from. (a) In the Ericksen–Leslie theory of nematic liquid crystals with director inertia σ_1, one-constant Frank energy K, frozen flow and no dissipation, the director $\boldsymbol{n} \in S^2$ obeys (7.1) with wave speed $c^2 = K/\sigma_1$; unequal elastic constants lead to the variational wave equations of Hunter and Saxton [74]. (b) Inertial magnetisation dynamics (nutation) has been observed in ferromagnetic films at sub-THz frequencies [104]; in antiferromagnets the leading-order dynamics of the Néel vector is a nonlinear σ-model, i.e. a wave map into S^2. (c) In physics, the $O(4)$ σ-model ($m = 3$, $d = 3$) describes cosmological textures; its collapse is the explicit self-similar solution of Shatah [111] and Turok–Spergel [116].

Honest assessment. In display-type liquid crystals the director motion is overdamped and σ_1 is negligible, so (7.1) is a formal inertial limit. A damping term $\gamma\,\partial_t \boldsymbol{u}$ is, however, a lower-order perturbation in similarity variables (relative size $\gamma(T-t)$, cf. Section 3.4): *if* a singularity forms in the inertial regime, damping is asymptotically irrelevant, and the inertial self-similar regime is entered only when $T - t \ll \sigma_1/\gamma$. The scaling $\boldsymbol{u} \mapsto \boldsymbol{u}(\mu t, \mu x)$ ($s = 0$) leaves (7.1) invariant and multiplies the energy by μ^{2-d}, so $d = 2$ is energy critical and $d \geq 3$ supercritical.

- $d = 2$ (films, planar magnets; target S^2): there is no self-similar blowup; energy cannot concentrate in the interior of the backward light cone **(K)** [112], and equivariant blowup happens by bubbling off a harmonic map [115], at non-self-similar (type II) rates constructed in [88] and shown to be stable within co-rotational symmetry in [109]. Donninger's method is *not* the right tool there.

- $d = 3$, target S^3: self-similar blowup is generic and stable (Section 7.5). For the physically most relevant case $d = 3$, *target* S^2 (bulk nematics, magnets) the co-rotational reduction below is unavailable; self-similar wave maps can still be sought as harmonic maps from hyperbolic space [14], but we are not aware of a stability theory for S^2 targets. What transfers is the method (similarity variables, mode stability without symmetry as in [119, 34]); what does not transfer is the explicit profile.

7.2 Co-rotational wave maps and similarity variables

For $m = d$ and the co-rotational ansatz $\boldsymbol{u}(t,x) = (\sin u(t,r)\,\frac{x}{r}, \cos u(t,r))$, $r = |x|$, (7.1) becomes

$$u_{tt} - u_{rr} - \frac{d-1}{r}u_r + \frac{d-1}{2r^2}\sin(2u) = 0. \tag{7.2}$$

Proposition 7.1 (**K**; verified **D**). *For $d \geq 3$, $u_T(t,r) = f_0\big(\frac{r}{T-t}\big)$ with $f_0(\rho) = 2\arctan\big(\rho/\sqrt{d-2}\big)$ solves (7.2); equivalently f_0 solves $(\rho^2-1)f'' + (2\rho - \frac{d-1}{\rho})f' + \frac{d-1}{2\rho^2}\sin 2f = 0$ and is analytic on $[0,\infty)$, in particular across the light cone $\rho = 1$.*

This is due to Shatah and Turok–Spergel for $d = 3$ and to Bizoń–Biernat [8] for $d \geq 4$; we checked it symbolically for $3 \leq d \leq 10$ (`geom_wavemaps_symbolic.py`). For $d = 3$, f_0 is the ground state of a family of smooth self-similar solutions; numerically it attracts generic large data, and the first excited one, which has exactly one unstable mode, sits at the threshold of blowup [9].

Reduction to a semilinear radial equation. Writing $u = r\psi$, (7.2) becomes (checked symbolically)

$$\psi_{tt} - \Delta_{\mathbb{R}^{d+2}}\psi + \frac{d-1}{2r^3}\big(\sin(2r\psi) - 2r\psi\big) = 0, \tag{7.3}$$

a radial wave equation on $\mathbb{R}^{d+2}$ with a smooth nonlinearity and scaling exponent $s = 1$. With $\psi_1 = (T-t)\psi$, $\psi_2 = (T-t)^2\psi_t$ and $\xi \in \mathbb{B}^{d+2}$, the conventions of Section 3 give an *autonomous* system, because $(T-t)^3\frac{d-1}{2r^3}(\sin 2r\psi - 2r\psi) = \frac{d-1}{2|\xi|^3}(\sin 2|\xi|\psi_1 - 2|\xi|\psi_1)$. The profile is $\psi_1 = f_0(|\xi|)/|\xi|$. Linearising, $\partial_\tau\Phi = (\mathbf{L}_0 + \mathbf{L}')\Phi$ with $\mathbf{L}'(\phi_1,\phi_2) = (0, -V\phi_1)$ and

$$V(\rho) = \frac{(d-1)(\cos 2f_0(\rho) - 1)}{\rho^2} = -\frac{8(d-1)(d-2)}{(d-2+\rho^2)^2} \qquad (V = -\tfrac{16}{(1+\rho^2)^2} \text{ for } d = 3). \tag{7.4}$$

$\mathbf{L}'$ is compact on radial $\mathcal{H}_k := H^k \times H^{k-1}(\mathbb{B}^{d+2})$. For the free operator, the known Lumer–Phillips estimates (e.g. [26, 17, 61]) give a growth bound $-\omega_k$ which, by scaling, should have $\omega_k = k - \frac{d+2}{2} + 1 = k - \frac{d}{2}$ (the decay rate of Section 3 with $s = 1$ in dimension $d+2$). **(D)** This value is sharp: for $\operatorname{Re}\lambda < -\omega_k$ both Frobenius branches at $\rho = 1$, 1 and $(1-\rho^2)^{\frac{d-1}{2}-\lambda}$, lie in H^k near $\rho = 1$ (the second iff $\frac{d-1}{2} - \operatorname{Re}\lambda > k - \frac{1}{2}$), so the free mode regular at $\rho = 0$ is an eigenfunction in $\mathcal{H}_k$ and every such λ is in the spectrum. Hence the spectrum of $\mathbf{L}$ in $\{\operatorname{Re}\lambda > \frac{d}{2} - k\}$ consists of isolated eigenvalues, and the regularity k determines how much of it is visible.

7.3 The spectral problem and how mode stability is proved

Modes of the linearisation of (7.3), $\delta\psi = (T-t)^{-1-\lambda}g(\rho)$ (equivalently $u - u_T \propto (T-t)^{-\lambda}\rho\, g(\rho)$, with the same λ), solve

$$(1-\rho^2)g'' + \Big(\frac{d+1}{\rho} - 2(\lambda+2)\rho\Big)g' - (\lambda+1)(\lambda+2)g - V(\rho)g = 0, \qquad 0 < \rho < 1. \tag{7.5}$$

The Frobenius exponents are $\{0, -d\}$ at $\rho = 0$ and $\{0, \sigma_2\}$, $\sigma_2 = \frac{d-1}{2} - \lambda$, at $\rho = 1$. We call λ an *eigenvalue* if (7.5) has a nonzero solution in $C^\infty[0,1]$; for $\operatorname{Re}\lambda > \frac{d}{2} - k$ this coincides with the eigenvalues of $\mathbf{L}$ on $\mathcal{H}_k$. Generically the condition at $\rho = 1$ excludes the $(1-\rho)^{\sigma_2}$ branch; when $\sigma_2 \in \mathbb{N}$ the exponent-0 branch may carry a logarithm, and if it does not, the point $\rho = 1$ imposes no condition at all.

Symmetry mode. Time translation gives $\partial_T u_T = -(T-t)^{-1}\rho f_0'(\rho)$, i.e. $\lambda = 1$ with $g_1 = f_0' \propto (d-2+\rho^2)^{-1}$ (checked). Space translations and Lorentz boosts leave the co-rotational class and appear only in the non-symmetric problem. *Mode stability* means: no eigenvalue with $\operatorname{Re}\lambda \geq 0$ except $\lambda = 1$. The instability $\lambda = 1$ is removed in the nonlinear argument by adjusting the blowup time T.

Proposition 7.2 (Hyperbolic reformulation; **D**). *Put $\rho = \tanh\eta$, $g = \cosh^{1+\lambda}(\eta)\,\chi(\eta)$, $\kappa = \lambda - \frac{d-1}{2}$ and $\psi_{1\mathrm{D}} = \sinh^{\frac{d+1}{2}}(\eta)\,\chi$. Then (7.5) is equivalent to*

$$\Big(-\Delta_{\mathbb{H}^{d+2}} - \tfrac{(d+1)^2}{4} + W\Big)\chi = -\kappa^2\chi \iff H\psi_{1\mathrm{D}} := -\psi_{1\mathrm{D}}'' + Q\,\psi_{1\mathrm{D}} = -\kappa^2\psi_{1\mathrm{D}}, \tag{7.6}$$

with $W(\eta) = V(\tanh\eta)/\cosh^2\eta$ *and* $Q = \frac{(d+1)(d-1)}{4\sinh^2\eta} + W$*, a self-adjoint operator on* $L^2(0,\infty)$ independent of λ. *Smoothness of* g *at* $\rho = 1$ *means* $\psi_{1\mathrm{D}} = e^{-\kappa\eta}F(e^{-2\eta})$ *with* F *analytic at* 0*, i.e.* $\psi_{1\mathrm{D}}$ *is the (analytically continued) Jost solution.*

Sketch. Inside the backward light cone write $T - t = e^{-\sigma}\cosh\eta$, $r = e^{-\sigma}\sinh\eta$; then the Minkowski metric equals $e^{-2\sigma}(-d\sigma^2 + g_{\mathbb{H}^{d+2}})$. Conformal covariance of the conformal wave operator in dimension $d+3$, with scalar curvature $-(d+2)(d+1)$ of $\mathbb{H}^{d+2}$, turns $\psi_{tt} - \Delta\psi + V(\rho)(T-t)^{-2}\psi = 0$ into $\varphi_{\sigma\sigma} - \Delta_{\mathbb{H}}\varphi - \frac{(d+1)^2}{4}\varphi + W\varphi = 0$ for $\varphi = e^{-\sigma(d+1)/2}\psi$; a mode is $\varphi = e^{\kappa\sigma}\chi(\eta)$. We verified the resulting identity for (7.5) symbolically and at random complex λ (`geom_wavemaps_symbolic.py`). □

Consequences. (i) If $\operatorname{Re}\lambda > \frac{d-1}{2}$ then $\psi_{1\mathrm{D}} \in L^2$, so $-\kappa^2$ is an eigenvalue of H and λ is *real*; on $\operatorname{Re}\kappa = 0$, $\kappa \neq 0$ the Jost function does not vanish. For $d = 3$ the zero-energy solution regular at $\eta = 0$ is the gauge mode itself, $\psi_{1\mathrm{D}} = \sinh^2\eta/\cosh 2\eta > 0$ (a threshold resonance), so $H \geq 0$ by Sturm's theorem: *no eigenvalue with* $\operatorname{Re}\lambda \geq 1$ *other than* $\lambda = 1$. For $d \geq 4$ the gauge mode is a positive solution of $H\psi = -\kappa_1^2\psi$, $\kappa_1 = \frac{3-d}{2}$, whence $H \geq -\kappa_1^2$ (Allegretto–Piepenbrink) and there are no eigenvalues $\lambda > d-2$; for $\frac{d-1}{2} < \lambda \leq d-2$ we counted the nodes of the zero-energy solution numerically (none for $4 \leq d \leq 8$). (ii) In the strip $0 \leq \operatorname{Re}\lambda < \frac{d-1}{2}$ the eigenvalues are *resonances* of H (zeros of the continued Jost function with $\operatorname{Re}\kappa < 0$); no self-adjointness argument applies, and this is where the difficulty of mode stability lies. For $d \geq 4$ the gauge mode lies inside this strip.

Proposition 7.3 (Supersymmetric removal of the gauge mode; **D**)**.** *Let* $\psi_\star = \sinh^{\frac{d+1}{2}}\eta\,/((d-2)\cosh^2\eta + \sinh^2\eta) > 0$ *(the gauge mode) and* $w = -\psi_\star'/\psi_\star$*. Then* $H + \kappa_1^2 = (-\partial_\eta + w)(\partial_\eta + w)$*, and the partner* $\widetilde{H} = (\partial_\eta + w)(-\partial_\eta + w) - \kappa_1^2 = H + 2w'$ *is, after the substitutions of Proposition 7.2 read backwards with* $d+4$ *in place of* $d+2$ *and* $\lambda+1$ *in place of* λ*, the mode problem*

$$(1-\rho^2)h'' + \Big(\frac{d+3}{\rho} - 2(\lambda+3)\rho\Big)h' - (\lambda+2)(\lambda+3)h + \frac{4(d-1)}{d-2+\rho^2}\,h = 0. \tag{7.7}$$

The intertwiner $\partial_\eta + w$ *maps smooth modes of* (7.5) *to smooth modes of* (7.7) *and annihilates the gauge mode; since* $(-\partial_\eta + w)(\partial_\eta + w) = \kappa_1^2 - \kappa^2$ *on modes, it is injective for* $\lambda \notin \{1, d-2\}$*, and* $\lambda = d-2$ *is an eigenvalue of neither problem. Hence* $\operatorname{spec}(7.7) = \operatorname{spec}(7.5) \setminus \{1\}$*, and mode stability is equivalent to the absence of eigenvalues of* (7.7) *with* $\operatorname{Re}\lambda \geq 0$*.*

The partner potential $-4(d-1)/(d-2+\rho^2)$ was found and checked symbolically for $3 \leq d \leq 8$; the numerical spectra of (7.5) and (7.7) agree to 20 digits except for $\lambda = 1$ (Table 8). Removing the gauge mode by a Darboux transformation in this spirit is the standard first step in the rigorous proofs; we have not checked whether the cited works use exactly this partner.

How the remaining strip is handled (K). Donninger–Aichelburg excluded real unstable eigenvalues other than $\lambda = 1$ [30]; Donninger–Schörkhuber–Aichelburg confined possible unstable eigenvalues to a compact set [42]; Costin–Donninger–Xia proved mode stability for $d = 3$ [22] and Costin–Donninger–Glogić for all $d \geq 4$ [20]. In outline (as we understand it), the *quasi-solution method* of the latter works with the power series of the solution regular at the origin (after removal of the gauge mode): λ is an eigenvalue iff that series converges beyond $\rho = 1$. The coefficients satisfy a finite-term recurrence; one writes down an explicit approximate solution (the quasi-solution) for the ratio of consecutive coefficients and controls the error rigorously, showing that the ratio tends to 1 for all $\operatorname{Re}\lambda \geq 0$. The radius of convergence is then 1, and if $\rho = 1$ is the only singular point of the equation on the unit circle, the solution must be singular there. *Observation (D):* this last step is where $d = 3$ differs: the potential (7.4) has poles at $\rho = \pm i$, on the unit circle, whereas for $d \geq 4$ the poles $\pm i\sqrt{d-2}$ lie outside it (the same holds for the partner potential in (7.7)).

7.4 Numerically computed spectrum

We used three independent methods (`geom_modes.py`, `geom_wavemaps_spectrum.py`): (1) Chebyshev collocation in $y = \rho^2$ with $N = 40, \dots, 120$ nodes, without boundary conditions (both ends are singular points), linearising the quadratic pencil; (2) Frobenius series about $y = 0$ and $y = 1$ (400 terms, 30–35 digits), whose Wronskian at $y = \frac{1}{2}$, multiplied by $1/\Gamma(1-\sigma_2)$, is an *entire* function of λ whose zeros are exactly the eigenvalues (the factor removes the poles at $\sigma_2 \in \mathbb{N}$, and log-free resonances are zeros automatically), refined by Newton's method; (3) the argument principle for this Wronskian on the rectangle $[-1.9, 4] \times [-15, 15]$ (1200 points, maximal phase step < 0.6 rad).

	$d = 3$	$d = 4$	$d = 5$
gauge	1	1	1
λ_1	$-0.54246\,63534\,31726\,5370$	$-0.56361\,26748\,07079\,4374$	$-0.57231\,57218\,74117\,6664$
λ_2	-2 (exact)	$-2.10913\,10073\,42525\,1316$	$-2.16301\,19522\,28281\,9288$
winding (7.5)/(7.7)	2 / 1	2 / 1	2 / 1

Table 8: Eigenvalues of (7.5) with $\mathrm{Re}\,\lambda \geq -2.2$ (Wronskian method; digits stable under changes of truncation and precision). Collocation reproduces λ_1 to 10^{-8} for $N = 40$–80 (round-off grows for larger N). The last row is the winding number of the Wronskian on $[-1.9, 4] \times [-15, 15]$ for the original and the SUSY-partner problem.

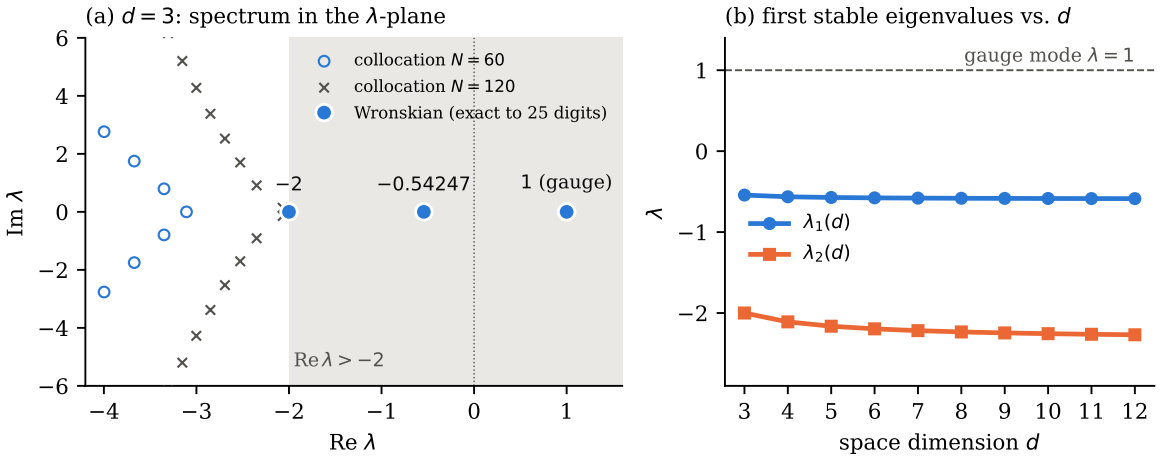


Figure 5: (a) $d = 3$: collocation eigenvalues for $N = 60$ (circles) and $N = 120$ (crosses) and the eigenvalues confirmed by the Wronskian (filled). Collocation values with $\mathrm{Re}\,\lambda \lesssim -2$ move with N; they come from the branch $(1-\rho^2)^{\sigma_2}$ with $\mathrm{Re}\,\sigma_2 \gtrsim 3$, which polynomials cannot distinguish from smooth functions at finite N, and they are rejected by the Wronskian. They mirror the fact that on $\mathcal{H}_k$ the half-plane $\mathrm{Re}\,\lambda < \frac{d}{2} - k$ is spectrum. (b) The first two stable eigenvalues for $3 \leq d \leq 12$.

Findings (D). (1) For $d = 3, 4, 5$ the only eigenvalues in $[-1.9, 4] \times [-15, 15]$ are the gauge mode and one real stable eigenvalue $\lambda_1(d)$, and the only one in $[-2.25, -1.95] \times [-6, 6]$ is $\lambda_2(d)$ (argument principle); together with consequence (i) of Proposition 7.2 for large real parts and the known exclusion of large imaginary parts [42] this is a numerical confirmation of mode stability with spectral gap $|\lambda_1|$. (2) $\lambda_1(d)$ decreases slowly from -0.5425 ($d = 3$) to -0.5865 ($d = 12$). (3) For $d = 3$, $\lambda = -2$ is an eigenvalue with the explicit, analytic eigenfunction (in

the u-picture, $u - u_T \approx \varepsilon(T-t)^2\phi(\rho)$)

$$\phi(\rho) = \frac{3(1+\rho^2)^3 \arctan\rho - 3\rho - 4\rho^3 - 9\rho^5}{\rho^2(1+\rho^2)},$$

checked symbolically; here $\sigma_2 = 3$ and the Frobenius recursion at $\rho = 1$ turns out to be log-free, so $\rho = 1$ imposes no condition. We know no symmetry interpretation. (4) Consequently, in $\mathcal{H}_k$ with $k > \frac{d}{2} + |\lambda_1|$, perturbations (after fixing T) should decay like $e^{-\omega\tau} = (T-t)^\omega$ for every $\omega < |\lambda_1|$, i.e. $(T-t)^{0.54}$ for $d = 3$. For $k = 2$ and $d = 3$ the free decay rate $\omega_2 = \frac{1}{2}$ is the binding constraint, since $|\lambda_1| > \frac{1}{2}$.

7.5 Known stability theorems

Theorem 7.4 (**K**: Donninger [26]; Donninger–Schörkhuber–Aichelburg [42]; Costin–Donninger–Xia [22])**.** *Let $d = 3$. There are $\varepsilon, \delta > 0$ such that for co-rotational data $(u(0,\cdot), u_t(0,\cdot))$ that are ε-close to $(f_0(r), r f_0'(r))$ (the profile with $T = 1$) in a radial Sobolev norm of sufficiently high regularity on the ball $\{r < 1+\delta\}$, there is $T \in (1-\delta, 1+\delta)$ such that the solution blows up at $(T, 0)$ and, in similarity variables on the backward light cone of $(T,0)$, converges to the profile f_0 exponentially in $\tau = -\log(T-t)$.*

[26] proves this assuming mode stability, [42] develops the linear theory, and [22] proves mode stability, which makes the theorem unconditional. Further results (all **K**):

- odd d: nonlinear stability of f_0 [17], using mode stability from [20];
- $d = 3$, stability in the whole space (hyperboloidal similarity coordinates, evolution followed past the blowup time almost up to the Cauchy horizon) [6]; all $d \geq 3$, global-in-space stability [61];
- stability under perturbations small only in the critical Sobolev space: $d = 4$ [44]; $d = 3$, fractional critical space, via Strichartz estimates in similarity coordinates [45];
- without any symmetry: mode stability for $d = 3$ [119] and $d \geq 4$ [34] (quasi-solution method with two extra parameters), and asymptotic stability [33] (preprints).

7.6 Relativistic membranes and the scalar Born–Infeld equation

A timelike hypersurface $M^{d+1} \subset \mathbb{R}^{1,d+1}$ with vanishing mean curvature (Nambu–Goto membrane; $d = 1$ strings, $d = 2$ membranes) is, as a graph $z = u(t,x)$, a critical point of $-\int L$, $L = \sqrt{1 - u_t^2 + |\nabla u|^2}$:

$$\partial_t\Big(\frac{u_t}{L}\Big) - \nabla\cdot\Big(\frac{\nabla u}{L}\Big) = 0 \iff (1+|\nabla u|^2)u_{tt} - 2u_t\nabla u\cdot\nabla u_t - L^2\Delta u + u_i u_j u_{ij} = 0, \tag{7.8}$$

the scalar Born–Infeld equation (the equivalence, after multiplication by L^3, was checked in the radial case). It is quasilinear, hyperbolic while $L^2 > 0$, and degenerates when M becomes null. The scaling $u \mapsto \mu^{-1}u(\mu t, \mu x)$ gives $s = -1$, so self-similar solutions are $u = (T-t)U(x/(T-t))$: bounded slopes, curvature $\sim (T-t)^{-1}$.

Literature (K). Eggers–Hoppe derived self-similar string solutions in a graph representation near singularities (of swallowtail type) and their extension to point singularities of membranes [46]; Eggers–Hoppe–Hynek–Suramlishvili relate membrane singularities to caustics of the eikonal equation and observe them numerically, e.g. at the poles of collapsing ellipsoids [47]. Nguyen–Tian show that timelike maximal cylinders in $\mathbb{R}^{1+2}$ become singular in finite time and, near a generic singularity, move rigidly or self-similarly [105]. Wong proved that axially symmetric timelike minimal submanifolds (in the class considered there) cannot exist globally in

time, and analysed the limiting geometry [121]; Cai–Liu construct radial singularities through loss of hyperbolicity (the surface turns null) [12]. On the stability side, Donninger–Krieger–Szeftel–Wong proved codimension-one stability of the (static, non-singular) catenoid [32], and Bahouri–Marachli–Perelman constructed type II blowup near the Simons cone, adapting the Krieger–Schlag–Tataru method to this quasilinear setting [2]. Yan proposed explicit "lightlike" self-similar solutions $u = \pm(T-t)\sqrt{1-\rho^2}$ and stability statements in a proper subregion of the backward light cone [122, 124] (and for the 1+1 Born–Infeld equation [123]); see Proposition 7.5(c) for what these solutions are. We found no Donninger-type stability result for a *timelike* self-similar membrane singularity, nor for Born–Infeld blowup beyond these preprints (literature search of October 2026, which had to stop before every lead was exhausted).

Proposition 7.5 (Self-similar membranes are minimal cones; **D**)**.** *Let $p = (T, 0, 0)$. Inside the backward light cone of p the map $(t, x, z) \mapsto (x, z)/(T-t) \in \mathbb{B}^{d+1}$ is the Beltrami–Klein model of $\mathbb{H}^{d+1}$, and these are exactly the similarity coordinates.*

(a) The graph of $u = (T-t)U(\xi)$ is, inside the light cone of p, the cone $C(\Sigma)$ with vertex p over $\Sigma = \{(\xi, U(\xi))\} \cap \mathbb{B}^{d+1}$; it solves (7.8) iff Σ is a minimal hypersurface of $\mathbb{H}^{d+1}$.

(b) Normal perturbations $\varphi = \ell^a \chi(\omega)$, ℓ = proper time to p, solve the Jacobi equation $\Box_C \varphi + |\mathrm{II}_C|^2 \varphi = 0$ iff $J_\Sigma \chi = (a-1)(a+d)\chi$, where $J_\Sigma = \Delta_\Sigma + |A_\Sigma|^2 - d$ is the Jacobi operator of $\Sigma \subset \mathbb{H}^{d+1}$. In similarity variables $\lambda = 1 - a$:

$$-J_\Sigma \chi = \lambda(d + 1 - \lambda)\chi.$$

Translations give $\lambda = 1$ ($\chi = \langle \nu, e \rangle$, $J_\Sigma \chi = -d\chi$); Lorentz transformations fixing p give $\lambda = 0$ (Killing–Jacobi fields).

(c) Consequences: for $d = 1$ every self-similar string is a plane (geodesics of $\mathbb{H}^2$); every radial *self-similar graph regular at $\xi = 0$ is $U \equiv c$, a plane moving with constant velocity (a minimal hypersurface of revolution meeting the axis is totally geodesic; the Taylor recursion of the radial ODE confirms $U''(0) = U^{(4)}(0) = \cdots = 0$ for $d = 1, 2, 3$). The solutions $\pm(T-t)\sqrt{1 - |x|^2/(T-t)^2}$ correspond to $\Sigma = \partial_\infty \mathbb{H}^{d+1}$: they satisfy the non-divergence form of (7.8) but have $L \equiv 0$, i.e. they are* null *cones, where the equation is not hyperbolic (checked).*

Sketch. The induced metric of $C(\Sigma)$ is $-d\ell^2 + \ell^2 g_\Sigma$, the radial direction is geodesic and $\mathrm{II}_C = \ell\, \mathrm{II}_\Sigma$ on $T\Sigma$, so $H_C = \ell^{-1} H_\Sigma$ and $|\mathrm{II}_C|^2 = \ell^{-2}|A_\Sigma|^2$. With $\Box_C = -\ell^{-d}\partial_\ell(\ell^d \partial_\ell) + \ell^{-2}\Delta_\Sigma$ one gets $(\Delta_\Sigma + |A_\Sigma|^2)\chi = a(a+d-1)\chi$, and $\mathrm{Ric}_{\mathbb{H}^{d+1}}(\nu, \nu) = -d$. For $\Sigma = \mathbb{H}^d$ (the flat membrane) this reproduces the free eigenvalues $\lambda \in \{1, 0, -1, \dots\}$ of Section 3 with $s = -1$, $V_0 = 0$. This is the membrane analogue of the correspondence between self-similar wave maps and harmonic maps from $\mathbb{H}^d$ [14] and of Proposition 7.2. □

Shrinking spheres (D). A round k-sphere $R(t)\omega$ in $\mathbb{R}^{k+1}$ ($k = 1$ string, $k = 2$ membrane) has Lagrangian $-R^k\sqrt{1 - R_t^2 + |\nabla_{S^k} R|^2/R^2}$ (determinant of the induced metric checked for $k = 2$). Homogeneous solutions conserve $E = R^k/\sqrt{1-\dot{R}^2}$; for $k = 1$, $R = E\cos(t/E)$. Every such sphere collapses at a finite time T with $\dot{R} \to -1$ and Lorentz factor $E/R^k \to \infty$:

$$R(t) = (T-t) - \frac{(T-t)^{2k+1}}{2(2k+1)E^2} + O\big((T-t)^{4k+1}\big).$$

In similarity variables $R/(T-t) \to 1$: the collapse is *lightlike*; the limiting profile is the null cone of Proposition 7.5(c), sitting on the boundary $|\xi| = 1$ of the similarity domain. Linearising about it, $R = R_0 + \varepsilon \sum \delta_\ell(t) Y_\ell$ gives (derived and checked)

$$\ddot{\delta}_\ell - 2k\frac{\dot{R}_0}{R_0}\dot{\delta}_\ell + \frac{\ell(\ell+k-1) - k}{E^2} R_0^{2k-2}\delta_\ell = 0, \tag{7.9}$$

whose exponents at $t = T$ are 0 and $2k+1$. Hence generically $\delta_\ell(T) \neq 0$ and $\delta_\ell/R_0 \sim (T-t)^{-1}$: in similarity variables *every* harmonic (ℓ, m) has a mode with $\lambda = 1$ (and one with $\lambda = -2k$). Only $\ell = 0$ (time shift) and $\ell = 1$ (translations) are symmetries; for instance, for $k = 2$, $E = 1$, data $(\delta_\ell, \dot{\delta}_\ell) = (0, 1)$ at rest we find $\delta_\ell(T) = 0.5,\ 0.437,\ 0.328,\ 0.2,\ 0.083$ for $\ell = 0, \dots, 4$ (`geom_membrane.py`). Geometrically, each ray of the ultra-relativistic membrane reaches the centre at its own time: the round focus is an infinitely degenerate caustic which generic perturbations unfold, in line with the caustic picture of [47]. Spherical collapse is therefore unstable in infinitely many directions, and a Donninger-type argument (finite-dimensional instabilities removed by modulation) cannot apply to it.

7.7 Assessment and open problems

1. **Wave maps (K + D).** The Donninger programme is complete for co-rotational wave maps into S^d, $d \geq 3$, and is now being extended beyond symmetry. Our computations reproduce the spectral picture (gauge mode $\lambda = 1$, gap $|\lambda_1| \approx 0.54$–0.59), give a self-contained proof of the absence of eigenvalues with $\operatorname{Re}\lambda \geq \frac{d-1}{2}$ other than the gauge mode for $d = 3$ (consequence (i) of Proposition 7.2), and give an explicit SUSY partner (7.7). The open cases relevant for applications are $d = 2$ with target S^2 (bubbling, not self-similar, so a different theory) and $d = 3$ with target S^2, where neither the profile nor its stability is settled as far as we know.

2. **Membranes: obstacles (D).** (a) *Quasilinearity:* the perturbation $\mathbf{L}'$ contains second derivatives, so it is not compact relative to $\mathbf{L}_0$, and the domain of dependence moves with the solution. One needs energy estimates adapted to the background's characteristic cone; this has not been done in similarity variables. (b) *Characteristic degeneracy:* the singularities found so far in analysis and numerics (strings, spheres, axisymmetric membranes) are lightlike, and then the similarity variables put the profile on the boundary $|\xi| = 1$. (c) *Gauge:* the graph gauge fails where the surface turns vertical (swallowtails); normal graphs over the cone are the natural choice in Proposition 7.5.

3. **Where a Donninger-type theory is plausible (C).** Timelike self-similar membranes are cones over minimal hypersurfaces $\Sigma \subset \mathbb{H}^{d+1}$. Such Σ with prescribed ideal boundary exist in abundance by classical results on the asymptotic Plateau problem in hyperbolic space. The questions are whether some non-flat Σ extends *smoothly* across the ideal boundary in the Klein coordinates, i.e. across the light cone of p, which is the analogue of the selection of f_0 by smoothness at $\rho = 1$, and whether the resonances of J_Σ in Proposition 7.5(b) satisfy mode stability. Boundary expansions of minimal submanifolds in hyperbolic space suggest that a generic Σ has a non-smooth term of order $(1 - |\xi|^2)^{(d+1)/2}$ (with a logarithm for odd d), so smooth cones, if they exist, are isolated. We have not verified this.

4. **Industrial relevance (honest).** Nambu–Goto and Born–Infeld are physics models (strings and branes, nonlinear electrodynamics). The engineering large-deflection membrane $\rho_h u_{tt} = \mathcal{T}\nabla\cdot(\nabla u/\sqrt{1+|\nabla u|^2})$ shares the scaling $s = -1$ but is not Lorentz invariant and is hyperbolic for every slope, with squared wave speeds $\mathcal{T}/(\rho_h W)$ and $\mathcal{T}/(\rho_h W^3)$, $W = \sqrt{1+|\nabla u|^2}$, so there is no null degeneration (the speeds degenerate only as $|\nabla u| \to \infty$). We know of no self-similar blowup result for it; in MEMS the dominant membrane nonlinearity is usually stretching, not slope.

8 Nonlinear acoustics and derivative nonlinearities

High-intensity ultrasound (HIFU tissue ablation, shock-wave lithotripsy, ultrasonic cleaning and welding) and sonic-boom propagation are described, to second order in the acoustic Mach number, by the Westervelt and Kuznetsov equations [67, 78]. Their nonlinearities are quadratic

in *derivatives* of the unknown and they are quasilinear. We ask which of these models, and two semilinear caricatures, show ODE-type self-similar blowup, and how far the method of Section 3 carries. As in Section 7 we mark statements as **(K)** known (with reference), **(D)** derived or proved here (proof or proof sketch given, formulas checked in `code/acou_symbolic.py`), **(C)** conjectural, formal or numerical only.

8.1 Models

(K) With acoustic pressure p, velocity potential ψ, ambient density ρ_0, sound speed c, coefficient of nonlinearity $\beta = 1 + B/(2A)$ and diffusivity of sound b, the Westervelt equation in pressure form reads

$$(1 - 2kp)\, p_{tt} - c^2 \Delta p - b \Delta p_t = 2k\, (p_t)^2, \qquad k = \frac{\beta}{\rho_0 c^2}, \tag{8.1}$$

i.e. $p_{tt} - c^2\Delta p - b\Delta p_t = k(p^2)_{tt}$. It is obtained by applying ∂_t to the potential form $\psi_{tt} - c^2\Delta\psi - b\Delta\psi_t = \beta c^{-2}\partial_t(\psi_t^2)$ and setting $p = \rho_0\psi_t$. The Kuznetsov equation is

$$\psi_{tt} - c^2\Delta\psi - b\Delta\psi_t = \partial_t(K\psi_t^2 + |\nabla\psi|^2), \qquad K = \frac{B}{2A}\, c^{-2} \tag{8.2}$$

[89, 120, 78]. **(D)** The two right-hand sides differ exactly by $\partial_t(|\nabla\psi|^2 - c^{-2}\psi_t^2)$, the time derivative of the (negative) *null form* $Q_0(\psi,\psi)$: the classical argument that "local nonlinear effects" are negligible for progressive waves is the statement that Q_0 vanishes on plane waves.

Nondimensionalisation. With $u = 2kp$, $t \mapsto t_0 t$, $x \mapsto c t_0 x$, (8.1) becomes

$$(1 - u)\, u_{tt} - \Delta u - \varepsilon \Delta u_t = (u_t)^2, \qquad \varepsilon = \frac{b}{c^2 t_0}. \tag{8.3}$$

For water ($\rho_0 c^2 \approx 2.2\,\mathrm{GPa}$, $\beta \approx 3.5$, $b \approx 4 \times 10^{-6}\,\mathrm{m^2/s}$; standard handbook values, cf. [67]) the degeneracy $u = 1$ corresponds to $p^* = \rho_0 c^2/(2\beta) \approx 3 \times 10^2\,\mathrm{MPa}$, and the viscous time scale is $b/c^2 \approx 2 \times 10^{-12}\,\mathrm{s}$.

Two semilinear caricatures. Freezing the coefficient $1 - u$ to 1 and setting $\varepsilon = 0$ in (8.3) gives the *Riccati wave equation*; replacing $(u_t)^2$ by the null form gives Nirenberg's example:

$$\text{(RW)} \quad u_{tt} - \Delta u = (u_t)^2, \qquad\qquad \text{(NF)} \quad u_{tt} - \Delta u = (u_t)^2 - |\nabla u|^2. \tag{8.4}$$

(RW) keeps the "cumulative" part $2kp_t^2$ of $k(p^2)_{tt} = 2kpp_{tt} + 2kp_t^2$ and drops the quasilinear part; (NF) is the semilinear analogue of the local nonlinear effects.

8.2 Shocks versus ODE-type blowup

(K) For propagating waves the generic singularity is a shock: in 1D, simple waves of (8.1) obey Burgers' equation and steepen at the shock-formation distance while p stays bounded [67]; in several dimensions shock formation for quasilinear wave equations is a different, geometric theory [19]. Shocks are not ODE-type blowup and lie outside the semigroup method. ODE-type blowup instead requires a spatially *coherent* drive (in phase over a region larger than $c(T - t)$), such as a uniform compression. **(D)** The spatially homogeneous dynamics are:

- (RW), (NF): $u_T(t) = -\log(T - t) + c_0$ (exact), so that u and u_t blow up.
- Westervelt, $b = 0$: with $q := 1 - 2kp$, (8.1) is equivalent to

$$(q^2)_{tt} = 2c^2\Delta q. \tag{8.5}$$

Homogeneous solutions have $q^2 = A(T - t)$ (plus a constant), so $p \to 1/(2k)$ stays bounded while $p_t = \sqrt{A}/(4k\sqrt{T - t})$. This is a *degeneration*, not a blowup of p: the coefficient of p_{tt} vanishes and the local sound speed $c/\sqrt{q}$ becomes infinite.

- Kuznetsov, $b = 0$: $\psi = \psi(t)$ forces $(1-2K\psi_t)\psi_{tt} = 0$, so there is *no* homogeneous blowup, Riccati-type or otherwise (the right-hand side of (8.2) is a total time derivative). The uniform compression $\psi = f(t) + B_0|x|^2/(2dc^2)$, whose velocity field is linear in x, solves (8.2) if and only if $(f' - Kf'^2)' = B_0$, and it reaches the same degeneracy $\psi_t \to 1/(2K)$ with $\psi_{tt} \sim (T-t)^{-1/2}$.

The remaining subsections treat (RW) (Sections 8.3 to 8.5), (NF) (Section 8.6), the degenerate Westervelt blowup (Section 8.7) and the role of b (Section 8.8).

8.3 The Riccati wave equation: symmetries, tilted blowup, similarity variables

(D) (RW) is invariant under translations, rotations, the scaling $u \mapsto u(\mu t, \mu x)$ and the *shift* $u \mapsto u + c_0$. It is *not* Lorentz invariant, because $(u_t)^2$ is not.

Lemma 8.1 (tilted ODE blowup)**.** *For $a \in \mathbb{R}^d$, $|a| < 1$, and $c_0 \in \mathbb{R}$,*

$$u_{T,a}(t,x) = -(1-|a|^2)\log\left(T - t + a\cdot x\right) + c_0$$

solves (RW) on $\{T - t + a\cdot x > 0\}$. It blows up on the spacelike hyperplane $t = T + a\cdot x$, with $\partial_t u_{T,a} = (1-|a|^2)/(T-t+a\cdot x)$. Among the functions $-C\log(T-t+a\cdot x)$ with $C \neq 0$, the coefficient $C = 1-|a|^2$ is the only one that gives a solution.

Proof. With $f = T - t + a\cdot x$ one has $\Box(-C\log f) = C(1-|a|^2)f^{-2}$ and $(\partial_t(-C\log f))^2 = C^2 f^{-2}$. □

So the pointwise blowup rate depends on the slope of the blowup surface: if a solution blows up on $t = T(x)$, the local ansatz $u \approx -C(x)\log(T(x) - t)$ gives $C(x) = 1 - |\nabla T(x)|^2$. For (NF) the analogous family is $-\log(T - t + a\cdot x)$ with coefficient 1, the Lorentz boosts of u_T.

Similarity variables. With τ, ξ as in Section 3 let

$$\psi_1 = u + \log(T-t), \qquad \psi_2 = (T-t)\,\partial_t u.$$

Then (RW) is equivalent to

$$\partial_\tau \psi_1 = -\Lambda\psi_1 + \psi_2 - 1, \qquad \partial_\tau \psi_2 = \Delta\psi_1 - \Lambda\psi_2 - \psi_2 + \psi_2^2, \tag{8.6}$$

with equilibria $(c_0, 1)$ (the ODE blowup). In these variables the tilted solutions read, with $T = 1$,

$$\psi_1 = -|a|^2\tau - (1-|a|^2)\log(1 + a\cdot\xi) + c_0, \qquad \psi_2 = \frac{1-|a|^2}{1+a\cdot\xi}. \tag{8.7}$$

They are stationary in $(\nabla\psi_1, \psi_2)$, while ψ_1 drifts linearly in τ. Since (8.6) contains ψ_1 only through $\nabla\psi_1$, the natural phase space is the quotient by constants

$$\mathcal{H}^k := (H^k \times H^{k-1}(B^d))/\mathbb{R}(1,0), \qquad \|[\Phi]\| := \|\nabla\phi_1\|_{H^{k-1}} + \|\phi_2\|_{H^{k-1}}.$$

In $\mathcal{H}^k$ the tilted solutions form a d-parameter family of equilibria Φ_a.

Linearisation. Writing $\Psi = (c_0, 1) + \Phi$,

$$\partial_\tau \Phi = \mathbf{L}\Phi + \mathbf{N}(\Phi), \qquad \mathbf{L}\Phi = \begin{pmatrix} -\Lambda\phi_1 + \phi_2 \\ \Delta\phi_1 - \Lambda\phi_2 + \phi_2 \end{pmatrix}, \qquad \mathbf{N}(\Phi) = \begin{pmatrix} 0 \\ \phi_2^2 \end{pmatrix}. \tag{8.8}$$

Thus $\mathbf{L} = \mathbf{L}_0^{(0)} + \mathbf{L}'$ with $\mathbf{L}'\Phi = (0, 2\phi_2)$. Unlike $(0, V_0\phi_1)$ for $F = F(u)$, the operator $\mathbf{L}'$ is bounded but *not compact* on $H^k \times H^{k-1}$, because it gains no derivative. This is the generic obstruction for derivative nonlinearities, and Step 2 of Section 3.3 fails as stated. For (RW) the problem disappears because of an exact identity.

Lemma 8.2 (exact conjugation)**.** *Let* $\mathbf{S}(f_1, f_2) = (f_1, f_1 + f_2)$*, which is bounded with bounded inverse on* $H^k \times H^{k-1}(B^d)$*. Then* $\mathbf{L} = \mathbf{S}\,\mathbf{L}_0^{(-1)}\mathbf{S}^{-1}$*. In physical variables,* v *solves the linearised equation* $v_{tt} - \Delta v = 2(T-t)^{-1}v_t$ *if and only if* $w = (T-t)v$ *solves* $\Box w = 0$*.*

Proof. $\mathbf{LS}(f_1, f_2) = (-\Lambda f_1 + f_1 + f_2,\ \Delta f_1 - \Lambda f_1 + f_1 - \Lambda f_2) = \mathbf{S}\mathbf{L}_0^{(-1)}(f_1, f_2)$. In physical variables, $(T-t)\big(v_{tt} - \Delta v - \frac{2}{T-t}v_t\big) = \Box\big((T-t)v\big)$. □

At the linear level, (RW) is therefore a free wave equation with *effective scaling exponent* $s = -1$. This is precisely the borderline case excluded in Theorem 3.2: at $s = -1$ the second family $-2s - 2 - \ell - 2n$ reaches 0 at $\ell = n = 0$, and this is the shift mode. The derivative nonlinearity has moved the effective exponent, not added a potential.

8.4 Free flow, spectrum, and the neutral modes

Let $\mathcal{P}_m$ be the polynomials of degree $\le m$ on $\mathbb{R}^d$ and $X_k = \mathcal{P}_{k-1} \times \mathcal{P}_{k-2}$.

Lemma 8.3 (free flow)**.** *Let* $s \in \mathbb{R}$ *and* $k \ge 1$*. (i)* $\mathbf{L}_0^{(s)}$ *generates a* C_0*-semigroup* $\mathbf{S}_0(\tau)$ *on* $H^k \times H^{k-1}(B^d)$ *(standard, cf. [41, 59, 107]). (ii)* X_k *is invariant, and* $\mathbf{L}_0^{(s)}|_{X_k}$ *is diagonalisable with eigenvalues* $-s - m$*,* $m = 0, \dots, k-1$*, whose eigenvectors are the similarity data of the homogeneous polynomial free waves of degree* m *in* $(T-t, x)$*. (iii) The semigroup induced on* $(H^k \times H^{k-1})/X_k$ *satisfies* $\|\bar{\mathbf{S}}_0(\tau)\| \le Me^{-(k-\frac{d}{2}+s)\tau}$*.*

Proof of (ii)–(iii). (ii) Polynomial data produce polynomial free waves, which split into homogeneous parts in $(T-t, x)$. (iii) For the system with exponent s' and $E = \int_{B^d} |\nabla f_1|^2 + f_2^2$ one has the identity

$$\tfrac{d}{d\tau}E = (d - 2 - 2s')E - \int_{\partial B^d} \big(|\nabla f_1|^2 + f_2^2 - 2f_2\,\partial_r f_1\big) \le (d - 2 - 2s')E.$$

Apply it to $\partial^\beta\Psi$ with $|\beta| = k-1$. Since $\partial^\beta\Lambda = (\Lambda + |\beta|)\partial^\beta$, these solve the system with $s' = s + k - 1$. Summing over β gives a seminorm that vanishes exactly on X_k and decays like $e^{-(k-d/2+s)\tau}$. By the Bramble–Hilbert lemma this seminorm is equivalent to the quotient norm. □

Proposition 8.4 (spectrum of (RW) at the ODE blowup)**.** *Let* $k > \frac{d}{2} + 1$ *and* $\omega_k = k - \frac{d}{2} - 1$*. On* $H^k \times H^{k-1}(B^d)$*:*

(i) $\sigma(\mathbf{L}) \cap \{\operatorname{Re}\lambda > -\omega_k\} = \{\lambda_j = 1 - j :\ j \in \mathbb{N}_0,\ j < k - \frac{d}{2}\}$*;*

(ii) the eigenfunctions of λ_j *are* $(g, (\lambda_j + \Lambda)g)$ *with* $g(\xi) = P(1, \xi)$*, where* $P(\rho, x)$ *is a homogeneous polynomial solution of* $P_{\rho\rho} = \Delta_x P$ *of degree* j*. The multiplicity is* $N_j + N_{j-1}$*, where* $N_m = \binom{m+d-1}{d-1}$ *for* $m \ge 0$ *and* $N_{-1} = 0$*. In the sector of spherical harmonics of degree* ℓ *the eigenvalues are* $\lambda = 1 - \ell - 2n$ *(*P *even in* ρ*) and* $\lambda = -\ell - 2n$ *(*P *odd in* ρ*),* $n \in \mathbb{N}_0$*. Equivalently,* $g = |\xi|^\ell Y_\ell\, {}_2F_1\big(\frac{\ell+\lambda}{2}, \frac{\ell+\lambda-1}{2}; \ell + \frac{d}{2}; |\xi|^2\big)$ *terminates;*

(iii) every λ_j *in (i) is semisimple, i.e. there are no Jordan chains;*

(iv) if $\mathbf{P}$ *is the Riesz projection onto* $\{1, 0\}$*, then* $\|\mathbf{S}(\tau)(\mathbf{I} - \mathbf{P})\| \le M_\omega e^{-\omega\tau}$ *for every* $\omega < \min\{1, \omega_k\}$*.*

Proof. (i) By Lemma 8.2 it suffices to treat $\mathbf{L}_0^{(-1)}$. By Lemma 8.3, $\sigma(\mathbf{L}_0^{(-1)})$ is contained in $\{1 - m\}_{m<k} \cup \{\operatorname{Re}\lambda \le -\omega_k\}$, and each $1 - m$ is an eigenvalue. No hypergeometric analysis is needed. As an independent check, the ℓ-sector ODE reduces to the stated ${}_2F_1$ (with $c - \alpha - \beta = \frac{d+1}{2} - \lambda$, verified symbolically); the connection argument of Theorem 3.2 reproduces (i)–(ii). (ii) Since $v = w/(T-t)$ and $w = P$, we get $\phi_1 = e^{(1-j)\tau}P(1, \xi)$. (iii) A Jordan chain at λ_j would give a solution $\phi_1 = e^{\lambda_j\tau}(f_1 + \tau f_0)$ with $f_0 \ne 0$, i.e. a free wave $w = \rho^j\big(f_1(\xi) - f_0(\xi)\log\rho\big)$, $\rho = T - t$. Extend the data to $H^k(\mathbb{R}^d)$. Since $k > \frac{d}{2} + j$, $w \in C^j$ up to the tip of the cone, so

$w(\rho,\rho\xi)=\sum_{i\le j}\rho^i P_i(1,\xi)+o(\rho^j)$, which excludes the $\rho^j\log\rho$ term. A direct ODE computation confirms this: the would-be generalised eigenfunctions for $\lambda=1$ ($\ell=0$) and $\lambda=0$ ($\ell=0,1$) contain $(1-|\xi|)^{\frac{d+1}{2}-\lambda}\log(1-|\xi|)$ (script, check B5). (iv) This follows from (i), (iii) and $\lambda_j\le -1$ for $j\ge 2$. □

λ	ℓ	$P(\rho,x)$	origin
1	0	1	time translation, $\partial_T u_T=-(T-t)^{-1}$; $g=(1,1)$
0	0	ρ	shift $u\mapsto u+c_0$; $g=(1,0)$
0	1	x_i	tilt, $\partial_{a_i}u_{T,a}\|_{a=0}=-\xi_i$; $g=-(\xi_i,\xi_i)$
-1	$0,1,2$	$\rho^2+\frac{\|x\|^2}{d}$, ρx_i, $x_ix_j-\frac{\delta_{ij}}{d}\|x\|^2$	stable

Table 9: Unstable and neutral modes of (RW) at the ODE blowup, and the first stable level.

Remark 8.5 (the $\lambda=0$ modes: no Jordan block, but a nonlinear drift). **(D)** The eigenvalue 0 has multiplicity $d+1$ and is semisimple. One mode is the shift, an exact symmetry. The other d modes are tangent to the tilted family, which is a manifold of equilibria in $\mathcal{H}^k$. In the unreduced space the tilted solutions drift along the shift direction at speed $-|a|^2$ (by (8.7)). This is a *second-order resonance*, not a Jordan block. Insert $\Phi=\varepsilon\Phi^{(1)}+\varepsilon^2\Phi^{(2)}$ with $\Phi^{(1)}=-a\cdot(\xi,\xi)$. The second-order equation $\partial_\tau\Phi^{(2)}=\mathbf{L}\Phi^{(2)}+(0,(a\cdot\xi)^2)$ has a forcing with nonzero component along the $\lambda=0$ shift eigenvector, and indeed $\phi_1^{(2)}=-|a|^2\tau+\frac12(a\cdot\xi)^2$ (check B6). The drift is harmless for stability, since $\mathbf{N}$ does not see ϕ_1 and the drift is invisible in $\mathcal{H}^k$. It does change the asymptotics: $u+\log(T-t)$ is *unbounded* in the nonradial case, $u+\log(T-t)\approx|a|^2\log(T-t)$.

8.5 Nonlinear stability of the Riccati blowup

Theorem 8.6 (radial stability; **(D)**, proof sketch)**.** *Let $d\ge 1$, $k\in\mathbb{N}$ with $k>\frac{d}{2}+1$, and $0<\omega<\min\{1,k-\frac{d}{2}-1\}$. There are $\delta_0,M>0$ such that the following holds for radial data (u_0,u_1) (even data if $d=1$) with $\|\nabla u_0\|_{H^{k-1}(B_2)}+\|u_1-1\|_{H^{k-1}(B_2)}\le\delta\le\delta_0$. There exist $T\in[1-M\delta,1+M\delta]$ and $c_0\in\mathbb{R}$ such that the solution of (RW) exists in the backward light cone Γ_T of $(T,0)$ and, for $0\le t<T$ and $\rho=T-t$,*

$$\begin{aligned}\rho^{j-\frac{d}{2}}\|\nabla^j u(t)\|_{L^2(B_\rho)}&\le M\delta\rho^\omega\quad(1\le j\le k),\\ \rho^{j+1-\frac{d}{2}}\|\nabla^j(\partial_t u(t)-\rho^{-1})\|_{L^2(B_\rho)}&\le M\delta\rho^\omega\quad(0\le j\le k-1),\\ \sup_{|x|\le\rho}|u(t,x)+\log\rho-c_0|&\le M\delta\rho^\omega.\end{aligned}$$

In particular $\partial_t u(t,0)=(T-t)^{-1}\big(1+O((T-t)^\omega)\big)$. For $d=3$ one may take $k=4$ and any $\omega<1$.

Proof sketch. Work in $\mathcal{H}^k_{\mathrm{rad}}$, where by Proposition 8.4 the spectrum in $\{\mathrm{Re}\,\lambda>-\min\{1,\omega_k\}\}$ is $\{1\}$: the $\ell=0$, $\lambda=0$ mode is the quotiented shift and $\ell=1$ is absent. Since $\mathbf{L}$ annihilates $(1,0)$, the semigroup descends to the quotient, and $\|\bar{\mathbf{S}}(\tau)(\mathbf{I}-\bar{\mathbf{P}}_1)\|\le Me^{-\omega\tau}$. Because $k-1>\frac{d}{2}$, $H^{k-1}(B^d)$ is an algebra, so $\|\mathbf{N}(\Phi)-\mathbf{N}(\Psi)\|\lesssim(\|\Phi\|+\|\Psi\|)\|\Phi-\Psi\|$. The data in similarity variables are $[\Phi(0)]=[(u_0(T\cdot),Tu_1(T\cdot)-1)]=(T-1)[(0,1)]+O(\delta)$, and $[(0,1)]=[(1,1)]$ is the $\lambda=1$ eigenvector. The Lyapunov–Perron fixed point and the choice of T (here one-dimensional: intermediate value theorem) are then verbatim those of Steps 4–5 in Section 3.3 [38, 41]. This gives $\|\nabla\psi_1(\tau)\|_{H^{k-1}}+\|\psi_2(\tau)-1\|_{H^{k-1}}\lesssim\delta e^{-\omega\tau}$. Rescaling gives the first two estimates. For the third, $\partial_\tau\psi_1(\tau,0)=\psi_2(\tau,0)-1=O(e^{-\omega\tau})$ by Sobolev embedding, so $\psi_1(\tau,0)\to c_0$; then use $\|\nabla\psi_1\|_{L^\infty}\lesssim\delta e^{-\omega\tau}$. □

Theorem 8.7 (nonradial stability; **(C)**, sketch only)**.** *Under the assumptions of Theorem 8.6 without symmetry, there exist T, $a \in \mathbb{R}^d$ with $|a| \le M\delta$, and c_0 such that in Γ_T*

$$\big\|\rho\,\partial_t u(t,\rho\,\cdot) - \tfrac{1-|a|^2}{1+a\cdot\xi}\big\|_{H^{k-1}(B^d)} + \big\|\rho\nabla u(t,\rho\,\cdot) + \tfrac{(1-|a|^2)a}{1+a\cdot\xi}\big\|_{H^{k-1}(B^d)} \le M\delta\rho^{\omega},$$

and $u = u_{T,a} + O(\rho^{\omega})$ uniformly in Γ_T. In particular $\partial_t u(t,0) \sim (1-|a|^2)/(T-t)$. The rate of the ODE solution is attained only where the blowup surface is horizontal, $a = \nabla T(0) = 0$.

Sketch and status. In $\mathcal{H}^k$ the unstable direction ($\lambda = 1$) is removed by T as before. The neutral directions ($\lambda = 0$, $\ell = 1$) are tangent to the manifold of equilibria $\{\Phi_a\}$. Write $\Phi(\tau) = \Phi_{a(\tau)} + \varphi(\tau)$ and fix $a(\tau)$ by requiring $\varphi \in \ker \mathbf{P}_{a(\tau)}$. Here $\mathbf{L}_a = \mathbf{L} + (0, 2(q_a - 1)\cdot)$ with $q_a = (1-|a|^2)/(1+a\cdot\xi)$ is a small bounded perturbation of $\mathbf{L}$. By Proposition 8.4(iii) and perturbation theory, 0 remains a semisimple eigenvalue of $\mathbf{L}_a$ of multiplicity exactly d, and the spectral projections depend smoothly on a. The modulation equation is then $\dot a = O(\|\varphi\|^2)$, which is integrable, so $a(\tau) \to a$. This parallels the treatment of Lorentz boosts in [40, 107], with one difference: the family Φ_a is not the orbit of a symmetry group, so the linearised operator genuinely depends on a. We have not carried out the estimates. The 1D computation in Section 8.9 supports the statement quantitatively.

Relation to the literature. **(K)** Speck [113] proved, by energy methods and without symmetry assumptions, stable ODE-type blowup in three space dimensions for an open set of data, for a class of quasilinear wave equations whose semilinear part is of Riccati type $(\partial_t\Phi)^2$. His method is global in space and covers quasilinear perturbations of $\Box$. The Donninger approach is local in a light cone but gives the sharp spectral picture: rate ρ^{ω} for every $\omega < 1$, the tilted profile $u_{T,a}$, and the explanation of the drift. In 1D, Gough [65] and Liu–Raees [95] construct generalised self-similar solutions with logarithmic growth and prove stability of the relevant branches, including the ODE branch (see Section 2.4); the same literature treats the spatial-derivative analogue [55]. We expect their logarithmically growing self-similar family to contain the $d = 1$ tilted solutions $u_{T,a}$, which are of the form $-(1-a^2)\log(T-t) + \Phi(\xi)$, but we have not checked this. For small data (RW) still blows up in $d \le 3$ (John [75] for $d = 3$; Glassey's conjecture), but these solutions blow up at large times and are not described by Theorem 8.6.

8.6 The null-form model: blowup is touchdown of a free wave

(K/D) For any C^2 function u, $\Box e^{-u} = e^{-u}(u_t^2 - |\nabla u|^2 - \Box u)$. Hence u solves (NF) if and only if $w = e^{-u}$ solves $\Box w = 0$, with data $(e^{-u_0}, -u_1 e^{-u_0})$. Consequently:

(a) small, regular and localised data give global solutions, because w stays close to 1; this is the prototype of the null condition [86];

(b) blowup occurs exactly when the free wave w first touches 0. The blowup set is the zero set $\{w = 0\}$, which is a level set of a free wave and may contain characteristic or timelike pieces;

(c) at a first zero (T^*, x^*) with $\partial_t w \neq 0$ (the generic case) one has $w = |\partial_t w|(T^* - t) + O(\rho^2)$ in the backward cone, so $u = -\log(T^* - t) - \log|\partial_t w(T^*, x^*)| + O(T^* - t)$. This is ODE-type blowup with rate $e^{-\tau}$, i.e. exactly the gap of Proposition 8.4. At the ODE solution, the linearisations of (NF) and (RW) *coincide* because $\nabla u_T = 0$. At points of $\{w = 0\}$ where the zero set is spacelike, the profile is the Lorentz boost $-\log(T - t + a\cdot x)$. This boost is stationary in similarity variables and has no drift, in contrast to Remark 8.5;

(d) non-generic zeros give *non-ODE* self-similar blowup. If P is a homogeneous free wave of degree m that is positive on the closed cone minus its tip, then $u = -\log P(T-t, x)$ is an exact solution, e.g. $u = -\log\big((T-t)^2 + |x|^2/d\big)$, with $u + 2\log(T-t) = -\log(1 + |\xi|^2/d)$. These solutions are unstable: they require w and $\nabla_{t,x} w$ to vanish simultaneously.

So the "level-set" description of (NF) does not contradict ODE-type blowup. The generic first singularity is ODE-type in its light cone, and the level-set picture adds the global geometry

of the blowup set and the non-generic profiles. For (RW) no linearising transform is available: $w = e^{-u}$ only satisfies $w\Box w + |\nabla_x w|^2 = 0$.

8.7 Westervelt and Kuznetsov: degenerate self-similar blowup

(D) Equation (8.5) is invariant under the two-parameter group $q \mapsto (\theta^2/L^2)\, q(t/\theta, x/L)$. Along the homogeneous solution $q^2 = A(T-t)$ the local sound speed is $c\, q^{-1/2} \propto (T-t)^{-1/4}$, so the backward characteristic conoid of the degenerate point is

$$|x - x_0| \le \tfrac{4}{3}\, c\, A^{-1/4} (T-t)^{3/4}.$$

This set is *wider* than a light cone and has a horizontal tangent at its tip. The adapted similarity variables are

$$\eta = \frac{x - x_0}{(T-t)^{3/4}}, \qquad q = (T-t)^{1/2} Q(\tau, \eta),$$

and (8.5) becomes exactly $(\partial_\tau + \tfrac{3}{4}D)(\partial_\tau + \tfrac{3}{4}D - 1)Q^2 = 2c^2 \Delta_\eta Q$, where $D = \eta \cdot \nabla_\eta$. Linearising $Q^2 = A + g$ and rescaling the conoid to the unit ball $|\zeta| \le 1$ gives the mode equation

$$(\Lambda + \tfrac{4}{3}\lambda)(\Lambda + \tfrac{4}{3}(\lambda - 1))h = \Delta h \quad \text{in } B^d,$$

whose regular solutions are terminating ${}_2F_1(\frac{\ell}{2} + \frac{2\lambda}{3}, \frac{\ell}{2} + \frac{2(\lambda-1)}{3}; \ell + \frac{d}{2}; |\zeta|^2)$.

Proposition 8.8 (formal linear spectrum; **(D)** for the mode equation, **(C)** as a stability statement)**.** *The eigenvalues are* $\lambda \in \{1 - \frac{3}{4}(\ell + 2n)\} \cup \{-\frac{3}{4}(\ell + 2n)\}$, $\ell, n \in \mathbb{N}_0$*:*

$\lambda = 1$, $\ell = 0$	$\lambda = \frac{1}{4}$, $\ell = 1$	$\lambda = 0$, $\ell = 0$	$\lambda = -\frac{1}{2}$, $\ell = 0, 2$	$\lambda = -\frac{3}{4}$, $\ell = 1$
time translation	*tilt,* $\delta(q^2) = x_i$	*amplitude* A *(scaling)*	*curvature*	$\delta(q^2) = \rho x_i$

The tilt mode is unstable *($\lambda = \frac{1}{4}$): a tilted degeneracy set $q^2 \approx A(T-t) + b \cdot x$ meets the conoid of (T, x_0) before T. It is removed by choosing x_0 as the point of first degeneracy. The formal conclusion is stability modulo the $d+2$ parameters (T, x_0, A), with rate $(T-t)^{1/2}$ set by the curvature mode. Near the first degeneracy point,*

$$q^2 = A\rho + H|x - x_0|^2 + C\rho^{3/2} + O(\rho^2), \qquad C = \tfrac{8d}{3}\, \frac{c^2 H}{\sqrt{A}}.$$

So q^2 is C^1 but not C^2 in t at the degeneracy point.

Kuznetsov. Writing $r = 1 - 2K\psi_t$, the inviscid form of (8.2) becomes $(r^2)_{tt} = 2c^2 \Delta r - 8K \partial_t (\nabla\psi \cdot \nabla\psi_t)$. At a degeneracy point where the fluid is at rest ($\nabla\psi = O(|x - x_0|)$, as for the uniform compression) the extra term is $O((T-t)^{1/2})$ relative to the others in the variables (τ, η), so formally the Westervelt picture applies. With a nonzero background velocity, the convective term $\nabla\psi \cdot \nabla\psi_t$ enters the principal part and the analysis changes. We have not pursued this case.

Is this Donninger-tractable? (honest assessment). The linear theory has the same structure as in Section 3: a degenerate transport operator on a ball with a characteristic boundary, and explicit polynomial modes. The nonlinear problem, however, is *quasilinear*: $\Delta_\eta \sqrt{G}$ with $G = Q^2$ contains $(G^{-1/2} - A^{-1/2})\Delta_\eta G$. Hence $\mathbf{N}$ maps $H^k \times H^{k-1}$ only into $\{0\} \times H^{k-2}$, one derivative short, and the hyperbolic semigroup does not regularise. The Duhamel/Lyapunov–Perron scheme therefore does not close in $H^k \times H^{k-1}$. A proof would need quasilinear energy estimates in similarity variables (in the spirit of [113]) together with the spectral information above, and the conoid itself moves with A. We regard this as open. Physically, the degeneracy amplitude $p^* \approx 3 \times 10^2\,\mathrm{MPa}$ lies well outside the regime $p \ll \rho_0 c^2$ in which (8.1) is derived. The degeneracy is therefore best read as a signal of model breakdown, not as a physical event.

8.8 The viscous term is not a lower-order perturbation

(D) For (RW) with strong damping, $u_{tt} - \Delta u - b\Delta u_t = (u_t)^2$, the ODE solution $-\log(T-t)$ is still exact. The same holds for the homogeneous Westervelt degeneracy and for the Kuznetsov compression of Section 8.2, since $\Delta p_t = 0$ there. In the variables (8.6) the damping term becomes

$$(T-t)^2\, b\, \Delta u_t = \frac{b}{T}\, e^{\tau}\, \Delta_\xi \psi_2 .$$

It *grows* like $(T-t)^{-1}$ relative to the wave operator, in contrast with the damping γu_t of Section 3.4, which decays like $e^{-\tau}$. The same holds for (8.1) and (8.2), and also in the variables (τ, η) of Section 8.7. Consequences: (i) the light-cone localisation and the semigroup setting of Section 3 are lost, since $\partial_t^2 - \Delta - b\Delta\partial_t$ has infinite speed of propagation and is parabolic at high frequencies; (ii) **(C)** near the singularity the dominant balance for $w = u_t$ is $w_t \approx b\Delta w + w^2$: at the parabolic scale $|x| \sim \sqrt{b(T-t)}$ the term Δu is $O((T-t)/b)$ relative to the others. This is the semilinear heat equation with $p = 2$, and we conjecture the Giga–Kohn/Merle–Zaag behaviour [56, 97]

$$(T-t)\, u_t(t,x) \approx \Big(1 + \frac{|x|^2}{8b(T-t)|\log(T-t)|}\Big)^{-1},$$

with instabilities from translations ($\lambda = \frac{1}{2}$) as well as T; (iii) quantitatively, $b/c^2 \approx 2\,\mathrm{ps}$ for water, so the inviscid self-similar regime describes the dynamics until picoseconds before the singularity. In practice the weakly nonlinear model breaks down long before that. With $b > 0$, (8.1) is globally well posed for small data [79]. We know of no analysis of large-data singularities.

8.9 Numerical confirmation

(C) *Spectra.* We discretise the first-order eigenvalue problems of (RW) and of the Westervelt linearisation directly, without the hypergeometric reduction. We use $g = r^\ell h(r) Y_\ell$, Chebyshev collocation on even functions, and no boundary condition at the characteristic boundary $r = 1$ (`code/acou_eigen.py`). In 60-digit arithmetic with $N = 61$, every eigenvalue with $\mathrm{Re}\,\lambda > -3.2$ for $d = 1, 3$ and $\ell = 0, 1, 2$ coincides with the predictions to better than 10^{-40}, with no additional eigenvalues (Fig. 6). The collocation matrices are extremely non-normal. In double precision with $N \le 25$ the same eigenvalues are reproduced with errors between 10^{-6} and 10^{-2}, growing with $|\lambda|$; for $N \gtrsim 61$ spurious complex pairs appear from $\mathrm{Re}\,\lambda \approx -2$. These are rounding artefacts (pseudospectra), not eigenvalues. This is a warning for numerical mode-stability studies in similarity variables.

Tilted blowup in 1D. We solve (RW) with $u(0) = 0$, $u_t(0) = 2/(1+x^2)$ by the characteristic-diamond scheme, which respects the light cone exactly ($h = 10^{-4}$, `code/acou_tilt1d.py`). Fitting $u_t \approx C(x)/(T(x) - t)$ at each x gives a spacelike blowup curve ($\max_{|x|\le 1.2} |T'| = 0.52$, $T_{\min} = 0.5445$). The fitted coefficient satisfies $|C(x) - (1 - T'(x)^2)| \le 5 \times 10^{-4}$, as predicted by Lemma 8.1. At $x_1 = 0.8$, with $a = T'(x_1) = 0.404$, the rescaled time derivative $\rho\, u_t(T(x_1) - \rho, x_1 + \rho\xi)$ converges to $(1-a^2)/(1+a\xi)$. The sup-error is $6.1, 2.9, 1.3, 0.64 \times 10^{-2}$ for $\rho = 0.2, 0.1, 0.05, 0.025$, i.e. $O(\rho)$, matching the gap 1. Its distance to the ODE profile 1 stays at 0.41 (Fig. 7).

Westervelt degeneracy in 1D. We solve (8.5) ($c = 1$, periodic) with $q(0) = 1$, $q_t(0) = -(1 + 0.2\cos x)$ (`code/acou_westervelt1d.py`). The first degeneracy occurs at $x_0 = 0$, $T = 0.41991$. Fitting $q^2(t,0) = A\rho + C\rho^{3/2} + D\rho^2 + E\rho^{5/2}$ gives $A = 2.3092$ and $C = 0.1342$, while the prediction $C = \frac{8}{3}H/\sqrt{A}$ with the measured $H = 0.0766$ is 0.1345, an agreement of 0.2% that is stable under grid refinement and changes of the fit window. A fit with integer powers only leaves residuals 2×10^4 times larger, which confirms the half-integer expansion predicted by Proposition 8.8.

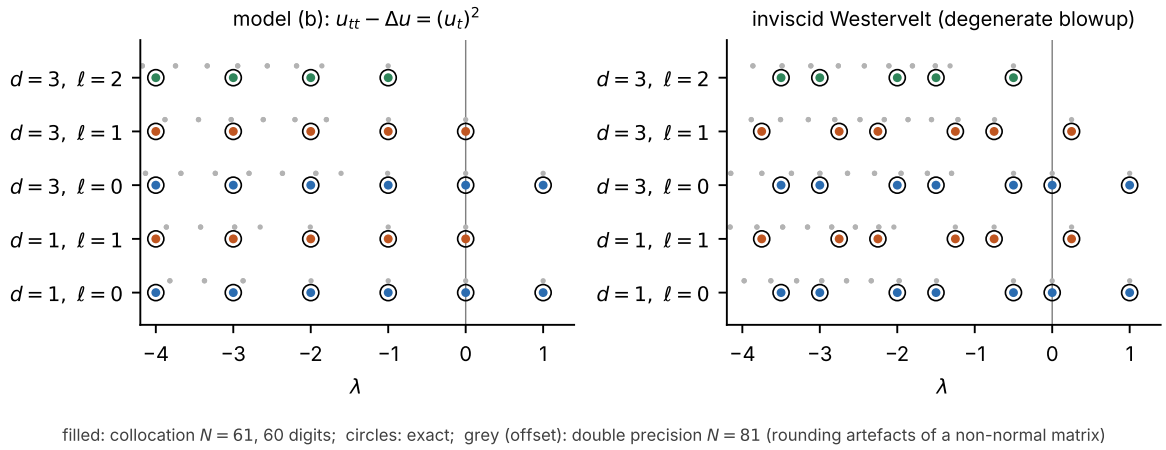


Figure 6: Spectra of the linearisation at the ODE blowup for (RW) (left; exact: $\lambda = 1 - j$, $j \geq \ell$) and for the inviscid Westervelt degeneracy (right; exact: $1 - \frac{3}{4}(\ell + 2n)$, $-\frac{3}{4}(\ell + 2n)$). Filled: collocation, $N = 61$, 60 digits. Circles: exact. Grey, offset: double precision, $N = 81$.

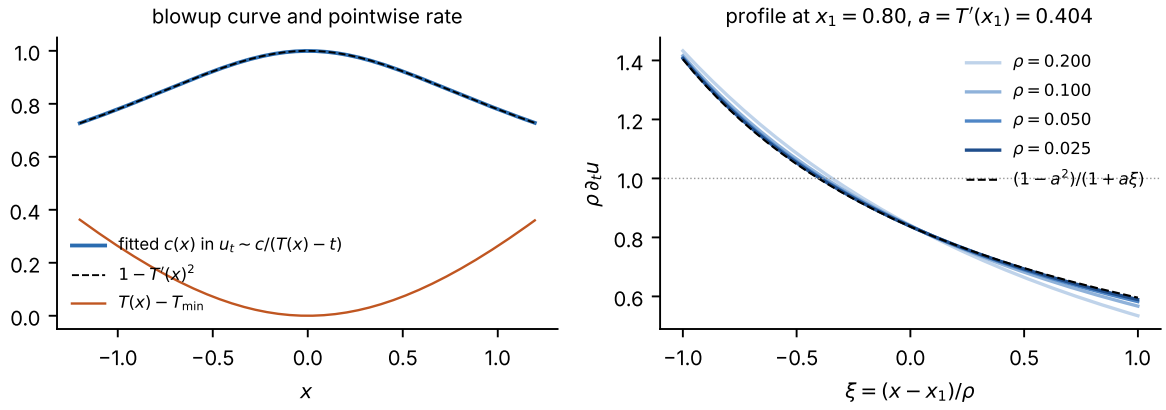


Figure 7: (RW) in 1D. Left: blowup curve and pointwise rate coefficient $C(x)$ versus $1 - T'(x)^2$. Right: convergence of $\rho\,\partial_t u$ to the tilted profile $(1 - a^2)/(1 + a\xi)$ at a point of the blowup curve with slope $a = 0.404$.

8.10 Summary and caveats

- **(D)** (RW): the linearisation at the ODE blowup is *exactly* conjugate to the free wave flow with $s = -1$. Its spectrum is $\{1 - j\}$, i.e. $1 - \ell - 2n$ and $-\ell - 2n$ in the ℓ-sector, and all eigenvalues are semisimple. The neutral $\lambda = 0$ modes are the shift ($\ell = 0$) and the tangents to the exact tilted family $u_{T,a}$ ($\ell = 1$). There is no Jordan block, only a quadratic drift $-|a|^2\tau$ of ψ_1. Radial stability (Theorem 8.6) holds with every rate $\omega < 1$ for $k > \frac{d}{2} + 2$ (proof sketch).
- **(C)** Nonradial stability converges to $u_{T,a}$, not to u_T (Theorem 8.7); this is supported by 1D numerics but not proved here.
- **(K/D)** (NF): explicit via $w = e^{-u}$. The generic first blowup is ODE-type with gap 1 and boosted profiles; non-generic zeros give self-similar profiles $-\log P$.
- **(D/C)** Westervelt/Kuznetsov ($b = 0$): degenerate blowup with p bounded and $p_t \sim (T - t)^{-1/2}$, scaling $x \sim (T - t)^{3/4}$, and a formal spectrum $\{1 - \frac{3}{4}(\ell + 2n)\} \cup \{-\frac{3}{4}(\ell + 2n)\}$. The linear structure is Donninger-like, but the nonlinear problem is quasilinear and open; it is physically outside the validity of the model.

- **(D/C)** The viscous term $b\Delta p_t$ dominates in similarity variables, so the light-cone theory does not apply for $b > 0$. Heat-type blowup is conjectured, but only on picosecond scales.
- Caveats: semigroup generation on $H^k \times H^{k-1}(B^d)$ for general d, k is quoted, not reproved. The nonlinear arguments are sketches. References marked `TODO verify` in the source could not be checked online.

9 Numerical verification

This section reports an *independent* numerical check of the spectral claims of Theorem 3.2 and of the physical picture behind them: that hyperbolic MEMS touchdown, started from physical data, is governed in the backward light cone by the ODE profile, at the rate predicted by the spectral gap. All computations are done from scratch from the coordinates of Section 3; no formula of the analytic sections is used as input except as the thing to be compared with. The scripts are listed in Section 9.5. In brief:

- **Spectra** (Section 9.1). For all 17 model/dimension combinations and $\ell = 0, 1, 2$ (46 sectors) the converged eigenvalues with $\operatorname{Re}\lambda \geq -3$ coincide with $\{a_1 - s - \ell - 2n\} \cup \{a_2 - s - \ell - 2n\}$ (to $5 \cdot 10^{-20}$ in 30-digit arithmetic) and do not depend on d; no other eigenvalue occurs. Double precision is *not* sufficient: a spurious branch, which moves to the right as N grows, swallows part of the window. Two discrepancies with the hints: *(i)* for MEMS with $q = 3$ (and for $p = 5$) the two families collide and the collision eigenvalues are 2×2 Jordan blocks, among them the gap eigenvalue $\lambda = -1$ for $q = 3$; *(ii)* the derivative model is not of the potential form (3.6); its eigenvalues are $\{1 - \ell - 2n\} \cup \{-\ell - 2n\}$.
- **MEMS touchdown** (Section 9.2). From the physical data $w(0) = 1$, $w_t(0) = 0$ in $d = 1, 2$, with and without damping, the minimum of the gap behaves like $c(T-t)^{2/3}$, local exponent 0.66666 and $c = (9\lambda/2)^{1/3}$ recovered to $2 \cdot 10^{-7}$ ($d = 1$), $4 \cdot 10^{-6}$ ($d = 2$); the light-cone profile converges to the constant c at the rate $(T-t)^{\omega}$, measured $\omega = 0.6666 \pm 0.0001$, in agreement with $\omega_0 = 2/3$; the slowest mode is the spatially constant "ODE-energy" mode $\lambda = -2/3$, the spatial variation decays like $(T-t)^1$. Power-law fits at moderate gaps ($w_{\min} \geq 0.02$) give exponents 0.50–0.64.
- **Asymmetric data** (Section 9.3). At the touchdown point the profile converges to the *flat* ODE profile ($a = 0$); a boosted profile with $a = T'(x_0) \neq 0$ appears at the other points x_0 of the blowup curve, which are not physically reached.
- **Exponential nonlinearity** in $d = 1$ (Section 9.4): convergence with $\omega = 1.000$, along the predicted eigenfunction $1 - \xi^2$.

9.1 Spectra of the linearised operators

Formulation. For $F = |u|^{p-1}u$ and $F = -\lambda w^{-q}$ we use the variables (3.2); for $F = e^u$ the logarithmic variables of Section 3; for the derivative model $u_{tt} - \Delta u = (u_t)^2$, whose ODE solution is $u = -\log(T-t)$, we set $\psi_1 = u + \log(T-t)$, $\psi_2 = (T-t)u_t$, which gives $\partial_\tau \psi_1 = -\Lambda\psi_1 + \psi_2 - 1$, $\partial_\tau \psi_2 = \Delta\psi_1 - \Lambda\psi_2 - \psi_2 + \psi_2^2$ with the fixed point $(0, 1)$ (up to the shift $\psi_1 \mapsto \psi_1 + \text{const}$). Linearising at the constant fixed point and inserting $\phi_j = e^{\lambda\tau}|\xi|^\ell Y_\ell(\xi/|\xi|)\, h_j(z)$, $z = |\xi|^2$, where $\Lambda \mapsto \ell + 2z\partial_z$ and $\Delta \mapsto 4z\partial_z^2 + (4\ell + 2d)\partial_z$, gives the eigenvalue problem on $z \in [0, 1]$

$$\lambda \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} = \begin{pmatrix} -(\ell + s) - 2z\partial_z & 1 \\ 4z\partial_z^2 + (4\ell + 2d)\partial_z + V_0 & -(\ell + s + 1) - 2z\partial_z \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} \tag{9.1}$$

with $(s, V_0) = (\frac{2}{p-1},\, ps(s+1))$, $(-\frac{2}{q+1},\, \frac{2q(q-1)}{(q+1)^2})$, $(0, 2)$ for the power, MEMS and exponential cases; for the derivative model the $(1, 1)$ entry is $-\ell - 2z\partial_z$, the $(2, 2)$ entry is $-(\ell - 1) - 2z\partial_z$ and $V_0 = 0$.

Discretisation. Chebyshev–Gauss–Lobatto collocation in $z \in [0,1]$ with $N+1$ nodes (Trefethen's differentiation matrix, mapped by $z = (1+x)/2$), applied to (9.1) *without* any boundary condition: at $z = 1$ (the light cone) none is admissible because all characteristics leave the cone, and at $z = 0$ regularity is built into the ansatz $|\xi|^\ell h(|\xi|^2)$ with h a polynomial. This is a standard eigenvalue problem of size $2(N+1)$ (no quadratic pencil is needed). It was solved (a) in double precision for $N = 20, 40, 80$ (and 120), (b) with `mpmath` at 30 digits for $N = 12$ and $N = 20$, and (c) exactly, in rational arithmetic, on polynomials of degree ≤ 5 (see below). An eigenvalue is called *converged* if (b) gives the same value for both N to 10^{-6} (simple eigenvalues in fact agree to $\le 5 \cdot 10^{-15}$; the looser tolerance accommodates the splitting of Jordan blocks, see (D1)).

Why the closed form is reproduced exactly. The computations revealed a simple structural fact that we record because it explains both the agreement and the numerical difficulties.

Proposition 9.1. *Let $\mathcal{P}_n$ be the polynomials of degree $\le n$ in z. The operator $\mathbf{A}_\ell$ on the right of (9.1) maps $\mathcal{P}_n \times \mathcal{P}_n$ into itself, and in the basis $\{(z^k,0),(0,z^k)\}_{k\le n}$ it is block upper triangular with diagonal blocks $\begin{pmatrix} -(\ell+2k+s) & 1 \\ V_0 & -(\ell+2k+s+1) \end{pmatrix}$, $k = 0,\dots,n$. Hence its eigenvalues are exactly $\lambda = a_i - s - \ell - 2k$, $i = 1,2$, where $a_{1,2}$ are the roots of $a^2 + a - V_0 = 0$. Collocation with $N+1$ nodes is the restriction of $\mathbf{A}_\ell$ to $\mathcal{P}_N \times \mathcal{P}_N$, so in exact arithmetic its eigenvalues are precisely these numbers with $k \le N$.*

Proof. $z\partial_z z^k = kz^k$, while $4z\partial_z^2 z^k + (4\ell+2d)\partial_z z^k = k(4k-4+4\ell+2d)z^{k-1}$ lowers the degree. The characteristic polynomial of the k-th block is $(\lambda+\ell+2k+s)(\lambda+\ell+2k+s+1) - V_0$. □

The Laplacian (and hence d) enters only through the off-diagonal coupling $k \to k-1$; it decides eigenvectors and Jordan structure, not eigenvalues. For the derivative model the blocks are $\begin{pmatrix} -(\ell+2k) & 1 \\ 0 & -(\ell+2k-1) \end{pmatrix}$, giving $\lambda \in \{1-\ell-2k\} \cup \{-\ell-2k\}$. The proposition says nothing about non-polynomial eigenfunctions; that these do not occur for $\operatorname{Re}\lambda > -\omega_k$ is the content of the connection-formula argument in Theorem 3.2.

Results. Table 10 lists the converged eigenvalues (46 sectors). Every simple one agrees with the closed form to $\le 5 \cdot 10^{-20}$, the double ones are discussed in (D1), the multiplicities are those predicted by Proposition 9.1, and the lists are identical for $d = 1,2,3$. No eigenvalue outside the closed-form list was found.

model	(s, V_0)	$\ell = 0$	$\ell = 1$	$\ell = 2$ $(d \ge 2)$	ω_0
MEMS $q = 2$	$(-\frac{2}{3}, \frac{4}{9})$	$1, -\frac{2}{3}, -1, -\frac{8}{3}, -3$	$0, -\frac{5}{3}, -2$	$-1, -\frac{8}{3}, -3$	$\frac{2}{3}$
MEMS $q = 3$	$(-\frac{1}{2}, \frac{3}{4})$	$1, -1^J, -3^J$	$0, -2^J$	$-1, -3^J$	1^J
MEMS $q = 4$	$(-\frac{2}{5}, \frac{24}{25})$	$1, -1, -\frac{6}{5}, -3$	$0, -2, -\frac{11}{5}$	$-1, -3$	1
e^u	$(0, 2)$	$1, -1, -2, -3$	$0, -2, -3$	$-1, -3$	1
u^3, $d = 3$	$(1, 6)$	$1, -1, -3$	$0, -2$	$-1, -3$	1
u^5, $d = 3$	$(\frac{1}{2}, \frac{15}{4})$	$1, -1, -3^J$	$0, -2$	$-1, -3$	1
u^7, $d = 3$	$(\frac{1}{3}, \frac{28}{9})$	$1, -1, -\frac{8}{3}, -3$	$0, -2$	$-1, -\frac{8}{3}, -3$	1
$(u_t)^2$, $d = 1,3$	—	$1, 0, -1, -2, -3$	$0, -1, -2, -3$	$-1, -2, -3$	1

Table 10: Converged eigenvalues with $\operatorname{Re}\lambda \ge -3$ of the linearisation at the ODE profile (all real), for $d = 1,2,3$ (MEMS, e^u), $d = 3$ (powers), $d = 1,3$ (derivative model); in $d = 1$ only $\ell = 0,1$ (even/odd) exist. Identical for all d. J: algebraic multiplicity 2, geometric multiplicity 1 (Jordan block). Last column: gap after removing $\lambda = 1$ (and $\lambda = 0$, and for $(u_t)^2$ the shift mode $\lambda = 0$, $\ell = 0$).

Discrepancies with the closed-form hint.

(D1) *Jordan blocks.* When $a_1 - a_2 \in 2\mathbb{N}$ the two families collide. Among the cases considered this happens for MEMS $q = 3$ ($a_1 - a_2 = 3 - \frac{4}{q+1} = 2$): $\lambda = -1 - \ell - 2n$ is double, and for $p = 5$ ($a_1 - a_2 = \frac{4}{p-1} + 3 = 4$): $\lambda = -3 - \ell - 2n$. Exact rational arithmetic on $\mathcal{P}_5 \times \mathcal{P}_5$ gives geometric multiplicity 1 in every collision with $\operatorname{Re}\lambda \geq -3$ (Table 10); in 30-digit arithmetic the double eigenvalue splits into a pair at distance $4 \cdot 10^{-12}$–$7 \cdot 10^{-10}$ whose mean agrees with the closed form to $5 \cdot 10^{-21}$, the signature of a 2×2 block (simple eigenvalues are accurate to 10^{-20} in the same computation). Analytically: at $q = 3$, $\lambda = -1$, $\ell = 0$ one has $\alpha = -1$, $\beta = 0$ in (3.7), so the equation reduces to $z(1-z)h'' + ch' = 0$, whose only polynomial solution is $h = 1$. Consequence: for $q = 3$ the gap eigenvalue -1 is *not* semisimple, and the linearised flow on the stable subspace decays like $\tau e^{-\tau}$, i.e. $(T-t)|\log(T-t)|$; the statement "rate $e^{-\omega\tau}$ for every $\omega < \omega_0$" is unaffected, but $\omega = \omega_0$ is not attained. (This agrees with the analytic treatment in Remark 4.8, which was written in parallel.)

(D2) *Derivative model.* (3.6) does not apply (F depends on u_t); naively taking $s = 0$, $V_0 = F'(u_T)(T-t)^2 = 0$ would give $\{-\ell - 2n\} \cup \{-1 - \ell - 2n\}$ and miss the time-translation eigenvalue. The correct list is $\{1 - \ell - 2n\} \cup \{-\ell - 2n\}$ (Proposition 9.1), i.e. the free list with $s = -1$, and it contains the additional symmetry eigenvalue $\lambda = 0$, $\ell = 0$, with eigenfunction $(1, 0)$ (the shift $u \mapsto u + \text{const}$).

(D3) *Space dependence of the list.* The closed form is a list of *polynomial* eigenfunctions; whether a listed number is an isolated eigenvalue on $\mathcal{H}^k$ depends on k, see the next paragraph. For instance $-2/3$ (MEMS, $q = 2$) is invisible in the energy space $k = 1$ when $d = 1$.

Eigenfunctions and symmetries. For all cases, the computed eigenvector of $\lambda = 1$, $\ell = 0$ has h_1 constant and $h_2/h_1 = 1+s$ (resp. 1 for e^u and $(u_t)^2$) to $\leq 2 \cdot 10^{-9}$, which is ∂_T of the ODE profile in the variables (3.2). The eigenvector of $\lambda = 0$, $\ell = 1$ is $\phi_1 \propto \xi_j$, $h_2/h_1 = 1+s$, which is $\partial_a \Psi_a|_{a=0}$ for the boost family (3.5) (for MEMS: $\partial_a \Psi_a = c\sigma(\xi_j, (1-\sigma)\xi_j)$). For the derivative model $\lambda = 0$, $\ell = 0$ has eigenvector $(1, 0)$. More generally, for every simple eigenvalue $\lambda \geq -1.5$ the eigenvector h_1 coincides with the terminating hypergeometric polynomial ${}_2F_1(\alpha, \beta; \frac{d}{2} + \ell; z)$ of (3.7) and $h_2 = (\lambda + s + \Lambda)h_1$, to $\leq 10^{-7}$ (double precision, $N = 12$, 70 checks; for the Jordan eigenvalue -1 of $q = 3$ only to $3 \cdot 10^{-5}$, as expected for a defective eigenvalue). In particular the gap mode of MEMS $q = 2$ ($\lambda = -\frac{2}{3}$) is the *constant* $h_1 \equiv 1$, $h_2 = -\frac{4}{3}$, which is the linearisation of the energy of the ODE $w'' = -\lambda w^{-2}$: a solution with energy E near touchdown satisfies $w = c(T-t)^{2/3}(1 + \frac{9E}{10c^2}(T-t)^{2/3} + \dots)$.

The spurious branch and the Sobolev index. In double precision the picture is less clean (Fig. 8): only the top few eigenvalues converge, and the rest form a curve of complex eigenvalues which moves *to the right* as N grows; for MEMS, $q = 2$, $d = 3$, $N = 80$ already shifts $-\frac{2}{3}$ to -0.6948, and at $N = 120$ this eigenvalue is swallowed. Across all cases the right edge of the spurious set is close to $\frac{d-1}{2} - s - K_N$ with $K_N \in [4, 5.2]$, $[3, 4]$, $[2.4, 3]$, $[2, 2.7]$ for $N = 20, 40, 80, 120$ (six cases tested in `num_spurious.py`, s replaced by -1 for $(u_t)^2$; for $(u_t)^2$, $d = 3$, $N = 120$ even the symmetry eigenvalue 0 is destroyed), whereas in 30-digit arithmetic the same matrices have exactly the closed-form eigenvalues (black dots in Fig. 8). Since the exact eigenvalues of the collocation matrix are known (Proposition 9.1), these are not discretisation errors but pseudospectral effects of the extreme non-normality of $\mathbf{A}_\ell$ (the eigenvectors are nearly parallel polynomials). They have a clear meaning. The second Frobenius solution at the light cone behaves like $(1 - |\xi|^2)^\gamma$, $\gamma = \frac{d-1}{2} - s - \lambda$, and it belongs to $H^k(B^d)$ iff $\operatorname{Re}\gamma > k - \frac{1}{2}$. Hence on $\mathcal{H}^k = H^k \times H^{k-1}$ *every* λ with $\operatorname{Re}\lambda < \frac{d}{2} - s - k = -\omega_k$

is an eigenvalue (with the regular-at-0 solution as eigenfunction), so the spectrum of $\mathbf{L}$ on $\mathcal{H}^k$ contains a half-plane and the growth bound on $\mathcal{H}^k$ can be no better than $-\omega_k$, consistently with Theorem 3.2. A degree-N polynomial approximates $(1-z)^\gamma$ to accuracy $\approx N^{-2\operatorname{Re}\gamma}$, so in working precision ε the matrix "sees" this continuum for $\operatorname{Re}\gamma \gtrsim \log(1/\varepsilon)/(2\log N)$: the spurious curve is the continuum of an effective, N- and ε-dependent Sobolev index, and its edge tends to $\frac{d-1}{2} - s$ (the L^2-type threshold) as $N \to \infty$ at fixed ε. Two practical consequences: (a) eigenvalue computations of this type must be done in extended precision or with a convergence check in N, and more resolution in double precision is *worse*; (b) to see the gap ω_0 in $\mathcal{H}^k$ one needs $k \geq \frac{d}{2} - s + \omega_0$, i.e. $k \geq 2, 3, 3$ for MEMS $q = 2$ in $d = 1, 2, 3$, $k \geq 2, 2, 3$ for e^u, $k \geq 2$ for u^3, u^5 and $k \geq 3$ for u^7 in $d = 3$, and $k \geq 3$ ($d = 1$), $k \geq 4$ ($d = 3$) for $(u_t)^2$.

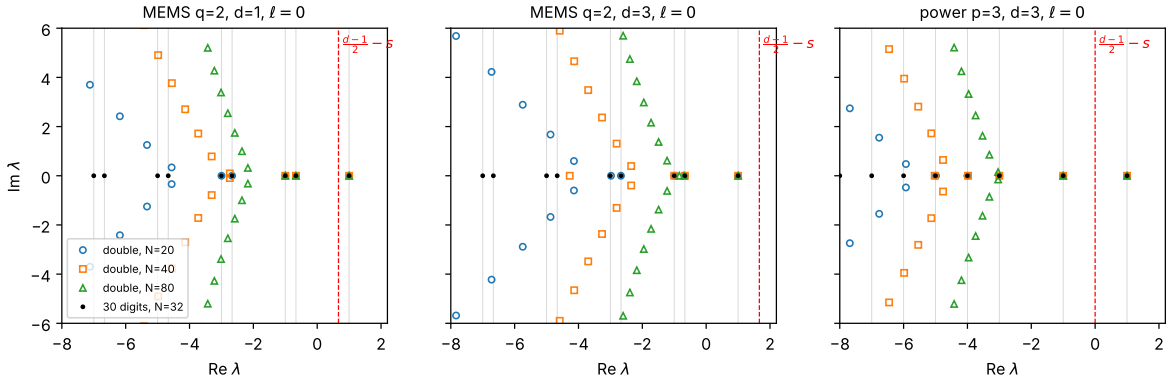


Figure 8: Eigenvalues of the collocation matrix of (9.1), $\ell = 0$. Coloured: double precision, $N = 20, 40, 80$. Black dots: 30-digit arithmetic, $N = 32$, which reproduce the closed form (grey vertical lines) exactly. The red dashed line is $\operatorname{Re}\lambda = \frac{d-1}{2} - s$; the spurious curve approaches it as N grows.

9.2 Touchdown dynamics of the hyperbolic MEMS equation

Problem and parameters. We solve

$$w_{tt} + \gamma w_t - \Delta w = -\lambda w^{-2} \ \text{ in } \Omega, \qquad w = 1 \text{ on } \partial\Omega, \qquad w(0) = 1,\ w_t(0) = 0, \tag{9.2}$$

for $\Omega = (-1, 1)$ and for radial solutions on the unit disk, with $\gamma \in \{0, 1\}$. The static pull-in value (fold of $\Delta w = \lambda w^{-2}$, $w|_{\partial\Omega} = 1$, by shooting) is $\lambda^* = 0.35000$ ($d = 1$) and 0.78923 ($d = 2$); the dynamic threshold, defined as the smallest λ with $\min w < 0.05$ before $t = 30$, lies in $[0.2895, 0.2961]$ ($d = 1$, $\gamma = 0$), $[0.3426, 0.3492]$ ($d = 1$, $\gamma = 1$), $[0.6406, 0.6492]$ ($d = 2$, $\gamma = 0$) and $[0.7438, 0.7523]$ ($d = 2$, $\gamma = 1$) (`num_mems_threshold.py`; for $\gamma = 0$ the finite-horizon value is an upper bound for the true threshold). We take $\lambda = 0.5$ ($d = 1$) and $\lambda = 0.8$ ($d = 2$), above the dynamic threshold in all four cases and small enough that the touchdown time exceeds 1: for $T \leq 1$ the solution in $|x| < 1 - t$ is the ODE solution, and the gap closes simultaneously on a whole interval (disk), a degenerate situation. With $T > 1$ the boundary has influenced the whole backward light cone of the touchdown point.

A weak singularity. The data are incompatible with the boundary condition at second order ($w_{tt}(0) = -\lambda \neq 0 = \partial_t^2 1$), so a jump of w_{tt} travels inwards from $\partial\Omega$ along characteristics. In $d = 2$ it focuses at $(t, r) = (1, 0)$, i.e. *inside* the backward light cone of the touchdown point. This is physical (a voltage switched on instantaneously), but it destroys the spectral accuracy of global methods: in a preliminary test, a Chebyshev method in physical space changed T by $\approx 10^{-7}$ between $N = 128$ and $N = 256$. We therefore use the following three-stage method.

Method. *Stage 1 (physical space, $0 \le t \le t_1$).* In $d = 1$: leapfrog with $\Delta t = \Delta x$ (exact for the free wave equation; the singular characteristics are grid diagonals), on $M = 4000, 8000, 16000$ cells, combined by Richardson extrapolation; in $d = 2$: the conservative second-order leapfrog for $r^{-1}\partial_r(r\partial_r)$ with $\Delta t = \Delta r/2$, $M = 4000, 8000$, followed by a least-squares fit of $w(t_1, \cdot)$, $w_t(t_1, \cdot)$ by even Chebyshev polynomials of degree 40 on $[0, R_0]$, which removes grid-scale noise ($\approx 10^{-4}$ in w_t) left by the focusing; R_0 is kept below the position of the singular front. The switching time t_1 is the multiple of 0.01 just before $\min w = 0.3$.

Stage 2 (light cone of $(T, 0)$, $t \ge t_1$). With $\tau = -\log \frac{T-t}{T-t_1}$, $\xi = \frac{x}{T-t}$, $\psi_1 = (T-t)^{-2/3} w$, $\psi_2 = (T-t)^{1/3} w_t$,

$$\partial_\tau \psi_1 = -\Lambda\psi_1 + \tfrac{2}{3}\psi_1 + \psi_2, \qquad \partial_\tau \psi_2 = \Delta\psi_1 - \Lambda\psi_2 - \tfrac{1}{3}\psi_2 - \lambda\psi_1^{-2} - \gamma(T-t_1)e^{-\tau}\psi_2, \quad (9.3)$$

discretised by Chebyshev collocation in $\xi \in [-1, 1]$ (21 nodes, $d = 1$) or in $z = |\xi|^2 \in [0, 1]$ (17 nodes, radial), without boundary conditions, and integrated with an adaptive 8th-order Runge–Kutta method (DOP853, relative tolerance 10^{-12}). Initial data are interpolated from stage 1 (local 10-point Lagrange interpolation in $d = 1$, the even fit in $d = 2$). The ODE profile is $(\psi_1, \psi_2) = (c, -\frac{2}{3}c)$, $c = (9\lambda/2)^{1/3}$.

Stage 3 (T by bisection). If T is too large, the solution quenches inside the cone at finite τ ($\min \psi_1 \to 0$); if T is too small, $\psi_1 \sim e^{2\tau/3} w(T, 0) \to \infty$. Bisection is run to relative width $4 \cdot 10^{-16}$. Because of the time-translation mode e^τ, the two trajectories belonging to the ends of the final bracket separate eventually; all rates are measured only on the τ-window where their values of $W(\tau, 0) - c$ agree to 0.1%.

Independent check. Stage 1 alone, continued in physical space ($M = 16000$, resp. 8000) down to $\min w = 0.02$ (resp. 0.03), and fitted with T as a free parameter.

Results. In all four cases the minimum of w is attained at the centre; the same holds for all values tested in `num_mems_threshold.py` ($\lambda = 0.35, 0.5, 1$ for $d = 1$, $\lambda = 0.7, \dots, 1.2$ for $d = 2$, $\gamma = 0, 1$): no off-centre or ring quenching occurred. The main numbers are collected in Table 11; Figs. 9 and 10 show the evolution.

case	T	$\min(T-t)$	ω_{fit}	ω, c free	$C_\infty/c - 1$	$\frac{\psi_2 + \frac{2}{3}c}{\psi_1 - c}$	rate of osc W
$d = 1$, $\gamma = 0$	1.75194263316	$1.2 \cdot 10^{-7}$	0.66663	0.6649	$2.1 \cdot 10^{-7}$	-1.3319	0.995
$d = 1$, $\gamma = 1$	2.40507517419	$4.0 \cdot 10^{-8}$	0.66662	0.6642	$2.6 \cdot 10^{-7}$	-1.3318	0.994
$d = 2$, $\gamma = 0$	1.41984985	$1.8 \cdot 10^{-7}$	0.66666	0.6627	$1.2 \cdot 10^{-6}$	-1.3312	0.991
$d = 2$, $\gamma = 1$	2.214154	$2.6 \cdot 10^{-7}$	0.66655	0.6597	$3.8 \cdot 10^{-6}$	-1.3296	0.985
prediction			$\frac{2}{3}$	$\frac{2}{3}$	0	$-\frac{4}{3}$	1

Table 11: Hyperbolic MEMS touchdown, $q = 2$, $\lambda = 0.5$ ($d = 1$), $\lambda = 0.8$ ($d = 2$). T: bisection (digits shown are stable under changes of the light-cone resolution; in $d = 2$ the stage-1 resolution changes T by $8 \cdot 10^{-9}$ ($\gamma = 0$) and $2 \cdot 10^{-7}$ ($\gamma = 1$)). $\min(T-t)$: end of the validity window. ω_{fit}: fit of $W(\tau, 0) - c$ by $Ae^{-\omega\tau} + Be^{-\tau} + Ce^{-4\tau/3}$ on $\tau \ge 3$ ($e^{-4\tau/3}$ is the square of the gap mode); without the C term one gets 0.6656, 0.6651, 0.6643, 0.6625. Next column: fit $W(\tau, 0) = C_\infty + Ae^{-\omega\tau} + Be^{-\tau}$ with c unknown. Ratio column: at the end of the window, $\xi = 0$. Last column: decay rate of $\max_\xi W - \min_\xi W$.

(i) *Exponent and constant.* With T from the bisection, the local slope $d \log w_{\min} / d \log(T-t)$ for $T - t < 10^{-6}$ is 0.666656, 0.666655, 0.666640, 0.666627, and $w_{\min}(T-t)^{-2/3} \to c$; leaving c free in the fit returns $c = (9\lambda/2)^{1/3}$ to $2 \cdot 10^{-7}$ ($d = 1$) and $4 \cdot 10^{-6}$ ($d = 2$). The approach is slow: the relative correction is $(A/c)(T-t)^{2/3}$ with $A/c = -1.44, -2.18, -2.82, -3.87$, so that $w_{\min}$ is within 1% of $c(T-t)^{2/3}$ only for $T - t \lesssim 10^{-4}$–$6 \cdot 10^{-4}$, i.e. $w_{\min} \lesssim 3 \cdot 10^{-3}$–$10^{-2}$ (Fig. 9, middle; Fig. 10, right). Consequently, the independent physical-space fits

over $w_{\min} \in [0.02, 0.25]$ with T free give $\beta = 0.596, 0.559, 0.545, 0.500$ for $w = C(T-t)^{\beta}$, and still $\beta = 0.644, 0.627, 0.614, 0.589$ if the correction $(1+e(T-t)^{2/3})$ is included; their T deviates from the bisection value by $8 \cdot 10^{-5}$–$1.4 \cdot 10^{-3}$. *Exponents read off from moderate gaps are biased low*; this matters when comparing with experiments or with simulations that stop at $w_{\min} = O(10^{-2})$.

(ii) *Light-cone profile.* $W(t,\xi) = (T-t)^{-2/3} w(t, (T-t)\xi)$ converges uniformly on $|\xi| \le 1$ to the constant c (Fig. 10, left), with $\|W - c\|_{\infty} \sim (T-t)^{\omega}$, $\omega = 0.6666 \pm 0.0001$ when the quadratic term of the gap mode is included, in agreement with $\omega_0 = \frac{2}{3}$. The deviation is, to leading order, the spatially *constant* eigenfunction $(1, -\frac{4}{3})$ of $\lambda = -\frac{2}{3}$ (ratio column), i.e. the ODE-energy mode: $W(t,0) - c \approx A(T-t)^{2/3}$ with $A < 0$, as for an ODE solution of negative energy. The spatial oscillation $\operatorname{osc}_{\xi} W$ decays faster, like $(T-t)^1$, which is the next even eigenvalue $\lambda = -1$ (eigenfunction $1 + \frac{|\xi|^2}{3d}$ in the first component). Damping does not change the rate (it enters at order $e^{-\tau}$), only the coefficients.

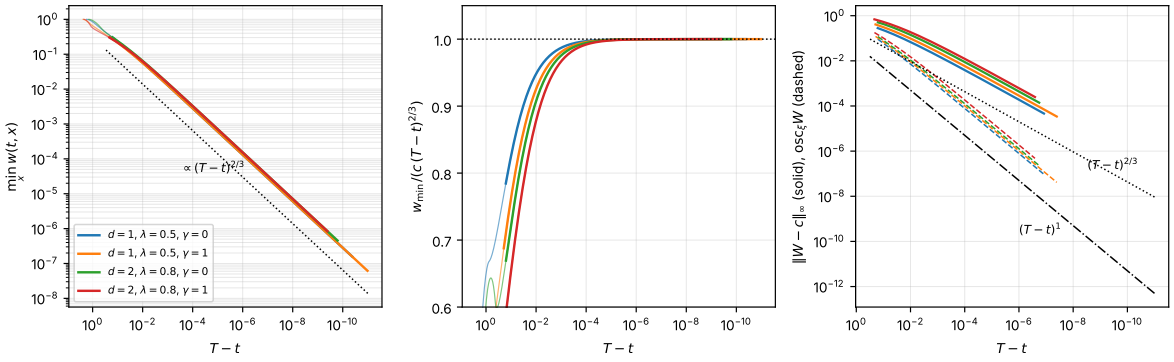


Figure 9: Hyperbolic MEMS touchdown. Left: $\min_x w$ versus $T-t$ (thin: physical-space leapfrog alone; thick: light-cone evolution). Middle: $w_{\min}/(c(T-t)^{2/3})$. Right: $\|W-c\|_{L^{\infty}(|\xi|\le 1)}$ (solid) and $\max_{\xi} W - \min_{\xi} W$ (dashed) with the slopes $\frac{2}{3}$ and 1.

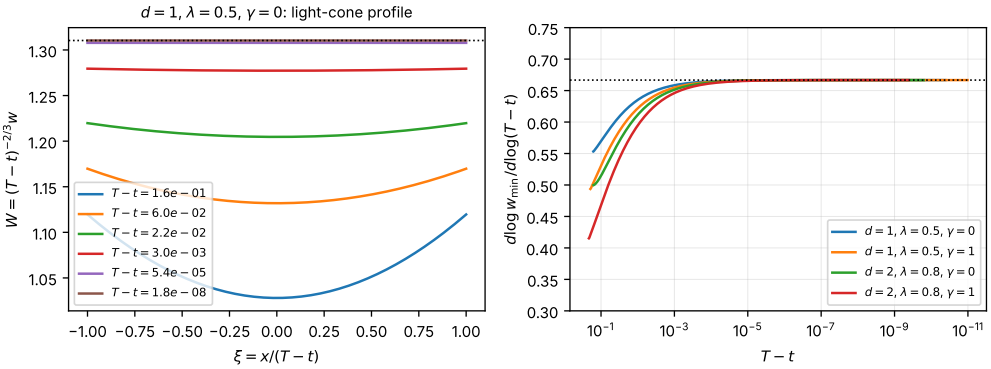


Figure 10: Left: light-cone profiles $W(t,\cdot)$, $d=1$, $\lambda=0.5$, $\gamma=0$, flattening to c (dotted). Right: local exponent $d\log w_{\min}/d\log(T-t)$.

Comparison with [57]. That work studies the same equation formally and numerically for radial data and predicts quenching at the origin with the ODE law $\kappa(T-t)^{2/(q+1)}$, stability of this profile up to time translation, and instability of the non-trivial self-similar profiles (as summarised in Section 2; we could not access the full text, so we do not compare individual

numbers). Our computations agree on all three qualitative points: quenching at the centre, exponent $\frac{2}{3}$ and constant $(9\lambda/2)^{1/3}$, and convergence to the ODE profile at the spectral-gap rate; starting from physical data we never observed an approach to a non-trivial profile. Our computations add two quantitative points: the identification of the slowest correction as the ODE-energy mode, and the warning in (i) about pre-asymptotic exponents.

9.3 Asymmetric data: flat or boosted profile?

The boosted family (3.5) reads, for MEMS, $w = c(1-a^2)^{-\sigma/2}(T - t + a\cdot x)^\sigma$, i.e. $W_a(\xi) = c(1-a^2)^{-\sigma/2}(1+a\cdot\xi)^\sigma$ in the light cone. We verified by substitution (`sympy`, $q = 2,3,4$) that it is an exact solution: with $Z = T - t + a\cdot x$, $w_{tt} - \Delta w = (1-|a|^2)W''(Z)$ and $(1-|a|^2)W'' + \lambda W^{-q} \equiv 0$ for $W = c(1-|a|^2)^{-\sigma/2}Z^\sigma$. As a test of the solver, $W_{0.4}$ stays stationary under (9.3) to $4\cdot 10^{-10}$ over $\tau \in [0,12]$ (the residual growth is the time-translation instability seeded by round-off).

Since W_a vanishes on the line $t = T + a(x - x_0)$, a boosted limit at a point x_0 of the blowup curve $x \mapsto T(x)$ should have $a = T'(x_0)$ (in one dimension and for power nonlinearities this is the Merle–Zaag picture of non-characteristic points [99]); at the *first* touchdown point, $T'(x^*) = 0$, and the profile should be flat. To test this we took the asymmetric data $w(0,x) = 1 + 0.25\,x(1-x^2)$, $w_t(0) = 0$, $\lambda = 0.5$, $\gamma = 0$, $d = 1$, and computed by bisection, for twelve base points x_0, the blowup time $T(x_0)$ of the backward light cone of (T, x_0); this is the blowup curve, which exists (and is 1-Lipschitz) also beyond the first touchdown, although the physical problem ends there.

Results. The blowup curve (Fig. 11, left) is smooth with a non-degenerate minimum at $x^* = 0.00860$, $T(x^*) = 1.7356545138$, $T''(x^*) \approx 1.76$: the asymmetric data shift the touchdown point (slightly to the right, although the initial gap is smaller on the left), and $|T'| \le 0.23 < 1$ on the computed range, i.e. these points are non-characteristic. A degree-6 fit through the twelve values of $T(x_0)$ (residual $7\cdot 10^{-9}$; changing the degree alters T' by $< 5\cdot 10^{-5}$) gives $T'(x_0)$. At five base points the boost parameter $a(t)$ of the light-cone profile, defined by $W(t,1)/W(t,-1) = \left(\frac{1+a}{1-a}\right)^{2/3}$, converges, and at $T(x_0) - t \approx 6\cdot 10^{-9}$:

$x_0 - x^*$	a (light cone)	$T'(x_0)$	$\|W - W_a\|_\infty$	rate of $\|W - W_a\|_\infty$
0	−0.0000112	−0.0000116	$7.3\cdot 10^{-6}$	0.654
−0.12	−0.1981299	−0.1981283	$6.2\cdot 10^{-6}$	0.649
−0.06	−0.1021323	−0.1021325	$6.2\cdot 10^{-6}$	0.650
+0.06	+0.1096470	+0.1096483	$8.1\cdot 10^{-6}$	0.649
+0.12	+0.2274879	+0.2274882	$8.7\cdot 10^{-6}$	0.647

(the rate is a straight-line fit of $\log\|W - W_{a(t)}\|_\infty$ against τ for $\tau > 3$, which is biased low by the same pre-asymptotic corrections as the two-term fits quoted in the caption of Table 11; the window ends where the trajectories for the two ends of the T-bracket differ by 10% of $\|W - W_a\|_\infty$). Thus $a_\infty = T'(x_0)$ to within $2\cdot 10^{-6}$ at every base point, and *at the touchdown point the limit profile is the flat ODE profile*: the residual $a \approx -1.1\cdot 10^{-5}$ there is exactly T' at our numerical x^*, which is off by $\approx 7\cdot 10^{-6}$. Boosted profiles appear only at points $x_0 \neq x^*$ of the blowup curve, which the physical (quenched) solution never reaches. For the stability theory this means: the boost parameters must be modulated (the limit a_∞ is not determined by the symmetry of the data), but at the first touchdown point the modulated parameter converges to 0.

9.4 Exponential nonlinearity in one dimension

The same three-stage method was applied to $u_{tt} - u_{xx} = e^u$ on $(-1,1)$, $u = 0$ at $x = \pm 1$, $u(0) = u_t(0) = 0$ (`num_exp_touchdown.py`; here $\psi_1 = u + 2\log(T-t)$, $\psi_2 = (T-t)u_t$,

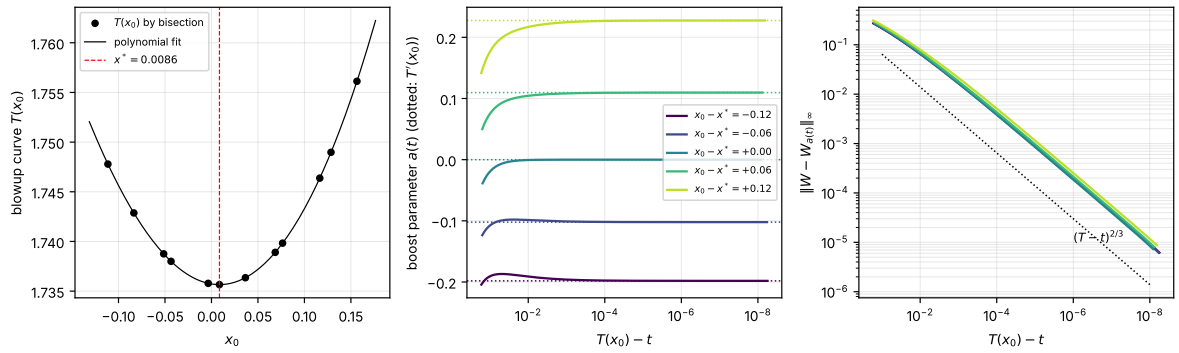


Figure 11: Asymmetric data $w(0,x) = 1 + 0.25\,x(1-x^2)$, $\lambda = 0.5$, $d = 1$. Left: blowup curve $T(x_0)$ (bisection) and the touchdown point x^*. Middle: boost parameter of the light-cone profile at five base points versus $T(x_0) - t$; dotted: $T'(x_0)$. Right: distance to the boosted family.

fixed point $(\log 2, 2)$; bisection flags: $\max \psi_1 \to \infty$ if T is too large, $\psi_1 \to -\infty$ if too small). Blowup occurs at $x = 0$ at $T = 2.91356673558862$ (switching time $t_1 = 2.68$; the value changes by $1.7 \cdot 10^{-13}$ if the light-cone resolution is increased). Here Table 10 predicts, in the even sector, the eigenvalues 1 (time translation), -1 with eigenfunction $(1-\xi^2, -(1+\xi^2))$ and -2 (constant, the ODE energy mode), so the deviation should decay like $(T-t)^1$ with the shape $1-\xi^2$. Measured: $\omega = 1.00000$ (fit $Ae^{-\omega\tau} + Be^{-2\tau}$ on $\tau \in [3, 12.1]$; late-window slope 0.99996); at $T - t = 1.3 \cdot 10^{-6}$ the deviation is $\psi_1 - \log 2 \propto 1 - 1.0004\,\xi^2 + 5 \cdot 10^{-7}\xi^4$, and $(\psi_2 - 2)/(\psi_1(\cdot,0) - \log 2) = -1.0008$ at $\xi = 0$ and -2.0012 at $\xi = \pm 1$ (predicted -1, -2; the small offsets are the time-translation mode $(1,1)$ seeded by round-off); see Fig. 12. Note that, unlike MEMS with $q = 2$, the slowest mode is *not* spatially constant: the ODE energy mode (-2) is faster than the first spatial mode (-1).

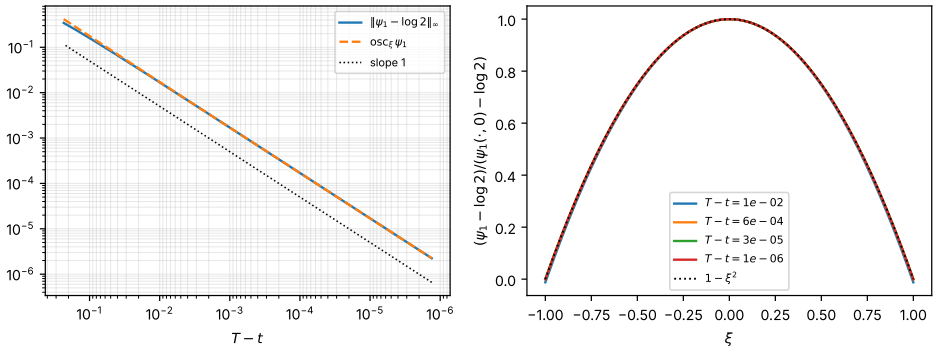


Figure 12: $u_{tt} - u_{xx} = e^u$: deviation from the ODE profile in the light cone (left) and its shape (right), normalised at $\xi = 0$, versus $1 - \xi^2$.

9.5 Discrepancies, caveats and reproducibility

Discrepancies with the analytic hints.

1. MEMS $q = 3$ (and $p = 5$): Jordan blocks at the collisions of the two eigenvalue families, including the gap eigenvalue $\lambda = -1$ for $q = 3$ ((D1)). Semisimplicity, which the hint asks to check, holds for $\lambda = 1$ and $\lambda = 0$ in all cases, but fails there.

2. The derivative model is outside the scope of (3.6) ((D2)); its spectrum is the free one with $s = -1$.
3. The closed-form lists are exact for polynomial eigenfunctions, but their meaning as $\mathcal{H}^k$-spectra requires $k \geq \frac{d}{2} - s + \omega_0$ ((D3)); the energy space is too small for MEMS in every dimension.
4. No discrepancy in the nonlinear tests: exponent $\frac{2}{3}$, constant $(9\lambda/2)^{1/3}$, rate $\omega_0 = \frac{2}{3}$ (MEMS), $\omega_0 = 1$ (e^u) and flat limit at the touchdown point are all confirmed. The hint that the profile "can also converge to a boosted profile" is correct only at points $x_0 \neq x^*$ of the blowup curve, with $a = T'(x_0)$.

Caveats.

- The eigenvalue computations only probe polynomial eigenfunctions (Proposition 9.1); absence of non-polynomial eigenfunctions in $\operatorname{Re}\lambda > -\omega_k$ rests on the connection-formula argument of Theorem 3.2, not on numerics.
- The nonlinear simulations cover one λ per geometry, $q = 2$, and data at rest; the light-cone stage is spectrally accurate, but the data at t_1 carry errors of 10^{-10}–$3 \cdot 10^{-8}$ ($d = 1$) and up to 10^{-7} (w) and $2 \cdot 10^{-5}$ (w_t) in $d = 2$ because of the corner singularity; these shift T (by up to $2 \cdot 10^{-7}$ in $d = 2$) but not the measured rates.
- Rates are measured on windows of 12–17 units of τ (5–7 decades of $T - t$), limited by the time-translation instability amplifying integration errors of size $\approx 10^{-14}$.
- The dynamic pull-in values for $\gamma = 0$ are finite-horizon ($t \leq 30$) values.
- We could not access the full text of [57] (network restrictions); the comparison above is with the summary of that work in Section 2.

Reproducibility. All scripts run from the project root with `python3 code/<name>.py` (numpy, scipy, sympy, mpmath, matplotlib); times are wall-clock times on a shared 4-core machine.

script	output	time
`num_spectra.py`	Table 10, Fig. 8, eigenvector checks	≈ 4 min
`num_spurious.py`	spurious-branch edges K_N (Section 9.1)	< 1 min
`num_mems_threshold.py`	pull-in values, location of the minimum	< 1 min
`num_mems_touchdown.py`	Table 11, Figs. 9 and 10	≈ 4–5 min
`num_mems_boost.py`	Fig. 11 and the boost table	≈ 4 min
`num_exp_touchdown.py`	Fig. 12	< 1 min
`num_common.py`	shared solvers (not run directly)	—

10 Summary, engineering consequences and open problems

10.1 What the analysis says about the devices

1. *Rates are universal and robust.* Once inertia is retained, the final approach to touchdown or runaway is governed by the spatially homogeneous ODE. For the Coulomb MEMS force the gap closes like $(9\lambda/2)^{1/3}(T-t)^{2/3}$, against $(3\lambda(T-t))^{1/3}$ in the parabolic (overdamped) model. Damping, restoring forces and smooth variations of the permittivity change only lower-order corrections (Section 4.7; proof sketches, not written out for the permittivity). They decay in similarity variables, so the hyperbolic rate is the true terminal one even for strongly damped devices. The parabolic rate is at most an intermediate regime.

Model (application)	Singularity	Non-negative modes	Gap ω_0	Status in these notes
Hyperbolic MEMS w^{-q}, $q>1$ (electrostatic, vdW, Casimir actuators)	touchdown $w \simeq c(T-t)^{\frac{2}{q+1}}$, flat in cone	1 (time), 0 (boosts)	$\min\{1, \frac{2(q-1)}{q+1}\}$	spectrum proved; stability modulo symmetries and with damping sketched for $d \le 3$ (Theorem 4.10 and Proposition 4.12); new
MEMS, focused profiles ψ_n, $d = 2, 3$	touchdown at a point, $\psi_n(0) \to 0$	n extra unstable	–	numerical; Conjecture 4.4
Reduced hyperbolic Gelfand $u_{tt} - \Delta u = \delta e^u$ (Cattaneo heat, FK source)	$u \simeq \log \frac{2}{\delta} - 2\log(T-t)$	1, 0	1 (sharp in $d = 1$)	stability sketched (Theorem 5.8); 1D exact check via Liouville; cf. [84]
Full Cattaneo–FK model	$\theta \simeq -\log(T-t)$ (parabolic rate)	1, 0 (tilts)	1 (Jordan block at -1)	sketched, less checked (Theorem 5.15)
Arrhenius source	no blowup	–	–	FK blowup only as an intermediate regime
Softening foundation / φ^4 / NLKG, $\lvert u\rvert^{p-1}u$	$u \simeq \kappa(T-t)^{-\frac{2}{p-1}}$	1, 0	1	known for $\gamma = m = 0$ [107]; lower-order lemma sketched (Theorem 6.12)
Co-rotational wave maps (inertial director dynamics)	$2\arctan \frac{r}{\sqrt{d-2}(T-t)}$	1	≈ 0.54 ($d = 3$)	known (Theorem 7.4); spectrum reproduced numerically
Relativistic membranes / Born–Infeld	shrinking spheres (lightlike collapse)	infinitely many	–	Donninger method not applicable to spheres; general case open (Section 7.6)
Acoustic toy model $u_{tt} - \Delta u = u_t^2$	$u \simeq -\log(T-t)$	1, 0 (shift; tilts)	1	radial sketched (Theorem 8.6); nonradial conjectural; cf. [113]
Inviscid Westervelt	degeneration $1-2kp \propto (T-t)^{1/2}$	1, $\frac{1}{4}$ (tilt), 0	formal	quasilinear, open (Section 8)

Table 12: Status of the models. "Non-negative modes" lists the eigenvalues $\lambda \ge 0$ of the linearisation in similarity variables together with their origin. "Sketched" means a proof sketch with all key estimates in the style of [41, 107].

2. *The terminal profile is flat.* In the backward light cone the stable profile is constant (or a Lorentz-tilted constant). The local analysis therefore gives a touchdown set that is a hypersurface $t = T^*(x)$, with the gap uniformly small on a region of size $\sim T^* - t$ around the first contact point (a minimum of T^*, where one expects $a^* = \nabla T^* = 0$, i.e. a flat profile), rather than the focused, point-like touchdown of the parabolic theory. The regularity of T^* is not proved (Remark 4.11). Focused, point-like touchdown profiles are found numerically for $d = 2, 3$, with extra unstable modes. A generic device is not expected to show them.

3. *Measured exponents converge slowly.* The approach to the self-similar regime is governed by the slowest stable mode, which for the Coulomb force is the spatially constant ODE-energy mode ($\lambda = -\frac{2}{3}$). In the simulations of Section 9.2, $w_{\min}$ is within 1% of $c(T-t)^{2/3}$ only once $w_{\min} \lesssim 10^{-2}$. Power-law fits over moderate gaps ($0.02 \le w_{\min} \le 0.25$) return exponents between 0.50 and 0.64 instead of $\frac{2}{3}$. Experiments and simulations that stop at gaps of a few percent should be compared with the corrected law $c(T-t)^{2/3}(1 + A(T-t)^{2/3})$, not with the bare power law.

4. *Boundaries do not matter locally.* The results are local in the light cone of the singular point. Clamping conditions, the geometry of the device and far-away material parameters enter only through the touchdown time T^* and the tilt a^*.

5. *Which singularity actually forms is model-dependent.* The full Maxwell–Cattaneo thermal-explosion model blows up at the *parabolic* logarithmic rate, not at the hyperbolic one of the frequently used reduced model. Arrhenius kinetics removes the blowup altogether. Viscous nonlinear acoustics leaves the light-cone setting entirely.

10.2 Open problems suggested by these notes

1. Turn the proof sketches of Theorems 4.10 and 5.8 and Proposition 4.12 into complete proofs. The new ingredients relative to [107] are the negative scaling exponent (handled by a finite-rank splitting of the free operator, Lemma 4.5, a soft version of the constructions in [107, 61]) and the need to stay away from $w = 0$, which the L^∞ control gives.

2. Prove the existence of the focused MEMS profiles ψ_n in $d = 2, 3$ (Conjecture 4.4) and determine their co-dimension. A shooting argument around the singular static solution $A|x|^{2/(q+1)}$, as in [11], looks feasible.

3. Go beyond the light cone: regularity of the touchdown hypersurface, $a^* = \nabla T^*$, and the hyperboloidal approach of [6, 18] for MEMS.

4. Models not covered by the method in its present form: MEMS plates (Δ^2, no light cone), nonlocal MEMS, the full Cattaneo model in higher regularity, and the quasilinear problems (Westervelt degeneration, membranes).

5. Compare the rigorous picture of Section 4 in detail with the formal analysis and numerics of [57, 73]. This could not be done here because the paper was not accessible.

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